



# Open Chiplet Atlas Harness (OCAH)

## Integrator Guide

Version 0.0.7, 2026-09-11

# Table of Contents

|                                                           |    |
|-----------------------------------------------------------|----|
| Revision History .....                                    | 2  |
| 1. SMU Top-Level Integration .....                        | 3  |
| 1.1. Address Map Integration .....                        | 3  |
| 1.1.1. Register collateral .....                          | 3  |
| 1.2. Immutable ROM FW Integration .....                   | 4  |
| 1.2.1. Weak-symbol override method .....                  | 4  |
| 1.2.2. SMC ROM FW integration .....                       | 4  |
| 1.2.3. SEP ROM FW integration .....                       | 5  |
| 1.2.4. Key Manager ROM FW integration .....               | 5  |
| 1.3. System Management Controller (SMC) Integration ..... | 5  |
| 1.3.1. Clock and Reset .....                              | 5  |
| 1.3.2. Address Map and Aperture .....                     | 6  |
| 1.3.3. AXI Interface Configuration .....                  | 7  |
| 1.3.4. GPIO Pad Integration .....                         | 7  |
| 1.3.5. Peripheral Interface Connections .....             | 7  |
| 1.3.6. Memory Subsystem Integration .....                 | 9  |
| 1.3.7. Memory Repair Integration .....                    | 9  |
| 1.3.8. Interrupt Integration .....                        | 10 |
| 1.3.9. Telemetry and Debug Integration .....              | 11 |
| 1.3.10. Bring-Up Sequence .....                           | 11 |
| 1.3.11. SMC Port Interface .....                          | 11 |
| 1.3.12. GPIO Pin Requirements .....                       | 16 |
| 1.4. Debug & Test Ports (DTP) Integration .....           | 19 |
| 1.4.1. Configuration .....                                | 20 |
| 1.4.2. Instantiation .....                                | 20 |
| 1.4.3. Clock and Reset .....                              | 20 |
| 1.4.4. Constraints .....                                  | 20 |
| 1.4.5. STAP Scan Chain Topology .....                     | 21 |
| 1.4.6. DTP Port Interface .....                           | 21 |
| 1.5. Security Processor (SEP) Integration .....           | 25 |
| 1.5.1. Module Overview .....                              | 25 |
| 1.5.2. Clock and Reset Requirements .....                 | 25 |
| 1.5.3. AXI Interfaces .....                               | 25 |
| 1.5.4. Memory Interfaces .....                            | 26 |
| 1.5.5. Cryptographic Subsystem Connections .....          | 26 |
| 1.5.6. eFuse and Lifecycle .....                          | 26 |
| 1.5.7. SPI Host Interface .....                           | 27 |
| 1.5.8. Bring-Up Considerations .....                      | 27 |
| 1.5.9. SEP Port Interface .....                           | 27 |
| 1.6. SMU Composition and Internal Wiring .....            | 31 |
| 1.6.1. Module Variants and IP Integration .....           | 31 |
| 1.6.2. Clock and Reset Architecture .....                 | 33 |

|                                                                                  |    |
|----------------------------------------------------------------------------------|----|
| 1.6.3. SMU AXI Crossbar Address Map                                              | 33 |
| 1.6.4. SMU Default Parameters                                                    | 34 |
| 1.6.5. IC_RESET TDR Structure                                                    | 34 |
| 1.6.6. DTP Parameter Pass-Through                                                | 36 |
| 1.6.7. DTP Port Mapping                                                          | 37 |
| 1.6.8. SMC Port Mapping                                                          | 41 |
| 1.6.9. SMC-DTP Internal Interconnection                                          | 46 |
| 1.6.10. SEP-Conditional Wiring                                                   | 47 |
| 1.6.11. SMU Port Interface                                                       | 47 |
| 2. System Management Network (SMN) Integration                                   | 55 |
| 3. Project Defines                                                               | 56 |
| 3.1. SIMULATION                                                                  | 56 |
| 4. Prim Cell Replacement and Don't-Touch Constraints                             | 57 |
| 4.1. Naming Requirement                                                          | 57 |
| 4.2. Constraint Requirement                                                      | 57 |
| 4.2.1. After Synthesis                                                           | 58 |
| 4.3. Simulation                                                                  | 59 |
| 5. CDC Timing Constraints                                                        | 60 |
| 5.1. Why crossings are bounded rather than timed                                 | 60 |
| 5.2. What is constrained                                                         | 60 |
| 5.2.1. Why protocol-protected data still needs a bound                           | 61 |
| 5.3. The two layers                                                              | 61 |
| 5.4. What is covered                                                             | 62 |
| 5.5. Integration checklist                                                       | 62 |
| 5.5.1. 1. Repoint the constraints at the pin of the cell inside the prim wrapper | 62 |
| 5.5.2. 2. Point the files at your hierarchy, or regenerate                       | 63 |
| 5.5.3. 3. Set the default bound                                                  | 64 |
| 5.5.4. 4. Declare your own clock groups                                          | 64 |
| 5.5.5. 5. Declare periods for generated clocks                                   | 64 |
| 5.5.6. Also check                                                                | 65 |
| 5.6. Confirming it worked                                                        | 65 |
| 5.7. Scope and limitations                                                       | 65 |

This guide provides instructions and best practices for integrating OCAH subsystems into your chiplet design. Each section covers integration considerations, interfaces, and configuration requirements for the respective subsystem.

# Revision History

This document tracks major revisions and updates to the OCAH Integrator Guide.

*Table 1. Revision History*

| Version | Date       | Author               | Description                                                                                                                                                                                                                                         |
|---------|------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0.1   | 2026-06-17 | Alessandro Ottaviano | Initial integrator guide draft                                                                                                                                                                                                                      |
| 0.0.2   | 2026-08-13 | Alessandro Ottaviano | Completed the documentation migration: added primitive-cell replacement and CDC constraint guidance, corrected DTP/JTAG integration, and aligned SMC/SEP paths, GPIO and DFX status descriptions, and cross-guide references with the current tree. |
| 0.0.3   | 2026-08-21 | Ben Melton           | Added a Project Defines section covering compile-time defines that apply across OCAH, starting with <b>SIMULATION</b> : how to set it, the testbench hooks and models it enables, and the flows it must not reach.                                  |
| 0.0.4   | 2026-08-31 | Ben Melton           | Documented the SMU IC_RESET TDR slice layout (SMC, SEP, external) and marked the external slice as adopter-defined.                                                                                                                                 |
| 0.0.5   | 2026-09-02 | Alessandro Ottaviano | Consolidated immutable ROM FW integration guidance, including the generic weak-symbol override method and SMC, SEP, and Key Manager ROM FW integration sections.                                                                                    |
| 0.0.6   | 2026-09-10 | Ben Melton           | Documented the external TRNG aperture as a full-window AXI-Lite passthrough whose intra-aperture decode (SLVERR on unmapped offsets, no aliasing onto live registers) is an adopter constraint.                                                     |
| 0.0.7   | 2026-09-11 | Alessandro Ottaviano | Noted that <code>smc_ext_interrupts_i</code> is an active-high level or a pulse long enough to be sampled on <code>clk_smc</code> .                                                                                                                 |

# Chapter 1. SMU Top-Level Integration

The System Management Unit (SMU) comprises three peer subsystems: the System Management Controller (SMC), the Debug & Test Ports (DTP), and the Security Processor (SEP). The subsections below describe each subsystem’s interfaces independently, followed by the [SMU Composition](#) section that explains how they are wired together inside the SMU and how to select between the `smu.sv` logic-level core and the `smu_wrapper.sv` reference integration.

## 1.1. Address Map Integration

The OCAH address map has three levels, and the assignment contract splits across the integrator, SEP firmware, and SMC firmware. Assign and program them from this section; the deeper subsystem and crossbar sections referenced below give the per-block detail.

- **Chiplet envelope (integrator).** The SoC/chiplet address map assigns each subsystem a global base and reserved size, and the adopter’s fabric decoder routes those ranges to the SMU. The SMU crossbar preserves offsets and does not constrain envelope size, so the integrator owns non-overlap and containment at the SoC level.
- **SEP and SMC chiplet apertures (firmware).** Each subsystem exposes a chiplet-visible aperture whose base and size are programmable at runtime. Inbound transactions to an aperture are remapped global-to-local before they reach the subsystem.
- **Local-alias views (fixed in RTL).** Internally each subsystem addresses its own resources through a fixed local map: the SMC at `0xC000_0000` (see [Address Map and Aperture](#)), the SEP through its own local map. These are RTL-fixed and not integrator-assigned.

Table 2. Address-map assignment contract

| Level                | Owner        | Assigned / programmed by                                                                       |
|----------------------|--------------|------------------------------------------------------------------------------------------------|
| Chiplet envelope     | Integrator   | SoC/chiplet address map + adopter fabric decode                                                |
| SEP chiplet aperture | SEP firmware | <code>SEP_GLOBAL_BASE_ADDR / SEP_REGION_SIZE</code> (the <code>sep_cpu_ctrl</code> CSRs)       |
| SMC chiplet aperture | SMC firmware | <code>GLOBAL_BASE / REGION_SIZE</code> (the SMC CPU Control CSRs at <code>0xC001_0000</code> ) |
| SEP / SMC local view | Fixed (RTL)  | Offsets within each subsystem’s local map (SMC local base <code>0xC000_0000</code> )           |

Programming order and constraints: firmware programs the SEP and SMC aperture CSRs during early boot, **before any cross-subsystem AXI traffic is issued**, and must keep the two apertures non-overlapping and inside the chiplet envelope the integrator assigned. The full runtime crossbar behaviour and the exact CSR plumbing are in [Firmware programming order](#) under [SMU Composition and Internal Wiring](#).

### 1.1.1. Register collateral

The machine-readable address map ships in the repository tree; integrators consume it from the release checkout or tarball rather than a separate package. The source of truth is distributed by ownership: each subsystem or IP keeps its canonical register sources and generated outputs beside its implementation. The `integration/` directory centralizes symlinks to that distributed collateral so an integrator does not have to discover every owner-local path.

`integration/rdl/<subsystem>/` is a flat, self-contained list of the SystemRDL files required by that subsystem. Filenames remain unchanged, and shared dependencies appear in every subsystem that needs them. Non-RDL support files needed by relative includes remain at the `integration/rdl/` root—for example, `integration/rdl/i3c_defines.svh` sits there because both `integration/rdl/smc/registers.rdl` and `integration/rdl/smu/registers.rdl` reach it through the I3C core’s own parent-relative include (`../i3c_defines.svh`), and that relative path only resolves if the header is one directory above every subsystem folder that needs it. The matching generated IEEE 1685-2014 component descriptions

use the same flat subsystem organization under `integration/ipxact/<subsystem>/`.

These directories are generated symlink indexes; run `python3 scripts/collect_integration.py` after adding or moving collateral. Their canonical targets remain under `hw/sys/<sys>/regs/`, `hw/ip/<ip>/regs/`, `hw/common/axi/`, and `vendor/...` —the `regs/` element of every component’s directory layout, described in the *Getting Started Guide*. Other generated outputs remain under each owner’s `regs/gen/{c,sv,svh,py,ral,adoc,html,ipxact}/` directory.

For the architecture-level responsibility split between the integrator, firmware, and the internal SEP/SMC maps, see the *OCAH Memory Map* section of the Technical Reference Manual.

## 1.2. Immutable ROM FW Integration

OCAH contains three immutable ROM FW images: the SMC production ROM FW, the SEP production ROM FW, and the Key Manager production ROM FW. Each image has two integration contracts. The adopter connects and configures the corresponding ROM macro, and the final ROM FW image must be present before its processor leaves reset. Where ROM FW must initialize or access an adopter-owned peripheral outside OCAH, the image may also expose build-time integration hooks.

### 1.2.1. Weak-symbol override method

An immutable ROM FW image may provide selected vendor-neutral integration hooks as weak symbols. Supply an adopter implementation as a strong definition of the same symbol before the final image is frozen. The strong implementation must program the same adopter-owned IP instantiated by the RTL integration, and its signature, ABI, return values, and documented semantics must match the weak declaration.

This mechanism applies only when the ROM FW needs to interact with a peripheral outside OCAH. It is a build-time image-composition mechanism, not a runtime extension mechanism. Include the adopter implementation in the final ROM FW build so that it replaces the weak definition at link time.

Inspect the final ELF or linker map after linking. For example, `riscv64-unknown-elf-nm` must report each selected implementation as a strong symbol rather than `W`; successful linking alone does not prove that the weak default was replaced.

### 1.2.2. SMC ROM FW integration

Connect `smc_rom_intf_req_o` and `smc_rom_intf_rsp_i` to the adopter’s ROM implementation. In the current four-core configuration, the request provides the ROM clock, enable, and 14-bit 64-bit-word address, and the response returns one 64-bit word. This is a direct memory interface, not AXI-Lite. The 128 KiB ROM FW image must be present before the first boot; its build is under `hw/sys/smc/bootrom/prod/`.

Connect the ROM macro configuration input reached as `rom_cfg_i` and `mem_cfg_i` according to the selected macro’s contract. OCAH forwards these 11 bits without assigning technology-specific semantics; the reference `smc_ip_integration.sv` ties them to zero. Set the eFuse shadow value that drives the CPU’s `rom_flip_endianness_i` path to match the byte order of the image and ROM macro. Deasserted passes each 64-bit read word unchanged; asserted reverses its eight bytes.

*Table 3. SMC ROM FW override inventory*

| Hook                                                                                                                                        | Weak default                                                                                                 | Rationale                                                                                                               | Override and integration action                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>smc_pll_init()</code>                                                                                                                 | Returns <code>SMC_PLL_SUCCESS</code> without programming PLL hardware.                                       | PLL implementation is technology- and vendor-specific, so OCAH does not include a vendor-specific PLL driver.           | Override when <code>BL0_PLLCLK</code> is asserted and the adopter's PLL requires ROM-time configuration. Provide the same <code>smc_pll_result_t smc_pll_init(void)</code> signature in the final ROM FW build. Any result other than <code>SMC_PLL_SUCCESS</code> is fatal: the ROM FW records an interface error and remains in <code>wfi</code> .                         |
| <code>i3c_release_reset()</code> ,<br><code>cfg_ps()</code> , <code>init_i3c_ctrl()</code> ,<br>and<br><code>I3C_GetDriverInstance()</code> | Permit the ROM FW to boot without an active I3C implementation; I3C data operations report a hardware error. | OCAH includes an I3C implementation, but the adopter may select a different I3C target and driver for the final system. | When the ROM FW must use an active I3C target, select the open HCI implementation with <code>I3C_CORE=swap</code> , or connect the adopter's I3C target through the SMC external AXI-Lite master ( <code>smc_external_req_o</code> and <code>smc_external_resp_i</code> ) and link strong implementations that conform to <code>drivers/include/i3c_target_driver.h</code> . |

### 1.2.3. SEP ROM FW integration

The SEP exposes the `sep_boot_rom` memory interface for its production ROM FW image. The detailed ROM-macro configuration, image-loading contract, and adopter override inventory are not yet documented here.

Table 4. SEP ROM FW override inventory

| Hook                    | Status                                               |
|-------------------------|------------------------------------------------------|
| <i>To be documented</i> | No adopter override requirement is defined here yet. |

### 1.2.4. Key Manager ROM FW integration

The Key Manager exposes the `km_rom_mem` interface through SEP for its production ROM FW image. The detailed ROM-macro configuration, image-loading contract, and adopter override inventory are not yet documented here.

Table 5. Key Manager ROM FW override inventory

| Hook                    | Status                                               |
|-------------------------|------------------------------------------------------|
| <i>To be documented</i> | No adopter override requirement is defined here yet. |

## 1.3. System Management Controller (SMC) Integration

The System Management Controller (SMC) is a subsystem responsible for power management, clock control, reset sequencing, and peripheral interface coordination. The `smc.sv` module exposes logic-level interfaces; all technology-specific components (PLLs, PVT sensors, eFuses, memory macros, GPIO padings) are instantiated externally by the adopter. See the [SMU Module Variants](#) section for how the reference wrappers demonstrate this integration.

For detailed architectural documentation about the SMC internal design, refer to [hw/sys/smc/doc/](#). The public integration examples are [hw/top/smc\\_wrapper.sv](#) and [hw/top/smc\\_ip\\_integration.sv](#).

### 1.3.1. Clock and Reset

The SMC requires three external clock inputs: `clk_smc_i` (primary system clock driving the CPU cluster, fabric, and peripherals), `clk_periph_i` (peripheral clock driving the peripherals), and `clk_ref_i` (reference clock for timing-sensitive operations). All must be generated externally by the adopter's PLL infrastructure; the SMC provides AXI-Lite control outputs for PLL configuration (see [Peripheral Interface Connections](#)). The peripheral clock is used to drive the I3C interface, I2C interface, UART interface, and AVSBus interface.

Hold `rst_cold_ni` asserted (low) while power supplies and clocks stabilize. Assert `powergood_i` once supplies reach nominal, then deassert `rst_cold_ni` (drive high) to start the boot sequence.

The SMC generates AASD (async-assert / sync-deassert) reset outputs: `rst_primary_smc_clk_no` for the main clock domain, `rst_primary_periph_clk_no` for the peripheral clock domain, and `rst_primary_ref_clk_no` for the reference clock domain. These ensure clean reset release without metastability across the system hierarchy.

Reset domain overrides are delivered through a single typed input, `jtag_reset_ctrl_i` (`smc_pkg::jtag_smc_reset_ctrl_t`), which carries all SMC reset override enables and active-low values. The DTP drives this input through the JTAG IC\_RESET mechanism (see [SMC-DTP Internal Interconnection](#)); most integrations leave all override enables deasserted.

### 1.3.2. Address Map and Aperture

The SMC has a chiplet-visible aperture and an internal local address view. Internally, the SMC uses a fixed local base address of `0xC000_0000` and a fixed local window size of 16 MiB. SMC-local addresses (registers, memories, remapped regions) are defined as offsets from that base. The chiplet-visible aperture is adopter-defined: the chiplet or SoC address map assigns the global base and size that external masters can use to reach SMC through the SMU crossbar.

When integrating the SMC into a chiplet or SoC, the adopter assigns a system base address and reserved size for the SMC window. The system base (e.g. `0x0800_0000` in a chiplet address map) may be different from the SMC-local base `0xC000_0000`. Firmware programs the SMC `GLOBAL_BASE` / `REGION_SIZE` CSRs during early boot to match that assignment. The adopter's fabric or address decoder must route the reserved system range to the SMC block so that offsets are preserved: system base + offset maps to SMC-local `0xC000_0000` + offset.

*Table 6. Address Base Mapping*

| View                 | Base Address             | Description                                                         |
|----------------------|--------------------------|---------------------------------------------------------------------|
| SMC-local (internal) | <code>0xC000_0000</code> | Local alias address (fixed). Used by SMC CPU, firmware, and fabric. |
| System (adopter)     | Adopter-defined          | Base of the chiplet/SoC aperture assigned to SMC.                   |

The SMC fabric uses three registers in the CPU Control block (`0xC001_0000`) to distinguish local from remote addresses:

- `LOCAL_BASE` (read-only, fixed at `0xC000_0000`): The local alias base. Together with `REGION_SIZE`, this is the address space the SMC CPU and internal fabric use to access SMC resources.
- `GLOBAL_BASE` (read-write, defaults to `0x4000_0000`): The system-level base address. Firmware must program this during early boot to match the adopter's system address assignment, enabling the SMC to issue outbound transactions using global system addresses.
- `REGION_SIZE` (read-write, resets to `0x0100_0000` / 16 MiB): The size of both the local alias window and the global aperture. It sizes the SMC input fabric's local/global split as well as SMU-level routing and SEP-to-SMC decode. Firmware must not program it to zero, and must keep `LOCAL_BASE + REGION_SIZE` inside the 32-bit SMC-local address space.

For full details on alias remapping and outbound routing, see the SMC Fabric documentation ([hw/sys/smc/doc/fabric.adoc](#)).

For a detailed breakdown of all components within the SMC local map, refer to the SMC Memory Map documentation ([hw/sys/smc/doc/memmap.adoc](#)).

### 1.3.3. AXI Interface Configuration

The SMC provides three distinct AXI interface groups serving different access paths. The `sys_axi_in` interface accepts requests from external system masters requiring access to SMC-managed resources. This interface uses a 56-bit address space with 64-bit data width and 6-bit transaction IDs. Connect this interface to your system fabric for normal operational access from application processors or DMA controllers.

The `sep_axi_in` interface provides dedicated access from the Security Processor (SEP) subsystem. This interface shares the same sizing parameters as the system input but operates independently to maintain security domain separation. Use this interface only when integrating alongside the SEP subsystem. The interface may remain unconnected if your integration does not include SEP.

The `output_axi` interface carries filtered and remapped requests from the SMC to external system resources. This interface uses an 8-bit transaction ID field to accommodate merged traffic from multiple input paths. Connect this interface to your system interconnect to allow SMC components access to external memory and peripherals. An AXI VIP or equivalent BFM is a suitable stand-in during integration verification before the real system fabric is available.

The `jtag_axi_in` interface provides debug access through JTAG mechanisms. This interface uses reduced 2-bit transaction IDs and connects to your JTAG debug infrastructure.

### 1.3.4. GPIO Pad Integration

GPIO pad integration requires careful attention to function assignment and pull requirements. The GPIO table below lists each signal, bump location, function, recommended pull, and integration notes. Default input/output direction is defined by `smc_pading_pkg::DefaultDirectionMap`, not by the table.

The public `hw/top/smc_wrapper.sv` and `hw/top/smc_ip_integration.sv` examples show the pad connections. For bidirectional pins, implement proper tri-state buffers controlled by the `core2pad_en_o` signals. Connect multiplexed UART, I2C, I3C, and SPI pins to external devices according to the function assignments in the GPIO table.

Configuration straps are captured separately from the GPIO table. Establish them before cold reset deassertion and keep them stable during capture. The generated reset-unit `STRAPS_LO` and `STRAPS_HI` register descriptions define the decoded bit assignments, including `BYPASS_SRAM_REPAIR` at `STRAPS_LO[13]` and `TEST_EN` at `STRAPS_LO[14]`.

### 1.3.5. Peripheral Interface Connections

The SMC exposes logic-level interfaces to several technology-specific subsystem components that the adopter instantiates externally. The subsections below describe each connection in turn.

#### Adopter Peripheral Window

The `smc_external_req_o` and `smc_external_resp_i` signals form a 32-bit AXI-Lite interface to the adopter's own peripheral register maps. Any technology- or product-specific block that SMC firmware needs to reach sits behind this window rather than on a dedicated per-peripheral port.

The window is the recommended home for the control/status registers of logic that the SMC must configure or monitor but that the SMC itself does not implement — pading configuration logic, clock generation, on-die sensors, memory-technology SHIMs, and similar adopter-owned blocks are typical examples. For example, runtime reconfiguration of pin directions, pull settings, and peripheral multiplexing is expected to be reachable here. The SMC decodes the aperture and forwards the transaction; what lives behind it is entirely adopter-defined.

The window is split into a mandatory region, whose contents and placement the SMC memory map fixes and every

integration must implement, and a supplementary region for anything else the adopter chooses to expose. For placement and the full register layout, refer to the SMC Memory Map documentation ([hw/sys/smc/doc/memmap.adoc](http://hw/sys/smc/doc/memmap.adoc)).

## eFuse Interface

### NOTE

This guide uses **eFuse** and **OTP** interchangeably for the same one-time programmable bank and the SHIM that fronts it. Port names mix both (`efuse_*`, `otp`); they refer to the same interface.

The eFuse interface consists of two parts:

- `efuse_bank_ctrl` — a 32-bit AXI-Lite interface used to access the SHIM control CSRs (for example, foundry-specific bank timing such as the initialization delay before sensing can begin).
- `efuse_shim_command` — the custom fuse command interface used to issue read, program, and program/read-back operations to the eFuse bank.

**Adopter responsibility.** The `efuse_shim_command` interface is intentionally generic and technology-neutral. The adopter must implement an Interface SHIM that converts these generic commands into the hardware signals required by their OTP macro. The conversion is driven by the adopter’s own read and program state machines inside the SHIM; OCAH only defines the command/response contract, not the macro-side signaling.

The command and response payloads are exchanged as packed structs:

```
// Controller -> SHIM
typedef struct packed {
    addr_t      address;           // Target address (in bits)
    logic        program_data;     // Data to program (when programming)
    access_length_words_t access_length_words; // Number of WORD_WIDTH words to access
    command_t    command;          // READ / PROGRAM / PROGRAM_READ_BACK
    logic        valid;            // Request valid
} fuse_command_req_t;

// SHIM -> Controller
typedef struct packed {
    data_t data; // Read data (one WORD_WIDTH word)
    logic status; // Operation status (0 = OK, 1 = error)
    logic valid; // Response valid (qualifies `data` for one cycle)
} fuse_command_resp_t;
```

**Fuse sensing protocol.** During automatic sensing the controller issues a single read with `access_length_words` set to the entire bank (all `NUM_WORDS` words). The SHIM is expected to stream the bank back one `WORD_WIDTH` word at a time: for each OTP access it drives the read `data` onto `fuse_command_resp_o.data` and asserts `valid` for **exactly one cycle**. The controller consumes one word per cycle that `valid` is high, advancing to the next eFuse word on each assertion.

### WARNING

`valid` must be asserted for only a single cycle per returned word. If the SHIM holds `valid` high for two cycles, the controller will sample the same `data` value a second time and count it as the data for the **next** eFuse word, shifting all subsequent words and corrupting the sensed shadow contents. The adopter’s SHIM FSM is responsible for producing a clean one-cycle `valid` strobe per `WORD_WIDTH` access.

The `shadow_regs_o` output provides read-only access to the sensed fuse values. During bring-up the SMC automatically senses the fuse contents and propagates them through the shadow registers; the `smc_fuse_sense_done_o` signal indicates completion of this sensing operation.

### 1.3.6. Memory Subsystem Integration

The SMC includes an embedded RISC-V CPU cluster with associated memory interfaces. See [Immutable ROM FW Integration](#) for the ROM FW integration contract.

#### SRAM and Cache Array Integration

Scratch RAM interfaces (`smc_scratch_ram_intf_req_o` and `smc_scratch_ram_intf_rsp_i`) connect to 32 independent RAM banks totaling the scratchpad memory space. Cache memory interfaces for L1 instruction and data caches (`smc_l1_icache_*` and `smc_l1_dcache_*`) connect to tag and data RAM banks as specified in the port table.

The `smc_disable_sram_auto_init_i` input controls automatic memory initialization behavior. When deasserted, the SMC automatically zeroes all SRAM contents during bring-up. The `smc_init_mem_done_o` output indicates completion of this initialization sequence. The `SRAM_AUTO_ZERO_DISABLE` strap (`STRAPS_LO[26]`) can disable this automatic zeroing behavior if your integration requires preserved SRAM contents across resets.

### 1.3.7. Memory Repair Integration

The SMC implements Built-In Redundancy Analysis (BIRA) memory repair to address manufacturing defects in SRAM arrays. Repair uses redundant memory elements configured with data stored in eFuse memory, replacing defective cells detected during manufacturing test.

Memory repair executes automatically during boot after eFuse sensing completes and before MBIST operations. The process must complete successfully before the CPU cluster can safely access SRAM arrays. Repair is required on every cold reset but can be bypassed during Function Level Reset (FLR) operations for faster recovery.

#### Control Signals:

- `smc_fuse_sense_done_o`: eFuse controller output indicating fuse sensing completion. External DFT logic monitors this signal to trigger repair operations.
- `skip_mem_repair_o`: Reset unit output indicating repair should be bypassed. It is driven by the isolation request inputs (`isolate_req_pin_sync_smc` or the SMC isolation register), not directly by the strap. When asserted, external repair logic should skip operations and immediately assert completion status.
- `ext_boot_seq_done_i`: External signal indicating the external boot sequence (memory repair and shadow register override) is complete. Held low to pause reset release until the external agent gives the go-ahead; required before the system can continue its boot sequence

#### Repair Data:

Repair data is stored in the eFuse BIRA shadow register as a 16-kilobit `repair_data` field. The eFuse controller reads this data during fuse sensing and makes it available through shadow registers. The external memory repair controller must read repair data from `shadow_regs_o` and apply it to SRAM arrays according to your memory architecture.

#### Status Monitoring:

Repair completion and success are reported through `DFX_CTRL_STATUS.STATUS_SMU` at `0xC000_B800`. The `mem_repair_done` bit indicates completion regardless of success, while `mem_repair_success` indicates successful repair. Both bits are set by the

external repair controller. The SMC ROM FW monitors these bits during initialization to verify repair completion.

### Bypass Configuration:

The `BYPASS_SRAM_REPAIR` configuration strap (`STRAPS_LO[13]`) is captured into the reset-unit strap registers. Keep these three mechanisms distinct:

- No RTL path drives `skip_mem_repair_o` from this strap directly. `skip_mem_repair_o` is driven from the isolation request path.
- Production ROM reads the strap and, when asserted, skips repair-status validation and writes `MBIST_STATUS_MEM_REPAIR_BYPASSED` to scratchpad register 15 (`SMC_SCRATCH_MBIST_STATUS, 0xC003_90F8`). It does not cause the external repair controller to skip physical repair.
- External repair logic independently honors `skip_mem_repair_o` to decide whether to bypass physical repair operations.

Use bypass mode only when SRAM arrays are guaranteed defect-free, such as during development or with known-good components.

### Integration Requirements:

Provide an external memory repair controller that monitors `smc_fuse_sense_done_o` to initiate repair, respects `skip_mem_repair_o` for bypass control, reads repair data from eFuse shadow registers, applies repair configuration to SRAM arrays, asserts `ext_boot_seq_done_i` and updates DFT Control Status registers with completion status. The controller operates independently of the SMC core and must complete before the CPU cluster accesses SRAM.

For direct access to the eFuse besides from the shim, i.e. from a DFT fuse controller, adopters must ensure the integrated logic will not interfere with accesses from the OCAH's efuse interface.

During FLR operations, the isolation request path asserts `skip_mem_repair_o` to bypass repair for rapid reset sequences. Normal cold reset sequences always execute repair unless isolation is already asserted.

## 1.3.8. Interrupt Integration

The SMC aggregates interrupts from multiple sources into `cpu_interrupts_o`, which feeds the PLIC of the embedded RISC-V CPU cluster. The interrupt vector is assembled in `hw/sys/smc/rtl/smc_base.sv` and `hw/sys/smc/rtl/smc_peripherals/rtl/smc_peripherals.sv`. The authoritative bit-level map (RTL trace and PLIC source ID note) is maintained in a single place:

See the *SMC CPU interrupt vector map* in the Technical Reference Manual.

The SMC PLIC and CPU interrupt vector are internal. `sync_irq_o` is **not** the interrupt aggregate—it is the software-controlled `SYNC_REG.sync` global synchronisation bit in the reset unit, unrelated to PLIC interrupt delivery. Do not connect an external interrupt controller to `sync_irq_o`.

External interrupt sources visible to the adopter are `smc_ext_interrupts_i` (SMC external interrupt input vector, width `NUM_INT_TO_SMC`), `smc_ext_mailbox_interrupts_o` (outbound mailbox notifications, width `NUM_MAILBOXES`), `gpio_interrupt_o` (GPIO edge/level events, width `NUM_GPIO_WRAPS`), and `uart_interrupt_o` (UART TX/RX events, width `NUM_UART`). Drive `smc_ext_interrupts_i` as an active-high level or as a pulse that stays high long enough to be sampled on `clk_smc`. The pin is three-flop synchronized; a short asynchronous pulse can be lost. There is no edge-triggered mode. For the full PLIC source map see the SMC Technical Reference Manual.

### 1.3.9. Telemetry and Debug Integration

For system-level debug and telemetry, the SMC implements ATB (Advanced Trace Bus) receiver interfaces. The telemetry interface group (`telemetry_atdata_i`, `telemetry_atid_i`, `telemetry_atvalid_i`, `telemetry_atready_o`) operates in a dedicated `clk_telemetry_i` clock domain with separate `rst_telemetry_ni` reset. Connect these signals to ATB trace sources for debug data collection.

At the SMC boundary, `ext_debug_bus_i` is 512 bits wide; the SMU concatenates the adopter-visible 128-bit input with the SEP's internal 384-bit bus before driving SMC. The typed `xtrigger_ss_i` / `xtrigger_ss_o` ports are the only cross-trigger signals at the SMC boundary. The adopter-visible SMU cross-trigger interfaces are `xtrig_ctm_*`, `xtrig_clk_stop_req_i`, and the CTP groups.

Trace memory interfaces (`trace_mem_req_o` and `trace_mem_resp_i`) connect to trace RAM instances for storing captured debug data. `smc_ip_integration.sv` shows an example set of RAM instances that can be adapted to your debug requirements.

### 1.3.10. Bring-Up Sequence

The recommended bring-up sequence begins with asserting `rst_cold_ni` while stabilizing power supplies and clocks. Once all supplies reach nominal voltages, deassert `rst_cold_ni` to initiate the SMC boot sequence. The SMC immediately begins fuse sensing to read configuration values and security credentials.

Monitor the `smc_fuse_sense_done_o` signal to detect completion of fuse sensing. Configuration straps sampled during this phase determine boot modes and operational settings. After fuse sensing completes, the SMC releases internal subsystem resets based on the programmed reset sequencing configuration.

The CPU cluster begins executing the ROM FW after reset release. The ROM FW performs additional initialization including peripheral configuration, memory testing (unless disabled), and system health checks. The `smc_init_mem_done_o` signal indicates completion of memory initialization if automatic SRAM zeroing is enabled.

To support Function Level Reset (FLR), assert `cfg_flr_pf_active_i` to trigger a warm reset sequence. The SMC asserts `isolate_req_o` bits for the affected subsystem indices. The adopter's power controller drives `ss_reset_complete_i` high once the subsystem reset is complete; the SMC synchronizes this into the readable `SS_RESET_COMPLETE` register. Isolation remains asserted until cleared by software: SMC firmware must poll `SS_RESET_COMPLETE` and then write `ISOLATE_REQ_SMC_REG` to clear the FLR isolation latch. Do not assume isolation releases automatically on `ss_reset_complete_i`.

See `hw/top/smu_wrapper.sv` and `hw/top/smc_ip_integration.sv` for reference bring-up wiring examples covering clock, reset, strap, and peripheral connections.

### 1.3.11. SMC Port Interface

The SMC exposes a comprehensive set of interfaces for system integration. The following table defines all SMC top-level ports including clocks, resets, AXI interfaces, peripheral connections, and control signals. Refer to the detailed SMC documentation for internal signal descriptions and timing requirements.

Table 7. SMC Port Declaration

| Interface Name         | Type  | Direction | Width | Description                                                                            | Connection | Default value (if applicable) |
|------------------------|-------|-----------|-------|----------------------------------------------------------------------------------------|------------|-------------------------------|
| <code>clk_smc_i</code> | logic | Input     | 1     | Primary SMC clock domain. All SMC internal logic operates synchronously to this clock. | PLL output | N/A                           |

| Interface Name            | Type                               | Direction | Width | Description                                                                                 | Connection                                | Default value (if applicable) |
|---------------------------|------------------------------------|-----------|-------|---------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------|
| clk_ref_i                 | logic                              | Input     | 1     | Reference clock domain. Used for timing-sensitive operations and peripheral interfaces.     | PLL reference clock                       | N/A                           |
| clk_periph_i              | logic                              | Input     | 1     | Peripheral clock domain.                                                                    | PLL peripheral clock                      | N/A                           |
| powergood_i               | logic                              | Input     | 1     | Power-good indication. Assert after supply voltages stabilize.                              | Power management controller               | N/A                           |
| powergood_stable_o        | logic                              | Output    | 1     | Stabilized power-good output. Debounced version of <code>powergood_i</code> .               | Reset unit → External systems             | N/A                           |
| rst_cold_ni               | logic                              | Input     | 1     | Cold reset input (active low). Asynchronous assertion, synchronous deassertion.             | System cold reset                         | N/A                           |
| rst_primary_ref_clk_no    | logic                              | Output    | 1     | Primary reset synchronized to reference clock domain (active low).                          | To subsystems using ref clock             | N/A                           |
| rst_primary_smc_clk_no    | logic                              | Output    | 1     | Primary reset synchronized to SMC clock domain (active low).                                | To subsystems using SMC clock             | N/A                           |
| sys_axi_in_req_i          | smc_sys_in_56_64_12_axi_req_t      | Input     | -     | System AXI input request (56-bit addr, 64-bit data, 6-bit ID, 12-bit user).                 | External system masters → Input Fabric    | N/A, tie to '0 if unused      |
| sys_axi_in_resp_o         | smc_sys_in_56_64_12_axi_resp_t     | Output    | -     | System AXI input response.                                                                  | Input Fabric → External system masters    | N/A, leave floating if unused |
| jtag_axi_in_req_i         | smc_jtag_56_64_12_axi_req_t        | Input     | -     | JTAG debug AXI request (56-bit addr, 64-bit data, 2-bit ID, 12-bit user).                   | JTAG debug port → Input Fabric            | N/A, tie to '0 if unused      |
| jtag_axi_in_resp_o        | smc_jtag_56_64_12_axi_resp_t       | Output    | -     | JTAG debug AXI response.                                                                    | Input Fabric → JTAG debug port            | N/A, leave floating if unused |
| axil_smc_otp_jtag_req_i   | smc_pkg::smc_axil_32_32_req_t      | Input     | -     | eFuse OTP/JTAG AXI-Lite request.                                                            | JTAG → eFuse controller                   | N/A, tie to '0 if unused      |
| axil_smc_otp_jtag_resp_o  | smc_pkg::smc_axil_32_32_resp_t     | Output    | -     | eFuse OTP/JTAG AXI-Lite response.                                                           | eFuse controller → JTAG                   | N/A, leave floating if unused |
| sep_axi_in_req_i          | smc_sep_in_56_64_12_axi_req_t      | Input     | -     | SEP AXI request (56-bit addr, 64-bit data, 6-bit ID, 12-bit user).                          | SEP → Input Fabric                        | N/A, tie to '0 if unused      |
| sep_axi_in_resp_o         | smc_sep_in_56_64_12_axi_resp_t     | Output    | -     | SEP AXI response.                                                                           | Input Fabric → SEP                        | N/A, leave floating if unused |
| output_axi_req_o          | smc_sys_out_56_64_8_12_axi_req_t   | Output    | -     | Filtered and remapped AXI output request (56-bit addr, 64-bit data, 8-bit ID, 12-bit user). | Output Fabric → External system resources | N/A, leave floating if unused |
| output_axi_resp_i         | smc_sys_out_56_64_8_12_axi_resp_t  | Input     | -     | Filtered and remapped AXI output response.                                                  | External system resources → Output Fabric | N/A, tie to '0 if unused      |
| efuse_bank_ctrl_req_o     | smc_pkg::smc_axil_32_32_req_t      | Output    | -     | eFuse bank control AXI-Lite request.                                                        | eFuse controller → eFuse Shim             | N/A                           |
| efuse_bank_ctrl_resp_i    | smc_pkg::smc_axil_32_32_resp_t     | Input     | -     | eFuse bank control AXI-Lite response.                                                       | eFuse Shim → eFuse controller             | N/A                           |
| efuse_shim_command_req_o  | smc_efuse_pkg::fuse_command_req_t  | Output    | -     | eFuse command interface request.                                                            | eFuse controller → eFuse Shim             | N/A                           |
| efuse_shim_command_resp_i | smc_efuse_pkg::fuse_command_resp_t | Input     | -     | eFuse command interface response.                                                           | eFuse Shim → eFuse controller             | N/A                           |
| shadow_regs_o             | smc_efuse_pkg::efuse_map_t         | Output    | -     | eFuse shadow register output.                                                               | eFuse controller → External systems       | N/A                           |
| pll_clk_obs_i             | logic                              | Input     | 1     | PLL clock observation input.                                                                | PLL → GPIO padding (muxed)                | N/A                           |
| pll_clk_obs_en_i          | logic                              | Input     | 1     | PLL clock observation enable.                                                               | Control logic → GPIO padding              | N/A                           |

| Interface Name             | Type                                     | Direction | Width                        | Description                                                                                                                                                                                                                                 | Connection                                | Default value (if applicable) |
|----------------------------|------------------------------------------|-----------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------|
| pvt_process_clk_obs_i      | logic                                    | Input     | 1                            | PVT process monitor clock observation input.                                                                                                                                                                                                | PVT → GPIO padding (muxed)                | N/A                           |
| pvt_process_clk_obs_en_i   | logic                                    | Input     | 1                            | PVT process monitor clock observation enable.                                                                                                                                                                                               | Control logic → GPIO padding              | N/A                           |
| cat_therm_i                | logic                                    | Input     | 1                            | CAT thermal sensor input.                                                                                                                                                                                                                   | CAT thermal sensor → PVT wrap             | N/A, tie to '0 if unused      |
| lsio_interface_select_o    | logic                                    | Output    | [NUM_GPIO_WRAPS-1:0]         | LSIO interface selection for each GPIO wrap.                                                                                                                                                                                                | GPIO padding → GPIO wraps                 | N/A                           |
| pad2core_i                 | logic                                    | Input     | [NUM_GPIO_WRAPS-1:0]         | GPIO input data from pad ring to core.                                                                                                                                                                                                      | GPIO padding → SMC                        | N/A                           |
| core2pad_o                 | logic                                    | Output    | [NUM_GPIO_WRAPS-1:0]         | GPIO output data from core to pad ring.                                                                                                                                                                                                     | SMC → GPIO padding                        | N/A                           |
| pad2core_en_o              | logic                                    | Output    | [NUM_GPIO_WRAPS-1:0]         | GPIO pad-to-core path enable.                                                                                                                                                                                                               | SMC → GPIO padding                        | N/A                           |
| core2pad_en_o              | logic                                    | Output    | [NUM_GPIO_WRAPS-1:0]         | GPIO core-to-pad path enable.                                                                                                                                                                                                               | SMC → GPIO padding                        | N/A                           |
| rst_cool_n_from_pin_i      | logic                                    | Input     | 1                            | Cool reset input from GPIO pin 61.                                                                                                                                                                                                          | GPIO pin 61 → Reset Unit                  | N/A                           |
| clk_telemetry_i            | logic                                    | Input     | 1                            | Telemetry clock domain. Separate clock for ATB telemetry data collection.                                                                                                                                                                   | Telemetry clock source                    | N/A                           |
| rst_telemetry_n_i          | logic                                    | Input     | 1                            | Telemetry reset (active low). Resets telemetry receiver logic.                                                                                                                                                                              | Telemetry reset source                    | N/A                           |
| telemetry_atdata_i         | telemetry_receiver_pkg::telemetry_data_t | Input     | [NUM_TELMETRY_RECEIVERS-1:0] | ATB telemetry data input array.                                                                                                                                                                                                             | ATB sources → Telemetry receivers         | N/A                           |
| telemetry_atid_i           | telemetry_receiver_pkg::atb_id_t         | Input     | [NUM_TELMETRY_RECEIVERS-1:0] | ATB telemetry ID input array.                                                                                                                                                                                                               | ATB sources → Telemetry receivers         | N/A                           |
| telemetry_atready_o        | logic                                    | Output    | [NUM_TELMETRY_RECEIVERS-1:0] | ATB telemetry ready output.                                                                                                                                                                                                                 | Telemetry receivers → ATB sources         | N/A                           |
| telemetry_atvalid_i        | logic                                    | Input     | [NUM_TELMETRY_RECEIVERS-1:0] | ATB telemetry valid input.                                                                                                                                                                                                                  | ATB sources → Telemetry receivers         | N/A                           |
| telemetry_atflushvalid_o   | logic                                    | Output    | [NUM_TELMETRY_RECEIVERS-1:0] | ATB flush valid output.                                                                                                                                                                                                                     | Telemetry receivers → ATB sources         | N/A                           |
| telemetry_atflushready_i   | logic                                    | Input     | [NUM_TELMETRY_RECEIVERS-1:0] | ATB flush ready input.                                                                                                                                                                                                                      | ATB sources → Telemetry receivers         | N/A                           |
| smc_cluster_double_error_o | logic                                    | Output    | 1                            | Cluster double error detection output.                                                                                                                                                                                                      | CPU cluster → External systems            | N/A                           |
| smc_wdt_first_timeout_o    | logic                                    | Output    | 1                            | First watchdog timeout signal.                                                                                                                                                                                                              | CPU cluster WDT → Reset Unit              | N/A                           |
| smc_wdt_second_timeout_o   | logic                                    | Output    | 1                            | Second watchdog timeout signal.                                                                                                                                                                                                             | Reset Unit → External systems             | N/A                           |
| smc_region_size_o          | logic                                    | Output    | [31:0]                       | SMC address region size output.                                                                                                                                                                                                             | SMC Internal Registers → External systems | N/A                           |
| smc_ext_interrupts_i       | logic                                    | Input     | [NUM_EXT_INTERRUPTS-1:0]     | External interrupt inputs. Width is 256. Active-high level or a pulse long enough to be sampled on <code>clk_smc</code> . Synchronized to <code>clk_smc</code> (3-stage) in <code>smc_base</code> ; a short asynchronous pulse can be lost. | External sources → Interrupt aggregator   | N/A, tie to '0 if unused      |

| Interface Name               | Type                             | Direction | Width                           | Description                                                                                                                                                                                                                      | Connection                                         | Default value (if applicable)                       |
|------------------------------|----------------------------------|-----------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| sep_mailbox_interrupts_i     | logic                            | Input     | [7:0]                           | SEP → SMC mailbox interrupt inputs (8 channels). Routed to internal interrupts to the SMC CPU.                                                                                                                                   | SEP → Interrupt aggregator                         | N/A, tie to '0 if unused                            |
| sep_wdt_reset_n_i            | logic                            | Input     | 1                               | SEP watchdog reset (active low). Active-low polarity is inverted before routing to internal interrupts to the SMC CPU.                                                                                                           | SEP → Interrupt aggregator                         | N/A                                                 |
| smc_fuse_sense_done_o        | logic                            | Output    | 1                               | Fuse sense completion output.                                                                                                                                                                                                    | eFuse controller → Memory repair logic             | N/A                                                 |
| smc_fuse_reset_n_delayed_o   | logic                            | Output    | 1                               | Delayed fuse reset output (after 16 pipeline stages).                                                                                                                                                                            | eFuse controller → External systems                | N/A                                                 |
| skip_mem_repair_o            | logic                            | Output    | 1                               | Skip memory repair output.                                                                                                                                                                                                       | Reset Unit → Memory repair logic                   | N/A                                                 |
| ext_boot_seq_done_i          | logic                            | Input     | 1                               | External boot sequence completion input (memory repair and shadow register override done); gates reset release.                                                                                                                  | External boot/repair controller → eFuse controller | N/A                                                 |
| lc_state_i                   | logic                            | Input     | [2*LC_STATE_WIDTH-1:0] (8 bits) | Lifecycle state input, differentially encoded.                                                                                                                                                                                   | Lifecycle controller → SMC Internal Registers      | From SEP; tie to 8'hf0 if unused (encoded TEST_DEV) |
| smc_ext_mailbox_interrupts_o | logic                            | Output    | [NUM_MAILBOXES-1:0]             | External mailbox interrupt outputs (32 interrupts, mapped to IDs 156-187).                                                                                                                                                       | Mailbox controller → External targets              | N/A                                                 |
| cfg_flr_pf_active_i          | logic                            | Input     | 1                               | Function Level Reset (FLR) PF active signal.                                                                                                                                                                                     | PCIe/System → Reset Unit                           | N/A, tie to '0 if unused                            |
| isolate_req_o                | logic                            | Output    | [31:0]                          | Isolation request output for 32 subsystems.                                                                                                                                                                                      | Reset Unit → 32 subsystems                         | N/A                                                 |
| ss_reset_complete_i          | logic                            | Input     | [31:0]                          | Subsystem reset completion status.                                                                                                                                                                                               | 32 subsystems → Reset Unit                         | 32'hFFFFFFFF                                        |
| ss_config_o                  | logic                            | Output    | [31:0]                          | Subsystem configuration output.                                                                                                                                                                                                  | Reset Unit → 32 subsystems                         | N/A                                                 |
| ss_reset_ctrl_o              | smc_reset_unit_pkg::reset_ctrl_t | Output    | [31:0]                          | Subsystem reset control output array.                                                                                                                                                                                            | Reset Unit → 32 subsystems                         | N/A                                                 |
| sync_irq_o                   | logic                            | Output    | 1                               | Software-controlled global synchronisation bit (SYNC_REG.sync in the reset unit). Not an interrupt aggregate output.                                                                                                             | Reset unit → External                              | N/A                                                 |
| gpio_interrupt_o             | logic                            | Output    | [NUM_GPIO_WRAPS-1:0]            | Raw GPIO interrupt vector from the padding, one bit per GPIO wrap (bonded + unbonded). Software reads GPIO status registers to identify the interrupt source. Already OR-reduced into peripheral_interrupts_o[30:29] internally. | smc_padding → external consumers                   | N/A, leave floating if unused                       |
| uart_interrupt_o             | logic                            | Output    | [NUM_UART-1:0]                  | Raw UART interrupt outputs, one bit per UART instance, in the peripheral clock domain. Already aggregated into peripheral_interrupts_o[21:18] internally via CDC.                                                                | uart_wrap → external consumers                     | N/A, leave floating if unused                       |
| smc_rom_intf_req_o           | rom_req_t                        | Output    | -                               | ROM memory interface request. Width depends on CPU cluster configuration                                                                                                                                                         | CPU cluster → Boot ROM                             | N/A                                                 |
| smc_rom_intf_rsp_i           | rom_rsp_t                        | Input     | -                               | ROM memory interface response. Width depends on CPU cluster configuration                                                                                                                                                        | Boot ROM → CPU cluster                             | N/A                                                 |
| smc_scratch_ram_intf_req_o   | scratch_ram_req_t                | Output    | [NUM_SRAM_BANKS2-1:0]           | Scratch RAM interface requests. Width depends on CPU cluster configuration                                                                                                                                                       | CPU cluster → Scratch RAM banks                    | N/A                                                 |
| smc_scratch_ram_intf_rsp_i   | scratch_ram_rsp_t                | Input     | [NUM_SRAM_BANKS2-1:0]           | Scratch RAM interface responses. Width depends on CPU cluster configuration                                                                                                                                                      | Scratch RAM banks → CPU cluster                    | N/A                                                 |

| Interface Name                       | Type                               | Direction | Width                       | Description                                                                                                                                                             | Connection                                                    | Default value (if applicable) |
|--------------------------------------|------------------------------------|-----------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------|
| smc_l1_ichache_tag_intf_req_o        | l1_ichache_tag_req_t               | Output    | [NUM_ICACHE_TAG_BANKS-1:0]  | L1 I-cache tag interface requests. Width depends on CPU cluster configuration                                                                                           | CPU cluster → L1 I-cache tag banks                            | N/A                           |
| smc_l1_ichache_tag_intf_rsp_i        | l1_ichache_tag_rsp_t               | Input     | [NUM_ICACHE_TAG_BANKS-1:0]  | L1 I-cache tag interface responses. Width depends on CPU cluster configuration                                                                                          | L1 I-cache tag banks → CPU cluster                            | N/A                           |
| smc_l1_ichache_data_intf_req_o       | l1_ichache_data_req_t              | Output    | [NUM_ICACHE_DATA_BANKS-1:0] | L1 I-cache data interface requests. Width depends on CPU cluster configuration                                                                                          | CPU cluster → L1 I-cache data banks                           | N/A                           |
| smc_l1_ichache_data_intf_rsp_i       | l1_ichache_data_rsp_t              | Input     | [NUM_ICACHE_DATA_BANKS-1:0] | L1 I-cache data interface responses. Width depends on CPU cluster configuration                                                                                         | L1 I-cache data banks → CPU cluster                           | N/A                           |
| smc_l1_dcache_tag_intf_req_o         | l1_dcache_tag_req_t                | Output    | [NUM_DCACHE_TAG_BANKS-1:0]  | L1 D-cache tag interface requests. Width depends on CPU cluster configuration                                                                                           | CPU cluster → L1 D-cache tag banks                            | N/A                           |
| smc_l1_dcache_tag_intf_rsp_i         | l1_dcache_tag_rsp_t                | Input     | [NUM_DCACHE_TAG_BANKS-1:0]  | L1 D-cache tag interface responses. Width depends on CPU cluster configuration                                                                                          | L1 D-cache tag banks → CPU cluster                            | N/A                           |
| smc_l1_dcache_data_intf_req_o        | l1_dcache_data_req_t               | Output    | [NUM_DCACHE_DATA_BANKS-1:0] | L1 D-cache data interface requests. Width depends on CPU cluster configuration                                                                                          | CPU cluster → L1 D-cache data banks                           | N/A                           |
| smc_l1_dcache_data_intf_rsp_i        | l1_dcache_data_rsp_t               | Input     | [NUM_DCACHE_DATA_BANKS-1:0] | L1 D-cache data interface responses. Width depends on CPU cluster configuration                                                                                         | L1 D-cache data banks → CPU cluster                           | N/A                           |
| smc_disable_sram_auto_init_i         | logic                              | Input     | 1                           | Disable SRAM automatic initialization.                                                                                                                                  | Control logic → CPU cluster                                   | 1'b0                          |
| smc_init_mem_done_o                  | logic                              | Output    | 1                           | Memory initialization completion signal.                                                                                                                                | CPU cluster → Control logic                                   | N/A                           |
| chiplet_is_primary_i                 | logic                              | Input     | 1                           | Chiplet is primary signal. Used for System Timer OCTS                                                                                                                   | GPIO Pading Straps (strap 25) → SMC                           | N/A                           |
| timer_count_o                        | logic                              | Output    | [63:0]                      | 64-bit system timer count output for Open Chiplet Time Synchronization (OCTS).                                                                                          | System Timer OCTS → External systems                          | N/A                           |
| jtag_reset_ctrl_i                    | smc_pkg::jtag_smc_reset_ctrl_t     | Input     | -                           | Typed reset-override control struct carrying all SMC reset override enables and active-low values. Driven by DTP via JTAG IC_RESET. Tie all enables to 0 when not used. | DTP → SMC reset unit                                          | All enables 0                 |
| cla_gpio_enable_o                    | logic                              | Output    | 1                           | CLA GPIO enable output.                                                                                                                                                 | DFD controller → CLA                                          | N/A                           |
| tdr_dbg_ctrl_clock_stop_en_i         | logic                              | Input     | 1                           | DEBUG_CONTROL CLA clock-stop enable.                                                                                                                                    | DTP → DFD                                                     | 1'b0                          |
| tdr_dbg_ctrl_clocks_stopped_by_cla_o | logic                              | Output    | 1                           | CLA halt-clock status under enable; also drives <code>cpu_interrupts_o[NUM_EXT_INTERRUPTS+64]</code> (internal interrupt 0), bit 320.                                   | DFD → DTP / interrupt                                         | N/A                           |
| xtrigger_ss_i                        | smc_pkg::xtrigger_t                | Input     | -                           | Cross-trigger subsystem input. SMU drives from <code>dtp_xtrig_ctm_src_req[1:0]</code> . Internal SMU connection; not an SMU external port.                             | SMU internal: DTP <code>xtrig_ctm_src_req_o[1:0]</code> → SMC | N/A                           |
| xtrigger_ss_o                        | smc_pkg::xtrigger_t                | Output    | -                           | Cross-trigger subsystem output. SMU routes to <code>dtp_xtrig_ctm_dst_req[1:0]</code> . Internal SMU connection; not an SMU external port.                              | SMU internal: SMC → DTP <code>xtrig_ctm_dst_req_i[1:0]</code> | N/A                           |
| trace_mem_req_o                      | dfd_trace_mem_pkg::SinkMemPktIn_s  | Output    | [TRC_RAM_INSTANCES-1:0]     | Trace memory interface requests.                                                                                                                                        | DFD controller → Trace memory                                 | N/A, leave floating if unused |
| trace_mem_resp_i                     | dfd_trace_mem_pkg::SinkMemPktOut_s | Input     | [TRC_RAM_INSTANCES-1:0]     | Trace memory interface responses.                                                                                                                                       | Trace memory → DFD controller                                 | N/A, tie to '0 if unused      |
| ext_debug_bus_i                      | logic                              | Input     | [511:0]                     | External debug bus input (512 bits, 16-bit aligned).                                                                                                                    | External debug sources → Debug bus aggregator                 | N/A, tie to '0 if unused      |

| Interface Name | Type  | Direction | Width | Description                                                                   | Connection                    | Default value (if applicable) |
|----------------|-------|-----------|-------|-------------------------------------------------------------------------------|-------------------------------|-------------------------------|
| test_en_i      | logic | Input     | 1     | DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs. | Test controller → All modules | 1'b0                          |
| scan_rst_ni    | logic | Input     | 1     | Scan reset (active-low). Bypasses reset synchronizers during DFT.             | Test controller → All modules | 1'b1                          |

### 1.3.12. GPIO Pin Requirements

The SMC provides GPIO functionality with multiplexed peripheral functions and test interfaces. The following table lists the signal and bump location, assigned function, recommended pull, and GPIO-specific notes. It does not define default direction or decode configuration straps; use `smc_padrng_pkg::DefaultDirectionMap` and the generated reset-unit `STRAPS_LO` and `STRAPS_HI` register descriptions for those.

Table 8. OCAH GPIO Requirements

| Index | Signal                   | Bump Loc | Function    | Recommended Pull | GPIO Notes |
|-------|--------------------------|----------|-------------|------------------|------------|
| 0     | smc_wrapper.GPIO_PAD[0]  | SMC      | SPI.DATA[0] | PD               |            |
| 1     | smc_wrapper.GPIO_PAD[1]  | SMC      | SPI.DATA[1] | PD               |            |
| 2     | smc_wrapper.GPIO_PAD[2]  | SMC      | SPI.DATA[2] | PD               |            |
| 3     | smc_wrapper.GPIO_PAD[3]  | SMC      | SPI.DATA[3] | PD               |            |
| 4     | smc_wrapper.GPIO_PAD[4]  | SMC      | SPI.DATA[4] | PD               |            |
| 5     | smc_wrapper.GPIO_PAD[5]  | SMC      | SPI.DATA[5] | PD               |            |
| 6     | smc_wrapper.GPIO_PAD[6]  | SMC      | SPI.DATA[6] | PD               |            |
| 7     | smc_wrapper.GPIO_PAD[7]  | SMC      | SPI.DATA[7] | PD               |            |
| 8     | smc_wrapper.GPIO_PAD[8]  | SMC      | SPI.CS      | PU               |            |
| 9     | smc_wrapper.GPIO_PAD[9]  | SMC      | SPI.CLK     | PD               |            |
| 10    | smc_wrapper.GPIO_PAD[10] | SMC      | SPI.DQS     | PD               |            |
| 11    | smc_wrapper.GPIO_PAD[11] | SMC      | UART[0].RX  | PU               |            |
| 12    | smc_wrapper.GPIO_PAD[12] | SMC      | UART[0].TX  | PU               |            |
| 13    | smc_wrapper.GPIO_PAD[13] | SMC      | UART[0].RTS | PD               |            |
| 14    | smc_wrapper.GPIO_PAD[14] | SMC      | UART[0].CTS | PD               |            |
| 15    | smc_wrapper.GPIO_PAD[15] | SMC      | UART[1].RX  | PU               |            |
| 16    | smc_wrapper.GPIO_PAD[16] | SMC      | UART[1].TX  | PU               |            |

| Index | Signal                   | Bump Loc | Function     | Recommended Pull | GPIO Notes                                                |
|-------|--------------------------|----------|--------------|------------------|-----------------------------------------------------------|
| 17    | smc_wrapper.GPIO_PAD[17] | SMC      | UART[1].RTS  | PD               |                                                           |
| 18    | smc_wrapper.GPIO_PAD[18] | SMC      | UART[1].CTS  | PD               |                                                           |
| 19    | smc_wrapper.GPIO_PAD[19] | SMC      | UART[2].RX   | PU               |                                                           |
| 20    | smc_wrapper.GPIO_PAD[20] | SMC      | UART[2].TX   | PU               |                                                           |
| 21    | smc_wrapper.GPIO_PAD[21] | SMC      | UART[2].RTS  | PD               |                                                           |
| 22    | smc_wrapper.GPIO_PAD[22] | SMC      | UART[2].CTS  | PD               |                                                           |
| 23    | smc_wrapper.GPIO_PAD[23] | SMC      | UART[3].RX   | PU               |                                                           |
| 24    | smc_wrapper.GPIO_PAD[24] | SMC      | UART[3].TX   | PU               |                                                           |
| 25    | smc_wrapper.GPIO_PAD[25] | SMC      | UART[3].RTS  | PD               |                                                           |
| 26    | smc_wrapper.GPIO_PAD[26] | SMC      | UART[3].CTS  | PD               |                                                           |
| 27    | smc_wrapper.GPIO_PAD[27] | SMC      | I3C[0] SCL   | PU               |                                                           |
| 28    | smc_wrapper.GPIO_PAD[28] | SMC      | I3C[0] SDA   | PU               |                                                           |
| 29    | smc_wrapper.GPIO_PAD[29] | SMC      | I3C[2] SCL   | PU               |                                                           |
| 30    | smc_wrapper.GPIO_PAD[30] | SMC      | I3C[2] SDA   | PU               |                                                           |
| 31    | smc_wrapper.GPIO_PAD[31] | SMC      | I3C[3] SCL   | PU               |                                                           |
| 32    | smc_wrapper.GPIO_PAD[32] | SMC      | I3C[3] SDA   | PU               |                                                           |
| 33    | smc_wrapper.GPIO_PAD[33] | SMC      | I3C[4] SCL   | PU               |                                                           |
| 34    | smc_wrapper.GPIO_PAD[34] | SMC      | I3C[4] SDA   | PU               |                                                           |
| 35    | smc_wrapper.GPIO_PAD[35] | SMC      | I3C[5] SCL   | PU               |                                                           |
| 36    | smc_wrapper.GPIO_PAD[36] | SMC      | I3C[5] SDA   | PU               |                                                           |
| 37    | smc_wrapper.GPIO_PAD[37] | SMC      | I2C[0] SCL   | PU               | Every I2C can be programmed to act as I2C, SMBus or PMBus |
| 38    | smc_wrapper.GPIO_PAD[38] | SMC      | I2C[0] SDA   | PU               |                                                           |
| 39    | smc_wrapper.GPIO_PAD[39] | SMC      | I2C[0] SMB_A | PU               | SMBus Alert                                               |
| 40    | smc_wrapper.GPIO_PAD[40] | SMC      | I2C[0] SMB_D | PU               | SMBus Suspend                                             |
| 41    | smc_wrapper.GPIO_PAD[41] | SMC      | I2C[1] SCL   | PU               |                                                           |

| Index | Signal                        | Bump Loc | Function                  | Recommended Pull | GPIO Notes                                                                                                                                                                                                                                       |
|-------|-------------------------------|----------|---------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 42    | smc_wrapper.GPIO_PAD[42]      | SMC      | I2C[1] SDA                | PU               |                                                                                                                                                                                                                                                  |
| 43    | smc_wrapper.GPIO_PAD[43]      | SMC      | I2C[1] SMB_A              | PU               | SMBus Alert                                                                                                                                                                                                                                      |
| 44    | smc_wrapper.GPIO_PAD[44]      | SMC      | I2C[1] SMB_D              | PU               | SMBus Suspend                                                                                                                                                                                                                                    |
| 45    | smc_wrapper.GPIO_PAD[45]      | SMC      | I2C[2] SCL                | PU               |                                                                                                                                                                                                                                                  |
| 46    | smc_wrapper.GPIO_PAD[46]      | SMC      | I2C[2] SDA                | PU               |                                                                                                                                                                                                                                                  |
| 47    | smc_wrapper.GPIO_PAD[47]      | SMC      | I2C[2] SMB_A              | PU               | SMBus Alert                                                                                                                                                                                                                                      |
| 48    | smc_wrapper.GPIO_PAD[48]      | SMC      | I2C[2] SMB_D              | PU               | SMBus Suspend                                                                                                                                                                                                                                    |
| 49    | smc_wrapper.GPIO_PAD[49]      | SMC      | AVS.CLOCK                 | PU               |                                                                                                                                                                                                                                                  |
| 50    | smc_wrapper.GPIO_PAD[50]      | SMC      | AVS.MDATA                 | PU               |                                                                                                                                                                                                                                                  |
| 51    | smc_wrapper.GPIO_PAD[51]      | SMC      | AVS.SDATA                 | PU               |                                                                                                                                                                                                                                                  |
| 52    | smc_wrapper.GPIO_PAD[52]      | SMC      | Thermal Emergency N       | PU               |                                                                                                                                                                                                                                                  |
| 53    | smc_wrapper.GPIO_PAD[53]      | SMC      | Isolate Request           | PD               |                                                                                                                                                                                                                                                  |
| 54    | smc_wrapper.GPIO_PAD[54]      | SMC      | SPI DQS Loopback          | PD               |                                                                                                                                                                                                                                                  |
| 55    | smc_wrapper.GPIO_PAD[55]      | SMC      | System Timer Sync Load    | PD               |                                                                                                                                                                                                                                                  |
| 56    | smc_wrapper.GPIO_PAD[56]      | SMC      | System Timer Credit Count | PD               |                                                                                                                                                                                                                                                  |
| 57    | smc_wrapper.GPIO_PAD[57]      | SMC      | Boot Stall                | PD               | Stall after fuse sensing to override shadow register values. Boot will resume when this GPIO is released. This value can be overridden by the value specified over JTAG using the boot_stall_jtag_ovrd and boot_stall_jtag_val signals from DTP. |
| 58    | smc_wrapper.GPIO_PAD[58]      | SMC      | OCCP Interface            | PD               |                                                                                                                                                                                                                                                  |
| 59    | smc_wrapper.GPIO_PAD[59]      | SMC      | Reserved                  | PD               |                                                                                                                                                                                                                                                  |
| 60    | smc_wrapper.GPIO_PAD[60]      | SMC      | Reserved                  | PD               |                                                                                                                                                                                                                                                  |
| 61    | smc_wrapper.UNBONDED_GPIO[61] | SMC      | Cool Reset In             | PU               | Not packaged out                                                                                                                                                                                                                                 |
| 62    | smc_wrapper.UNBONDED_GPIO[62] | SMC      | Cool Reset Out            | PU               | Not packaged out                                                                                                                                                                                                                                 |
| 63    | smc_wrapper.UNBONDED_GPIO[63] | SMC      | I3C[1] SCL                | PU               | Not packaged out                                                                                                                                                                                                                                 |
| 64    | smc_wrapper.UNBONDED_GPIO[64] | SMC      | I3C[1] SDA                | PU               | Not packaged out                                                                                                                                                                                                                                 |
| 65    | smc_wrapper.PAD_JTAG_IN_TCK   | ?        | PTAP TCK                  |                  |                                                                                                                                                                                                                                                  |

| Index | Signal                         | Bump Loc | Function          | Recommended Pull | GPIO Notes                |
|-------|--------------------------------|----------|-------------------|------------------|---------------------------|
| 66    | smc_wrapper.PAD_JTAG_IN_TRSTN  | ?        | PTAP TRST         |                  |                           |
| 67    | smc_wrapper.PAD_JTAG_IN_TDI    | ?        | PTAP TDI          |                  |                           |
| 68    | smc_wrapper.PAD_JTAG_IN_TMS    | ?        | PTAP TMS          |                  |                           |
| 69    | smc_wrapper.PAD_JTAG_IN_TDO    | ?        | PTAP TDO          |                  |                           |
| 70    | smc_wrapper.PAD_JTAG_OUT_TCK   | ?        | STAP TCK          |                  |                           |
| 71    | smc_wrapper.PAD_JTAG_OUT_TRSTN | ?        | STAP TRST         |                  |                           |
| 72    | smc_wrapper.PAD_JTAG_OUT_TDI   | ?        | STAP TDI          |                  |                           |
| 73    | smc_wrapper.PAD_JTAG_OUT_TMS   | ?        | STAP TMS          |                  |                           |
| 74    | smc_wrapper.PAD_JTAG_OUT_TDO   | ?        | STAP TDO          |                  |                           |
| 75    | smc_wrapper.PAD_REFCLK         | SMC      | BP_REFCLK_VDD_SYS |                  |                           |
| 76    | smc_wrapper.PAD_RESETN         | SMC      | BP_RESETN         |                  |                           |
| 77    | smc_wrapper.PAD_POWERGOOD      | SMC      | BP_POWERGOOD      |                  |                           |
| 78    |                                | SEP      | OTP[0]            |                  |                           |
| 79    |                                | SEP      | OTP[1]            |                  |                           |
| 80    |                                | SEP      | OTP[2]            |                  |                           |
| 81    |                                | SEP      | OTP[3]            |                  |                           |
| 82    |                                | SEP      | OTP[4]            |                  |                           |
| 83    |                                | SEP      | OTP[5]            |                  |                           |
| 84    |                                | SMC      | PVT[0]            |                  |                           |
| 85    |                                | SMC      | PVT[1]            |                  |                           |
| 86    |                                |          | VDDIO             |                  |                           |
| 87    | smc_wrapper.PLL_CLK_OBS_PAD    | SMC      | PLL clock OBS     | PD               | PLL clock observation pin |
| 88    | smc_wrapper.PVT_CLK_OBS_PAD    | SMC      | PVT RO            | PD               | Process sensor RO output  |
| 89    |                                |          |                   |                  |                           |

For multi-chiplet systems, the **PRIMARY\_CHIPLET** strap (**STRAPS\_LO[25]**) determines whether the chiplet operates as primary or secondary. The primary chiplet's System Timer OCTS generates synchronization pulses (**timer\_sync\_load\_o**, **timer\_cnt\_credit\_o**) that secondary chiplets receive to maintain coherent timestamps across the system.

## 1.4. Debug & Test Ports (DTP) Integration

### 1.4.1. Configuration

The cross-trigger topology is fixed at elaboration by `ntp_pkg`, a committed source file at `hw/sys/ntp/rtl/ntp_pkg.sv`. `DEFAULT_NUM_CTP`, `DEFAULT_NUM_INT_CT`, and `DEFAULT_NUM_CLK_STOP_REQ` there set the `XTRIG_NUM_CTP`, `XTRIG_NUM_INT_CT`, and `XTRIG_NUM_CLK_STOP_REQ` localparams of `ntp`, so a different port count is applied by editing that package rather than by overriding parameters at the instantiation. The remaining DTP parameters, including `XTRIG_INT_CT_MODE` and every `JTAG_*` enable, are set at the instantiation as shown below.

Include the DTP in your build using Bender target `ntp`, which pulls in the DTP RTL together with the JTAG and cross-trigger IPs it instantiates. The legacy filelist `hw/sys/ntp/rtl/config/ntp.f` covers only the JTAG portion of the hierarchy and is not a complete DTP recipe.

### 1.4.2. Instantiation

```
ntp #(
    // JTAG Configuration
    .JTAG_BSR_ENABLE      (1'b1),
    .JTAG_TMP_ENABLE      (1'b1),
    .JTAG_SMC_DBG_ENABLE  (1'b1),
    .JTAG_SEP_DBG_ENABLE  (1'b1),
    .JTAG_IDCODE_MFR_ID   (11'h15A),
    .JTAG_IDCODE_PART_NUM (16'h1234),
    .JTAG_IDCODE_SI_REV   (4'h1),
    // Cross Trigger Configuration
    .XTRIG_INT_CT_MODE    (10'b0000000000)
) u_ntp (
    .clk_i      (clk),
    .rst_n_i    (rst_n),
    .pwr_on_rst_ni (pwr_on_rst_n),
    // JTAG, Boundary Scan, Cross Trigger interfaces...
);
```

### 1.4.3. Clock and Reset

- **TCK Domain:** JTAG logic operates on external Test Clock
- **System Clock** (`clk_i`): Cross trigger logic and JTAG2AXI bridges
- **System Reset** (`rst_n_i`): Resets cross trigger and AXI; does not affect TAP controller
- **Power-On Reset** (`pwr_on_rst_ni`): Asynchronous reset for JTAG state machine

### 1.4.4. Constraints

- JTAG interface: 10-50 MHz TCK frequency
- Cross trigger signals synchronized to system clock
- DTP remains active during clock stop for debug access
- External CTP signals require pad ring integration with ESD protection

### 1.4.5. STAP Scan Chain Topology

The DTP secondary scan path chains multiple STAPs on a single PTAP-managed scan string. The fixed order on this string is: PTAP, I/O STAP (if `JTAG_STAP_IO_ENABLE=1`), SMC debug STAP (if `JTAG_SMC_DBG_ENABLE=1`), SEP debug STAP (if `SEP=1` in the SMU), extra STAP[N-1:0], and return to PTAP.

Both the SMC and SEP debug STAP host interfaces are internal to the SMU and are not exported on the SMU boundary. The SMU wires the SMC debug STAP host outputs to the SMC CPU/system JTAG ports (`smc_cpu_jtag_*`) when `JTAG_SMC_DBG_ENABLE=1`; when disabled, the DTP bypasses this STAP in the secondary scan chain and drives the internal host outputs low. The SEP debug STAP connects to the SEP JTAG debug port when `SEP=1`; when `SEP=0`, the SMU ties `sep_stap_tdo_to_dtp = 10b0` to bypass the SEP STAP return path.

Each STAP host interface is gated by a field of `dbg_disable_i`, synchronized to TCK inside the DTP. Each field is active-high (1 = path disabled).

| STAP                      | Disable Field                         |
|---------------------------|---------------------------------------|
| I/O STAP                  | <code>dbg_disable_i.stap_io</code>    |
| SMC debug STAP            | <code>dbg_disable_i.stap_smc</code>   |
| SEP debug STAP            | <code>dbg_disable_i.stap_sep</code>   |
| Extra STAPs               | <code>dbg_disable_i.stap_extra</code> |
| External STAP return path | <code>dbg_disable_i.stap_host</code>  |

The full set of debug-disable-gated paths, including the iJTAG chains and the JTAG2AXI bridges, is tabulated in the DTP JTAG integration section of the TRM.

### 1.4.6. DTP Port Interface

The DTP exposes interfaces for JTAG TAP control, boundary scan, multiple STAP host ports, iJTAG scan chains, AXI debug bridges, cross trigger networks, and CTP GPIO pads. The following table defines all DTP top-level ports. Refer to the dedicated DTP documentation in [hw/sys/dtp/doc/](#) for internal design details.

Table 9. DTP Port Declaration

| Interface Name                           | Type                                               | Direction | Width | Description                                                                                        | Connection         | Default value (if applicable) |
|------------------------------------------|----------------------------------------------------|-----------|-------|----------------------------------------------------------------------------------------------------|--------------------|-------------------------------|
| <code>clk_i</code>                       | logic                                              | Input     | 1     | System clock for cross trigger logic and JTAG2AXI bridges.                                         | System clock       | N/A                           |
| <code>rst_n_i</code>                     | logic                                              | Input     | 1     | System reset (active low). Resets cross trigger and AXI logic; does not affect the TAP controller. | System reset       | N/A                           |
| <code>pwr_on_rst_ni</code>               | logic                                              | Input     | 1     | Power-on reset for JTAG state machine (active low, asynchronous).                                  | Power-on reset     | N/A                           |
| <code>dbg_disable_i</code>               | <code>sep_lifecycle_ctrl_pkg::dbg_disable_t</code> | Input     | -     | Per-path debug/test disables. Active-high (1 = path disabled).                                     | SEP                | N/A                           |
| <code>jtag_ptap_client_tap_ctrl_i</code> | <code>jtag_tap_ctrl_t</code>                       | Input     | -     | Primary JTAG TAP control (TCK, TMS, TRST_N).                                                       | External JTAG pads | N/A                           |
| <code>jtag_ptap_client_tdi_i</code>      | logic                                              | Input     | 1     | Primary JTAG TDI.                                                                                  | External JTAG pads | N/A                           |
| <code>jtag_ptap_client_tdo_o</code>      | logic                                              | Output    | 1     | Primary JTAG TDO.                                                                                  | External JTAG pads | N/A                           |
| <code>jtag_ptap_client_tdo_oe_o</code>   | logic                                              | Output    | 1     | Primary JTAG TDO output enable.                                                                    | External JTAG pads | N/A                           |

| Interface Name                  | Type             | Direction | Width                           | Description                                                                                                  | Connection          | Default value (if applicable)           |
|---------------------------------|------------------|-----------|---------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------|
| jtag_bsr_host_scan_ctrl_o       | jtag_scan_ctrl_t | Output    | -                               | Boundary scan control. Tie scan_in to scan_out if BSR is disabled.                                           | Boundary scan chain | N/A                                     |
| jtag_bsr_host_scan_in_i         | logic            | Input     | 1                               | Boundary scan data in.                                                                                       | Boundary scan chain | N/A                                     |
| jtag_bsr_host_scan_out_o        | logic            | Output    | 1                               | Boundary scan data out.                                                                                      | Boundary scan chain | N/A                                     |
| jtag_stap_io_host_tap_ctrl_o    | jtag_tap_ctrl_t  | Output    | -                               | I/O STAP TAP control for chiplet-to-chiplet connectivity. .tck remains driven when the STAP generate is off. | I/O STAP host       | TMS/TRST idle if JTAG_STAP_IO_ENABLE=0  |
| jtag_stap_io_host_tdi_i         | logic            | Input     | 1                               | I/O STAP TDI.                                                                                                | I/O STAP host       | Tie to '0 if unused                     |
| jtag_stap_io_host_tdo_o         | logic            | Output    | 1                               | I/O STAP TDO.                                                                                                | I/O STAP host       | N/A                                     |
| jtag_stap_io_host_tdo_oen_o     | logic            | Output    | 1                               | I/O STAP TDO output enable.                                                                                  | I/O STAP host       | N/A                                     |
| jtag_stap_smc_host_tap_ctrl_o   | jtag_tap_ctrl_t  | Output    | -                               | SMC debug STAP TAP control. .tck remains driven when the STAP generate is off.                               | SMC JTAG port       | TMS/TRST idle if JTAG_SMC_DBG_ENABLE=0  |
| jtag_stap_smc_host_tdi_i        | logic            | Input     | 1                               | SMC debug STAP TDI (from SMC TDO).                                                                           | SMC JTAG port       | Tie to '0 if unused                     |
| jtag_stap_smc_host_tdo_o        | logic            | Output    | 1                               | SMC debug STAP TDO (to SMC TDI).                                                                             | SMC JTAG port       | N/A                                     |
| jtag_stap_smc_host_tdo_oen_o    | logic            | Output    | 1                               | SMC debug STAP TDO output enable.                                                                            | SMC JTAG port       | N/A                                     |
| jtag_stap_sep_host_tap_ctrl_o   | jtag_tap_ctrl_t  | Output    | -                               | SEP debug STAP TAP control. .tck remains driven when the STAP generate is off.                               | SEP JTAG port       | TMS/TRST idle if JTAG_SEP_DBG_ENABLE=0  |
| jtag_stap_sep_host_tdi_i        | logic            | Input     | 1                               | SEP debug STAP TDI (from SEP TDO).                                                                           | SEP JTAG port       | Tie to '0 if unused                     |
| jtag_stap_sep_host_tdo_o        | logic            | Output    | 1                               | SEP debug STAP TDO (to SEP TDI).                                                                             | SEP JTAG port       | N/A                                     |
| jtag_stap_sep_host_tdo_oen_o    | logic            | Output    | 1                               | SEP debug STAP TDO output enable.                                                                            | SEP JTAG port       | N/A                                     |
| jtag_stap_extra_host_tap_ctrl_o | jtag_tap_ctrl_t  | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TAP control array. .tck remains driven when the STAP generate is off.                             | Extra STAP hosts    | TMS/TRST idle if JTAG_NUM_EXTRA_STAPS=0 |
| jtag_stap_extra_host_tdi_i      | logic            | Input     | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDI array.                                                                                        | Extra STAP hosts    | Tie to '0 if unused                     |
| jtag_stap_extra_host_tdo_o      | logic            | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDO array.                                                                                        | Extra STAP hosts    | N/A                                     |
| jtag_stap_extra_host_tdo_oen_o  | logic            | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDO output enable array.                                                                          | Extra STAP hosts    | N/A                                     |
| jtag_stap_host_scan_ctrl_o      | jtag_scan_ctrl_t | Output    | -                               | Extended STAP scan control. Tie scan_in to scan_out if unused.                                               | STAP scan chain     | N/A                                     |
| jtag_stap_host_scan_in_i        | logic            | Input     | 1                               | Extended STAP scan data in.                                                                                  | STAP scan chain     | N/A                                     |
| jtag_stap_host_scan_out_o       | logic            | Output    | 1                               | Extended STAP scan data out.                                                                                 | STAP scan chain     | N/A                                     |
| jtag_dfd_host_scan_ctrl_o       | jtag_scan_ctrl_t | Output    | -                               | DFD iJTAG scan control. Tie scan_in to scan_out if unused.                                                   | DFD iJTAG chain     | N/A                                     |

| Interface Name                   | Type                | Direction | Width                  | Description                                                                                                                                                                                                              | Connection                   | Default value (if applicable) |
|----------------------------------|---------------------|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------|
| jtag_dfd_host_scan_in_i          | logic               | Input     | 1                      | DFD iJTAG scan data in.                                                                                                                                                                                                  | DFD iJTAG chain              | N/A                           |
| jtag_dfd_host_scan_out_o         | logic               | Output    | 1                      | DFD iJTAG scan data out.                                                                                                                                                                                                 | DFD iJTAG chain              | N/A                           |
| jtag_dft_secure_host_scan_ctrl_o | jtag_scan_ctrl_t    | Output    | -                      | Secure DFT iJTAG scan control. Tie scan_in to scan_out if unused.                                                                                                                                                        | Secure DFT iJTAG chain       | N/A                           |
| jtag_dft_secure_host_scan_in_i   | logic               | Input     | 1                      | Secure DFT iJTAG scan data in.                                                                                                                                                                                           | Secure DFT iJTAG chain       | N/A                           |
| jtag_dft_secure_host_scan_out_o  | logic               | Output    | 1                      | Secure DFT iJTAG scan data out.                                                                                                                                                                                          | Secure DFT iJTAG chain       | N/A                           |
| jtag_dft_host_scan_ctrl_o        | jtag_scan_ctrl_t    | Output    | -                      | Non-secure DFT iJTAG scan control. Tie scan_in to scan_out if unused.                                                                                                                                                    | Non-secure DFT iJTAG chain   | N/A                           |
| jtag_dft_host_scan_in_i          | logic               | Input     | 1                      | Non-secure DFT iJTAG scan data in.                                                                                                                                                                                       | Non-secure DFT iJTAG chain   | N/A                           |
| jtag_dft_host_scan_out_o         | logic               | Output    | 1                      | Non-secure DFT iJTAG scan data out.                                                                                                                                                                                      | Non-secure DFT iJTAG chain   | N/A                           |
| axi_smc_dbg_req_o                | smc_jtag_axi_req_t  | Output    | -                      | SMC fabric debug AXI request. Active when JTAG_SMC_DBG_ENABLE=1.                                                                                                                                                         | SMC jtag_axi_in_req_i        | Tie to '0 if disabled         |
| axi_smc_dbg_resp_i               | smc_jtag_axi_resp_t | Input     | -                      | SMC fabric debug AXI response.                                                                                                                                                                                           | SMC jtag_axi_in_resp_o       | Tie to '0 if disabled         |
| axil_smc_otp_jtag_req_o          | smc_otp_axil_req_t  | Output    | -                      | SMC OTP debug AXI-Lite request.                                                                                                                                                                                          | SMC axil_smc_otp_jtag_req_i  | Tie to '0 if disabled         |
| axil_smc_otp_jtag_resp_i         | smc_otp_axil_resp_t | Input     | -                      | SMC OTP debug AXI-Lite response.                                                                                                                                                                                         | SMC axil_smc_otp_jtag_resp_o | Tie to '0 if disabled         |
| axil_sep_otp_jtag_req_o          | sep_otp_axil_req_t  | Output    | -                      | SEP OTP debug AXI-Lite request.                                                                                                                                                                                          | SEP axil_sep_otp_jtag_req_i  | Tie to '0 if disabled         |
| axil_sep_otp_jtag_resp_i         | sep_otp_axil_resp_t | Input     | -                      | SEP OTP debug AXI-Lite response.                                                                                                                                                                                         | SEP axil_sep_otp_jtag_resp_o | Tie to '0 if disabled         |
| stop_clks_o                      | logic               | Output    | 1                      | Clock stop output from JTAG_DEBUG_CONTROL or CLA clock-stop requests via CTN.                                                                                                                                            | PLL clock gates              | N/A                           |
| cla_clock_stop_en_o              | logic               | Output    | 1                      | CLA clock stop enable from DEBUG_CONTROL register.                                                                                                                                                                       | Clock control logic          | N/A                           |
| jtag_boot_stall_ovrd_o           | logic               | Output    | 1                      | JTAG boot stall override enable.                                                                                                                                                                                         | Boot control                 | N/A                           |
| jtag_boot_stall_o                | logic               | Output    | 1                      | JTAG boot stall value when override is asserted.                                                                                                                                                                         | Boot control                 | N/A                           |
| jtag_ic_reset_smc_o              | ic_reset_smc_t      | Output    | \$bits(ic_reset_smc_t) | SMC IC_RESET override slice. Struct contains .ovrd (override enables) and .val (active-low override values), each sized to match the corresponding SMC reset control struct. Tied to '0 when JTAG_IC_RESET_SMC_ENABLE=0. | SMC jtag_smc_reset_ctrl_i    | N/A                           |
| jtag_ic_reset_sep_o              | ic_reset_sep_t      | Output    | \$bits(ic_reset_sep_t) | SEP IC_RESET override slice. Struct contains .ovrd (override enables) and .val (active-low override values), each sized to match the corresponding SEP reset control struct. Tied to '0 when JTAG_IC_RESET_SEP_ENABLE=0. | SEP jtag_sep_reset_ctrl_i    | N/A                           |

| Interface Name              | Type                       | Direction | Width                        | Description                                                                                                                                                                                           | Connection                                                   | Default value (if applicable) |
|-----------------------------|----------------------------|-----------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------|
| jtag_ic_reset_ext_o         | ic_reset_ext_t             | Output    | \$bits(ic_reset_ext_t)       | External IC_RESET override slice. Struct contains .ovrd (override enables) and .val (active-low override values), each sized per integrator-defined type. Tied to '0 when JTAG_IC_RESET_EXT_ENABLE=0. | Integrator-defined reset overrides for logic outside the SMU | N/A                           |
| jtag_ptap_state_o           | tap_state_e                | Output    | -                            | Current PTAP TAP state.                                                                                                                                                                               | Debug infrastructure                                         | N/A                           |
| jtag_ptap_inst_decoded_o    | jtag_instruction_decoded_e | Output    | -                            | Current decoded PTAP instruction.                                                                                                                                                                     | Debug infrastructure                                         | N/A                           |
| axil_xtrig_req_i            | xtrig_axil_req_t           | Input     | -                            | Cross trigger CSR AXI-Lite request.                                                                                                                                                                   | SMC axil_dtp_csr_req_o                                       | N/A                           |
| axil_xtrig_resp_o           | xtrig_axil_resp_t          | Output    | -                            | Cross trigger CSR AXI-Lite response.                                                                                                                                                                  | SMC axil_dtp_csr_resp_i                                      | N/A                           |
| xtrig_ctm_src_req_o         | logic                      | Output    | [XTRIG_NUM_IN_T_CT-1:0]      | Cross trigger source requests from CTM to destination sinks.                                                                                                                                          | Cross trigger network                                        | N/A                           |
| xtrig_ctm_src_ack_i         | logic                      | Input     | [XTRIG_NUM_IN_T_CT-1:0]      | Cross trigger source acknowledges. Unused in pulse-sync mode (mode bit = 0).                                                                                                                          | Cross trigger network                                        | Tie to '0 in pulse-sync mode  |
| xtrig_ctm_dst_req_i         | logic                      | Input     | [XTRIG_NUM_IN_T_CT-1:0]      | Cross trigger destination requests from source generators.                                                                                                                                            | Cross trigger network                                        | Tie to '0 if unused           |
| xtrig_ctm_dst_ack_o         | logic                      | Output    | [XTRIG_NUM_IN_T_CT-1:0]      | Cross trigger destination acknowledges. Unused in pulse-sync mode (mode bit = 0).                                                                                                                     | Cross trigger network                                        | N/A                           |
| xtrig_clk_stop_req_i        | logic                      | Input     | [XTRIG_NUM_CLK_STOP_REQ-1:0] | Clock stop requests from CLAs or other devices in the chiplet.                                                                                                                                        | Clock stop requesters                                        | Tie to '0 if unused           |
| xtrig_ctp_req_out_dout_o    | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP request-out data output (to GPIO pad ring).                                                                                                                                                       | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_req_out_dout_en_o | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP request-out data output enable.                                                                                                                                                                   | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_req_out_din_i     | logic                      | Input     | [XTRIG_NUM_CTP_P-1:0]        | CTP request-out data input (from GPIO pad ring).                                                                                                                                                      | GPIO pad ring                                                | Tie to '0 if unused           |
| xtrig_ctp_req_out_din_en_o  | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP request-out data input enable.                                                                                                                                                                    | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_req_in_dout_o     | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP request-in data output.                                                                                                                                                                           | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_req_in_dout_en_o  | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP request-in data output enable.                                                                                                                                                                    | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_req_in_din_i      | logic                      | Input     | [XTRIG_NUM_CTP_P-1:0]        | CTP request-in data input.                                                                                                                                                                            | GPIO pad ring                                                | Tie to '0 if unused           |
| xtrig_ctp_req_in_din_en_o   | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP request-in data input enable.                                                                                                                                                                     | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_ack_in_dout_o     | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP ack-in data output.                                                                                                                                                                               | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_ack_in_dout_en_o  | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP ack-in data output enable.                                                                                                                                                                        | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_ack_in_din_i      | logic                      | Input     | [XTRIG_NUM_CTP_P-1:0]        | CTP ack-in data input.                                                                                                                                                                                | GPIO pad ring                                                | Tie to '0 if unused           |
| xtrig_ctp_ack_in_din_en_o   | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP ack-in data input enable.                                                                                                                                                                         | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_ack_out_dout_o    | logic                      | Output    | [XTRIG_NUM_CTP_P-1:0]        | CTP ack-out data output.                                                                                                                                                                              | GPIO pad ring                                                | N/A                           |

| Interface Name              | Type  | Direction | Width               | Description                     | Connection    | Default value (if applicable) |
|-----------------------------|-------|-----------|---------------------|---------------------------------|---------------|-------------------------------|
| xtrig_ctp_ack_out_dout_en_o | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data output enable. | GPIO pad ring | N/A                           |
| xtrig_ctp_ack_out_din_i     | logic | Input     | [XTRIG_NUM_CTP-1:0] | CTP ack-out data input.         | GPIO pad ring | Tie to '0 if unused           |
| xtrig_ctp_ack_out_din_en_o  | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data input enable.  | GPIO pad ring | N/A                           |

## 1.5. Security Processor (SEP) Integration

The Security Processor (SEP) is the Root of Trust for the chiplet, providing a physically isolated execution environment for security-critical services such as secure boot, lifecycle control, device attestation, cryptographic acceleration, and key management. This section provides practical integration procedures for adopters integrating the SEP as part of the SMU or as a standalone out-of-context IP.

For detailed architectural documentation about the SEP internal design, refer to the dedicated SEP documentation located in [hw/sys/sep/doc/](#). The SEP hardware overview and interconnect fabric are described in [hw/sys/sep/doc/overview.adoc](#) and [hw/sys/sep/doc/fabric.adoc](#) respectively.

### 1.5.1. Module Overview

The [sep.sv](#) module provides the complete SEP functionality with logic-level interfaces. Technology-specific components—SRAM, ROM, and eFuse shims—are instantiated by the adopter outside SEP, following the same vendor-neutral philosophy used for the SMC.

Separately, SEP exposes interfaces for supplementary peripherals an adopter may add on rather than substitute: an external TRNG and an additional SPI controller. These are not technology-specific—SEP contains its own entropy source and SPI host—so the interfaces exist to supplement what SEP already provides, and may be left unused.

When integrated inside the SMU, the SEP's external interfaces are passed through the SMU boundary with a [sep\\_](#) prefix.

### 1.5.2. Clock and Reset Requirements

The SEP requires two clock inputs. Connect [clk\\_i](#) to the primary system clock driving all SEP internal logic. Connect [clk\\_wdt\\_i](#) to a 200 kHz clock source for the watchdog timer; this clock must be independent of the system clock so that the WDT can detect system clock failures.

The primary reset [rst\\_ni](#) follows the standard active-low, asynchronous assertion / synchronous deassertion convention. The [dbg\\_rstb\\_i](#) input provides a separate reset for the EL2 CPU debug module. The [wdt\\_rst\\_ni](#) input receives the watchdog reset generated by the SMC reset unit. The SEP outputs [wdt\\_timer\\_rst\\_req\\_o](#) when the WDT triggers a reset request.

### 1.5.3. AXI Interfaces

The SEP connects to the system through three AXI paths. The [smn\\_outbound\\_axi](#) master interface allows SEP firmware to access system resources outside the SEP domain, such as SMC registers or external memory. The [smn\\_inbound\\_axi](#) slave interface accepts requests from system masters requiring access to SEP-managed resources. The [sep\\_ext\\_to\\_smc\\_axi](#) output provides a dedicated AXI path for SEP-to-SMC transactions, connecting to the SMC's [sep\\_axi\\_in](#) port.

An additional AXI extension port ([axi\\_extension\\_axi](#)) enables adopters to attach custom peripherals to the SEP

interconnect fabric without modifying the SEP core. When unused, tie the response input to return DECERR.

### 1.5.4. Memory Interfaces

All SEP memory macros are instantiated externally and connected through dedicated request/response interfaces. The `sep_cpu_tcm` interface provides tightly-coupled memory (ICCM and DCCM) for the VeeR EL2 CPU. The `sep_sram` interface connects scratch SRAM used for firmware data. The `sep_boot_rom` interface provides the BL0 ROM containing the first-stage bootloader.

The crypto subsystem requires additional memory interfaces. `sep_crypto_pka_imem_sram` and `sep_crypto_pka_dmem_sram` connect to the OTBN public key accelerator instruction and data memories respectively. `km_rom_mem` and `km_sram_mem` connect to the Key Manager's dedicated ROM and SRAM macros.

`abr_mem` connects the Adams Bridge working SRAM. On an idle cycle (`re` low), ABR expects the channel's read data to return 0 the following cycle. `SEP_ABR_SRAM_LATENCY` must equal the macros' native registered-read latency (default 1). Word-write channels are full-word `we` / `re` / address / data. `sig_z` and `pk` carry one-bit-per-byte `wstrobe`; map those to the macros' byte-write enables. `SEP_ABR_MASKING_EN` (default 1) selects 2-share DOM: thirteen 1R1W macros versus nine, plus a second NTT. This changes ABR internal behavior (sequencing, SCA, memory topology), not merely throughput. Give the same value to SEP and to the SRAM instances. When it is 0, do not instantiate the four masked coefficient banks; their response fields read as 0.

### 1.5.5. Cryptographic Subsystem Connections

The SEP includes an internal entropy source that requires a ring-oscillator sample clock connected to `entropy_rosc_sample_clk_i`. This clock must be asynchronous to `clk_i` and typically runs between 100 and 400 MHz.

#### External TRNG

For adopters providing an external TRNG, the `ext_trng_axil` AXI-Lite interface connects to the external TRNG's control registers, while the `ext_trng_axis` AXI-Stream interfaces carry entropy data. The `ext_trng_irq_i` input is routed to the PIC; `ext_trng_alarm_i` is reserved for RAS.

SEP decodes the 4 kB TRNG aperture (`0x1091_7000` - `0x1091_7FFF`) as a single window and forwards every access on `ext_trng_axil`. It does not decode offsets inside that window: the endpoint is an adopter-defined IP.

The adopter must perform the intra-aperture decode. An address that owns no register must return SLVERR and must not alias onto a live register. When no external TRNG is connected, terminate `ext_trng_axil` with a DECERR slave.

### 1.5.6. eFuse and Lifecycle

SEP has its own eFuse bank. Instantiate a separate SHIM and physical eFuse bank; do not share the SMC's. The SHIM contract is [eFuse Interface](#). On `sep.sv` connect `efuse_bank_ctrl_*` and `efuse_shim_command_*`. Through the SMU those ports appear as `sep_efuse_bank_ctrl_*` and `sep_efuse_shim_command_*`.

SEP senses this bank after its own reset. Watch `sep_fuse_sense_done_o`. On `sep.sv`, drive `smc_fuse_sense_done_i` from the SMC so firmware can see when the SMC fabric is out of reset; the SMU wires that input internally.

Lifecycle and test-mode ports on `sep.sv`, and what to do with them. Behaviour is in the Technical Reference Manual, not here.

- `secure_tm_req_i` signals a request to enter secure test mode. SEP latches it into `secure_tm_o` when fuse sensing completes, or at cold-reset release if SEC\_DIS is already asserted. Secure test mode forces a no-secret, no-fuse-

program configuration; it is not an authorization bit. See [Test Mode Entry](#) and [secure\\_tm](#) and the [Secure Scan Chain](#) in the TRM. [secure\\_tm\\_o](#) is also an SMU output.

- [security\\_disable\\_o](#) is the SEC\_DIS token-match result (emergency A0 unlock). On [sep.sv](#), route it to SMC [sep\\_security\\_disable\\_i](#); the SMU does that internally. See [Security Disable](#) in the TRM.
- [lc\\_state\\_o](#) is the differentially encoded LC\_STATE shadow. [lc\\_sigint\\_err\\_o](#) is the decode integrity error (fail-closed). Both are SMU outputs. See [Life Cycle Controller](#) in the TRM.
- [dbg\\_disable\\_o](#) is the typed per-path debug/test disable vector (active-high: 1 = path disabled). On [sep.sv](#), route it to DTP [dbg\\_disable\\_i](#); the SMU does that internally.
- [lcc\\_demote\\_state\\_1\\_o](#) / [lcc\\_demote\\_state\\_2\\_o](#) are differentially encoded in-field debug demotion. They leave the SMU for adopter lifecycle logic and are not connected to the SMC. See [in-field debug demotion](#) in the TRM.
- [sep\\_fuse\\_dft\\_disable\\_o](#) / [smc\\_fuse\\_dft\\_disable\\_o](#) gate DFT-inserted OTP access. Leave them open if no such path is inserted. They are SMU outputs.

OTP debug AXI-Lite ([axil\\_sep\\_otp\\_jtag](#)) is connected to the DTP inside the SMU; when [SEP=0](#) the SMU instantiates an AXI-Lite error slave on that path.

## 1.5.7. SPI Host Interface

The SEP provides a SPI host controller through the [sep\\_io\\_spi](#) interface, supporting single and quad modes. The controller’s own interrupt leaves the module as the [irq](#) field of [sep\\_io\\_spi\\_req\\_o](#), so no separate interrupt output is needed.

## 1.5.8. Bring-Up Considerations

The SEP boot sequence begins after the SMC completes fuse sensing and releases the SEP reset. The SEP BL0 ROM FW verifies the BL1 software image using non-volatile public keys before transferring control. Feature controls derived from secure boot authentication and lifecycle state govern what operations are permitted after boot.

The [jtag\\_id](#) input must be tied to an appropriate constant in the top-level integration. A [rst\\_vec\\_i](#) port exists on [tb\\_top.sv](#) for DV reset vector injection; it is not present on [sep.sv](#) itself and requires no integration tie-off.

## 1.5.9. SEP Port Interface

The SEP exposes interfaces for clocks, resets, JTAG debug, CPU control, memory macros, AXI bus connections, cryptographic subsystem, eFuse, lifecycle, SPI host, and mailbox. The following table defines all SEP top-level ports.

Table 10. SEP Port Declaration

| Interface Name             | Type                  | Direction | Width | Description                                                                     | Connection                 | Default value (if applicable) |
|----------------------------|-----------------------|-----------|-------|---------------------------------------------------------------------------------|----------------------------|-------------------------------|
| <a href="#">clk_i</a>      | <a href="#">logic</a> | Input     | 1     | Primary SEP clock. All SEP internal logic operates synchronously to this clock. | System clock               | N/A                           |
| <a href="#">clk_wdt_i</a>  | <a href="#">logic</a> | Input     | 1     | 200 kHz clock for watchdog timer.                                               | Low-frequency clock source | N/A                           |
| <a href="#">rst_ni</a>     | <a href="#">logic</a> | Input     | 1     | Primary reset (active low). Asynchronous assertion, synchronous deassertion.    | System reset               | N/A                           |
| <a href="#">dbg_rstb_i</a> | <a href="#">logic</a> | Input     | 1     | EL2 debugger reset (active low).                                                | JTAG debug infrastructure  | N/A                           |
| <a href="#">wdt_rst_ni</a> | <a href="#">logic</a> | Input     | 1     | Watchdog timer reset from SMC (active low).                                     | SMC reset unit             | N/A                           |

| Interface Name           | Type                             | Direction | Width  | Description                                                                                                                                     | Connection                                   | Default value (if applicable) |
|--------------------------|----------------------------------|-----------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------|
| wdt_timer_rst_req_o      | logic                            | Output    | 1      | WDT reset request output. Triggers memory interface reset in <a href="#">sep_wrapper</a> .                                                      | SEP wrapper                                  | N/A                           |
| jtag_tck                 | logic                            | Input     | 1      | JTAG test clock.                                                                                                                                | JTAG pads (via DTP STAP)                     | N/A                           |
| jtag_tms                 | logic                            | Input     | 1      | JTAG test mode select.                                                                                                                          | JTAG pads (via DTP STAP)                     | N/A                           |
| jtag_tdi                 | logic                            | Input     | 1      | JTAG test data in.                                                                                                                              | JTAG pads (via DTP STAP)                     | N/A                           |
| jtag_trst_n              | logic                            | Input     | 1      | JTAG test reset (active low).                                                                                                                   | JTAG pads (via DTP STAP)                     | N/A                           |
| jtag_tdo                 | logic                            | Output    | 1      | JTAG test data out.                                                                                                                             | JTAG pads (via DTP STAP)                     | N/A                           |
| jtag_tdoEn               | logic                            | Output    | 1      | JTAG test data out enable.                                                                                                                      | JTAG pads (via DTP STAP)                     | N/A                           |
| axil_sep_otp_jtag_req_i  | sep_efuse_pkg::efuse_axil_req_t  | Input     | -      | OTP debug AXI-Lite request from JTAG.                                                                                                           | DTP <a href="#">axil_sep_otp_jtag_req_o</a>  | Tie to '0 if unused           |
| axil_sep_otp_jtag_resp_o | sep_efuse_pkg::efuse_axil_resp_t | Output    | -      | OTP debug AXI-Lite response to JTAG.                                                                                                            | DTP <a href="#">axil_sep_otp_jtag_resp_i</a> | N/A                           |
| mpc_debug_halt_req       | logic                            | Input     | 1      | Async halt request from MPC.                                                                                                                    | Debug controller                             | Tie to '0 if unused           |
| mpc_debug_run_req        | logic                            | Input     | 1      | Async run request from MPC.                                                                                                                     | Debug controller                             | Tie to '0 if unused           |
| mpc_reset_run_req        | logic                            | Input     | 1      | Run/halt selection after reset.                                                                                                                 | Debug controller                             | 1'b1                          |
| i_cpu_halt_req           | logic                            | Input     | 1      | Async halt request to CPU.                                                                                                                      | Debug controller                             | Tie to '0 if unused           |
| i_cpu_run_req            | logic                            | Input     | 1      | Async restart request to CPU.                                                                                                                   | Debug controller                             | Tie to '0 if unused           |
| test_en_i                | logic                            | Input     | 1      | DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.                                                                   | Test controller                              | 1'b0                          |
| scan_rst_ni              | logic                            | Input     | 1      | Scan reset (active-low). Bypasses reset synchronizers during DFT.                                                                               | Test controller                              | 1'b1                          |
| ext_boot_seq_done_i      | logic                            | Input     | 1      | External boot sequence completion (memory repair and shadow register override done); gates reset release.                                       | External boot/repair controller              | 1'b1                          |
| dmi_core_enable          | logic                            | Input     | 1      | Enables DMI accesses to the core debug module (addresses 0x00-0x4F). Tied high at SMU; TAP access is already lifecycle-gated.                   | Debug infrastructure                         | 1'b1                          |
| dmi_uncore_enable        | logic                            | Input     | 1      | Enables DMI accesses to an external uncore aperture (addresses 0x50+). Unused: the core debug module AXI master already reaches the SEP fabric. | Debug infrastructure                         | Tie to '0 if unused           |
| dmi_uncore_en            | logic                            | Output    | 1      | DMI uncore access enable.                                                                                                                       | Debug infrastructure                         | N/A                           |
| dmi_uncore_wr_en         | logic                            | Output    | 1      | DMI uncore write enable.                                                                                                                        | Debug infrastructure                         | N/A                           |
| dmi_uncore_addr          | logic                            | Output    | [6:0]  | DMI uncore address.                                                                                                                             | Debug infrastructure                         | N/A                           |
| dmi_uncore_wdata         | logic                            | Output    | [31:0] | DMI uncore write data.                                                                                                                          | Debug infrastructure                         | N/A                           |

| Interface Name               | Type                                       | Direction | Width                   | Description                                                                                                                                       | Connection                   | Default value (if applicable) |
|------------------------------|--------------------------------------------|-----------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------|
| dmi_uncore_read_data         | logic                                      | Input     | [31:0]                  | DMI uncore read data.                                                                                                                             | Debug infrastructure         | N/A                           |
| dmi_active                   | logic                                      | Output    | 1                       | DMI active indicator.                                                                                                                             | Debug infrastructure         | N/A                           |
| sep_cpu_trace                | sep_cpu_trace_t                            | Output    | -                       | CPU trace output for debug.                                                                                                                       | Trace infrastructure         | N/A                           |
| lockstep_ctrl_i              | sep_pkg::sep_lockstep_ctrl_t               | Input     | -                       | VeeR lockstep controls: disable corruption detection, and error-injection enable. Inert unless the core is built with <b>RV_LOCKSTEP_ENABLE</b> . | Chip-level RAS control       | Tie to '0 if unused           |
| lockstep_status_o            | sep_pkg::sep_lockstep_status_t             | Output    | -                       | VeeR lockstep status: corruption detected. Reads 0 unless the core is built with <b>RV_LOCKSTEP_ENABLE</b> .                                      | Chip-level RAS aggregation   | N/A                           |
| jtag_id                      | logic                                      | Input     | [31:1]                  | JTAG ID value.                                                                                                                                    | Configuration tie-off        | N/A                           |
| timer_int                    | logic                                      | Input     | 1                       | Timer interrupt.                                                                                                                                  | Internal timer               | Tie to '0 if unused           |
| soft_int                     | logic                                      | Input     | 1                       | Software interrupt.                                                                                                                               | Software IRQ source          | Tie to '0 if unused           |
| extintsrc_req                | logic                                      | Input     | [NUM_EXTERNAL_IRQS-1:0] | External interrupt sources for PIC.                                                                                                               | External IRQ sources         | Tie to '0 if unused           |
| sep_cpu_tcm_req_o            | sep_cpu_tcm_req_t                          | Output    | -                       | CPU tightly-coupled memory (TCM) request (ICCM + DCCM).                                                                                           | SRAM macros                  | N/A                           |
| sep_cpu_tcm_rsp_i            | sep_cpu_tcm_rsp_t                          | Input     | -                       | CPU TCM response.                                                                                                                                 | SRAM macros                  | N/A                           |
| sep_sram_req                 | sep_sram_req_t                             | Output    | -                       | Scratch SRAM request.                                                                                                                             | SRAM macros                  | N/A                           |
| sep_sram_rsp                 | sep_sram_rsp_t                             | Input     | -                       | Scratch SRAM response.                                                                                                                            | SRAM macros                  | N/A                           |
| sep_boot_rom_req             | sep_sram_req_t                             | Output    | -                       | Boot ROM (BL0) request.                                                                                                                           | ROM macro                    | N/A                           |
| sep_boot_rom_rsp             | sep_sram_rsp_t                             | Input     | -                       | Boot ROM (BL0) response.                                                                                                                          | ROM macro                    | N/A                           |
| smn_outbound_axi_req_o       | sep_system_peripherals_outbound_axi_req_t  | Output    | -                       | SMN outbound AXI request (system bus master).                                                                                                     | SMN fabric                   | N/A                           |
| smn_outbound_axi_resp_i      | sep_system_peripherals_outbound_axi_resp_t | Input     | -                       | SMN outbound AXI response.                                                                                                                        | SMN fabric                   | N/A                           |
| smn_inbound_axi_req_i        | sep_system_peripherals_internal_axi_req_t  | Input     | -                       | SMN inbound AXI request (system bus slave).                                                                                                       | SMN fabric                   | Tie to '0 if unused           |
| smn_inbound_axi_resp_o       | sep_system_peripherals_internal_axi_resp_t | Output    | -                       | SMN inbound AXI response.                                                                                                                         | SMN fabric                   | N/A                           |
| sep_ext_to_smc_axi_req_o     | sep_system_peripherals_internal_axi_req_t  | Output    | -                       | SEP-to-SMC AXI request. Dedicated path for SEP to access SMC resources.                                                                           | SMC <b>sep_axi_in_req_i</b>  | N/A                           |
| sep_ext_to_smc_axi_resp_i    | sep_system_peripherals_internal_axi_resp_t | Input     | -                       | SEP-to-SMC AXI response.                                                                                                                          | SMC <b>sep_axi_in_resp_o</b> | N/A                           |
| entropy_rosc_sample_clk_i    | logic                                      | Input     | 1                       | Ring-oscillator sample clock for entropy source (async vs <b>clk_i</b> ; ~100-400 MHz typical).                                                   | Ring oscillator              | N/A                           |
| sep_crypto_pka_imem_sram_req | sep_crypto_pka_imem_sram_req_t             | Output    | -                       | OTBN PKA instruction memory SRAM request.                                                                                                         | SRAM macro                   | N/A                           |
| sep_crypto_pka_imem_sram_rsp | sep_crypto_pka_imem_sram_rsp_t             | Input     | -                       | OTBN PKA instruction memory SRAM response.                                                                                                        | SRAM macro                   | N/A                           |

| Interface Name               | Type                                  | Direction | Width                      | Description                                                                                                                                                                                 | Connection            | Default value (if applicable)     |
|------------------------------|---------------------------------------|-----------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------|
| sep_crypto_pka_dmem_sram_req | sep_crypto_pka_dmem_sram_req_t        | Output    | -                          | OTBN PKA data memory SRAM request.                                                                                                                                                          | SRAM macro            | N/A                               |
| sep_crypto_pka_dmem_sram_rsp | sep_crypto_pka_dmem_sram_rsp_t        | Input     | -                          | OTBN PKA data memory SRAM response.                                                                                                                                                         | SRAM macro            | N/A                               |
| ext_trng_axil_req_o          | sep_32_32_axil_req_t                  | Output    | -                          | External TRNG AXI-Lite passthrough request (to <a href="#">sep_ip_integration</a> ).                                                                                                        | External TRNG shim    | N/A                               |
| ext_trng_axil_resp_i         | sep_32_32_axil_resp_t                 | Input     | -                          | External TRNG AXI-Lite passthrough response.                                                                                                                                                | External TRNG shim    | N/A                               |
| ext_trng_axis_req_i          | ext_trng_axis_req_t                   | Input     | [EXT_TRNG_NUM_AXIS-1:0]    | External TRNG AXI-Stream request array (from <a href="#">sep_ip_integration</a> ).                                                                                                          | External TRNG shim    | N/A                               |
| ext_trng_axis_rsp_o          | ext_trng_axis_rsp_t                   | Output    | [EXT_TRNG_NUM_AXIS-1:0]    | External TRNG AXI-Stream response array.                                                                                                                                                    | External TRNG shim    | N/A                               |
| ext_trng_irq_i               | logic                                 | Input     | 1                          | External TRNG interrupt (routed to PIC).                                                                                                                                                    | External TRNG shim    | Tie to '0 if unused               |
| ext_trng_alarm_i             | logic                                 | Input     | 1                          | External TRNG alarm (reserved for RAS).                                                                                                                                                     | External TRNG shim    | Tie to '0 if unused               |
| km_rom_mem_req_o             | km_intf_pkg::km_rom_mem_req_t         | Output    | -                          | Key Manager ROM memory request.                                                                                                                                                             | ROM macro             | N/A                               |
| km_rom_mem_rsp_i             | km_intf_pkg::km_rom_mem_rsp_t         | Input     | -                          | Key Manager ROM memory response.                                                                                                                                                            | ROM macro             | N/A                               |
| km_sram_mem_req_o            | km_intf_pkg::km_sram_mem_req_t        | Output    | -                          | Key Manager SRAM memory request.                                                                                                                                                            | SRAM macro            | N/A                               |
| km_sram_mem_rsp_i            | km_intf_pkg::km_sram_mem_rsp_t        | Input     | -                          | Key Manager SRAM memory response.                                                                                                                                                           | SRAM macro            | N/A                               |
| efuse_bank_ctrl_req_o        | sep_efuse_pkg::efuse_axil_req_t       | Output    | -                          | eFuse bank control AXI-Lite request.                                                                                                                                                        | eFuse shim            | N/A                               |
| efuse_bank_ctrl_resp_i       | sep_efuse_pkg::efuse_axil_resp_t      | Input     | -                          | eFuse bank control AXI-Lite response.                                                                                                                                                       | eFuse shim            | N/A                               |
| efuse_shim_command_req_o     | sep_efuse_pkg::fuse_command_req_t     | Output    | -                          | eFuse command interface request.                                                                                                                                                            | eFuse shim            | N/A                               |
| efuse_shim_command_resp_i    | sep_efuse_pkg::fuse_command_resp_t    | Input     | -                          | eFuse command interface response.                                                                                                                                                           | eFuse shim            | N/A                               |
| lcc_demote_state_1_o         | logic                                 | Output    | [1:0]                      | Lifecycle demote state output 1. Exported at SMU boundary; not currently connected to SMC.                                                                                                  | SMU external port     | N/A                               |
| lcc_demote_state_2_o         | logic                                 | Output    | [1:0]                      | Lifecycle demote state output 2. Exported at SMU boundary; not currently connected to SMC.                                                                                                  | SMU external port     | N/A                               |
| sep_io_spi_req_o             | sep_io_spi_req_t                      | Output    | -                          | SPI host controller request.                                                                                                                                                                | SPI pad interface     | N/A                               |
| sep_io_spi_rsp_i             | sep_io_spi_rsp_t                      | Input     | -                          | SPI host controller response.                                                                                                                                                               | SPI pad interface     | N/A                               |
| lc_state_o                   | logic                                 | Output    | [2*LC_STATE_BIT_WIDTH-1:0] | Lifecycle state output (differentially encoded). Routed to SMC and the SMU external output; debug-disable control to DTP is via <a href="#">dbg_disable_o</a> .                             | SMC, SMU external     | N/A                               |
| dbg_disable_o                | sep_lifecycle_ctrl_pkg::dbg_disable_t | Output    | -                          | Typed debug-disable struct routed by SMU to DTP debug-disable control.                                                                                                                      | SMU → DTP             | N/A                               |
| sep_fuse_dft_disable_o       | logic                                 | Output    | 1                          | Gate for the DFT-inserted access path into the SEP OTP, composed as Case 3 AND <a href="#">SEP_FUSE_DBG</a> . Unconnected in the functional design; an adopter's DFT insertion connects it. | Adopter DFT insertion | Leave open if no path is inserted |

| Interface Name                        | Type                                   | Direction | Width                            | Description                                                                                                                                                                              | Connection                         | Default value (if applicable)     |
|---------------------------------------|----------------------------------------|-----------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------|
| <code>smc_fuse_dft_disable_o</code>   | logic                                  | Output    | 1                                | Gate for the DFT-inserted access path into the SMC OTP, composed as Case 2 AND <code>SMC_FUSE_DBG</code> . Unconnected in the functional design; an adopter's DFT insertion connects it. | Adopter DFT insertion              | Leave open if no path is inserted |
| <code>lc_sigint_err_o</code>          | logic                                  | Output    | 1                                | Lifecycle signal integrity error.                                                                                                                                                        | SMU                                | N/A                               |
| <code>security_disable_o</code>       | logic                                  | Output    | 1                                | Security disable output.                                                                                                                                                                 | SMU                                | N/A                               |
| <code>secure_tm_o</code>              | logic                                  | Output    | 1                                | Secure test mode output.                                                                                                                                                                 | SMU                                | N/A                               |
| <code>smc_mailbox_interrupt_o</code>  | logic                                  | Output    | <code>[NUM_MAILBOXES-1:0]</code> | Mailbox interrupt outputs to SMC.                                                                                                                                                        | SMC interrupt aggregator           | N/A                               |
| <code>smc_fuse_sense_done_i</code>    | logic                                  | Input     | 1                                | SMC fuse sense done indication.                                                                                                                                                          | SMC <code>fuse_sense_done_o</code> | N/A                               |
| <code>sep_fuse_sense_done_o</code>    | logic                                  | Output    | 1                                | SEP fuse sense done output.                                                                                                                                                              | System integration                 | N/A                               |
| <code>secure_tm_req_i</code>          | logic                                  | Input     | -                                | Secure test mode request.                                                                                                                                                                | GPIO padding / tie-off             | N/A                               |
| <code>axi_extension_axi_req_o</code>  | <code>sep_32_64_6_12_axi_req_t</code>  | Output    | -                                | AXI extension port request for adopter peripheral expansion.                                                                                                                             | Adopter peripherals                | N/A                               |
| <code>axi_extension_axi_resp_i</code> | <code>sep_32_64_6_12_axi_resp_t</code> | Input     | -                                | AXI extension port response.                                                                                                                                                             | Adopter peripherals                | Tie to DECERR if unused           |
| <code>smc_global_base_addr_i</code>   | logic                                  | Input     | <code>[55:0]</code>              | SMC global base address for outbound address translation.                                                                                                                                | System configuration               | N/A                               |
| <code>smc_region_size_i</code>        | logic                                  | Input     | <code>[55:0]</code>              | SMC region size for outbound address translation.                                                                                                                                        | SMC <code>smc_region_size_o</code> | N/A                               |
| <code>km_unrecoverable_err_o</code>   | logic                                  | Output    | 1                                | Key Manager unrecoverable error.                                                                                                                                                         | RAS / error handling               | N/A                               |
| <code>km_recoverable_err_o</code>     | logic                                  | Output    | 1                                | Key Manager recoverable error.                                                                                                                                                           | RAS / error handling               | N/A                               |
| <code>ext_debug_bus_o</code>          | logic                                  | Output    | <code>[511:0]</code>             | External debug observation bus.                                                                                                                                                          | Debug infrastructure               | N/A                               |

## 1.6. SMU Composition and Internal Wiring

The System Management Unit (SMU) combines the SMC, DTP, and optionally the SEP into a unified subsystem. The interface descriptions in the preceding sections apply directly when integrating the SMU, since the SMU inherits all three subsystems' ports. This section covers the module variants available to adopters, how the SMU wires its subsystems together internally, and what additional configuration it exposes.

### 1.6.1. Module Variants and IP Integration

OCAH core modules (`smc.sv`, `sep.sv`, `dtp.sv`, and the composite `smu.sv`) expose logic-level interfaces only. All technology-specific components—PLLs, PVT sensors, eFuse shims, memory macros, GPIO padings, SPI controllers, and external TRNG—are instantiated outside the core modules by the adopter. This separation keeps the OCAH deliverable technology-neutral.

The repository provides reference wrappers that demonstrate how to bridge the gap between OCAH core interfaces and

adopter-specific IP:

- `smu_wrapper.sv` wraps `smu.sv` with pad-level signal handling for clocks, resets, GPIO, eFuse, and JTAG interfaces. It internally instantiates `smc_ip_integration.sv` for SMC-related vendor IP and (when `SEP=1`) `sep_ip_integration.sv` for SEP-related vendor IP.
- `smc_wrapper.sv` provides an equivalent reference wrapping `smc.sv` alone, useful when integrating the SMC without the DTP/SEP subsystems.

Neither the wrappers nor the IP integration modules are part of the OCAH deliverable; they exist as integration guides and for Tenstorrent internal usage. Adopters must develop their own equivalents following the interface patterns demonstrated in these modules. The interface contracts at the module boundaries ensure technology substitution without modifying the verified OCAH logic.

The figure below illustrates the relationship between the OCAH core modules, the wrapper layer, and the IP integration modules.

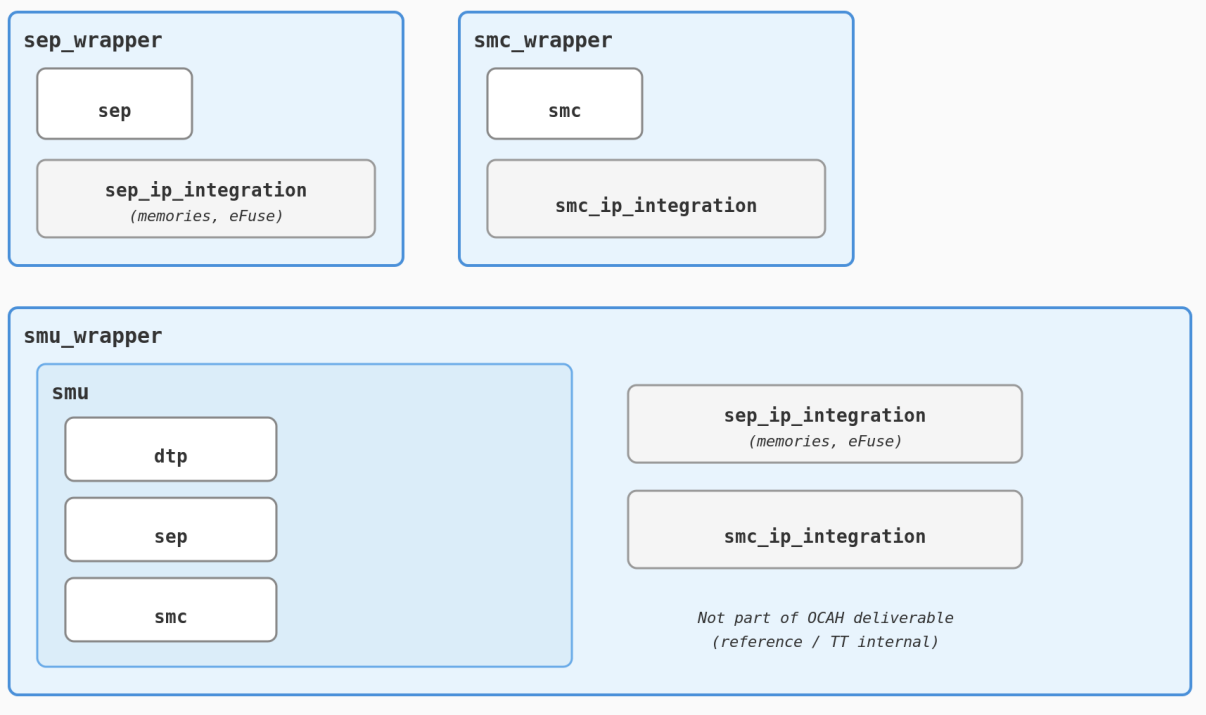


Figure 1. SMU Integration Components

## IP Integration Modules

`smc_ip_integration.sv` demonstrates integration of SMC-side technology components: PLL wrappers (generating `pll_cgm_clk_o` and `pll_awm_clk_o`), PVT sensor interfaces, eFuse shims, GPIO padding logic, memory macros (ROM, scratchpad RAM, cache tag and data arrays), and trace memory instances.

`sep_ip_integration.sv` demonstrates integration of SEP-side technology components: SEP memory macros (TCM, scratch SRAM, boot ROM, OTBN IMEM/DMEM, Key Manager ROM/SRAM), eFuse shim, external TRNG, and SPI controller selection.

## Wrapper Clock Architecture

The `smu_wrapper.sv` reference wrapper accepts pre-generated clocks directly: `clk_smu_i` (primary), `clk_ref_i` (reference),

and `clk_periph_i` (peripheral). The `smc_wrapper.sv` uses `clk_smc_i` for the primary clock. Clock generation is entirely the adopter's responsibility. The `pll_wrap` model included in the DV environment only terminates the SMC's AXI-Lite PLL control interface; it does not generate functional clocks and must not be used as a synthesis model.

## 1.6.2. Clock and Reset Architecture

Connect `clk_smu_i` to the primary system clock and `clk_ref_i` to a stable reference source. The DTP operates on TCK for JTAG logic and on the system clock for cross-trigger and AXI bridges. Reset sequencing follows the SMC flow; the DTP JTAG state machine uses `pwr_on_rst_ni`, which the SMU connects to `rst_cold_ni`. The SMU internally connects DTP debug paths to the SMC and optionally instantiates the SEP when the `SEP` parameter is set to 1. The full interconnection details are described in [SMC-DTP Internal Interconnection](#) and [SEP-Conditional Wiring](#) below.

## 1.6.3. SMU AXI Crossbar Address Map

Inside the SMU, the `smu_axi_xbar` module routes traffic between three AXI4 manager/subordinate pairs: the SEP outbound master (`sep_out`), the SMC outbound master (`smc_out`), and the external SMU AXI input (`ext_in` / `smu_axi_in`). The corresponding targets are the SEP and SMC inbound subordinates, and the external SMU AXI output (`ext_out` / `smu_axi_out`).

See the *SMU AXI Crossbar* and *OCAH Memory Map* sections of the Technical Reference Manual for connectivity and the responsibility split between hardware-fixed and firmware-programmable apertures.

The crossbar uses a **runtime** address map. Only the SEP and SMC apertures are programmable at runtime; the `ext_out` rule is static:

- `sep_global_base_addr_i` / `sep_region_size_i` come from the `sep_cpu_ctrl` CSRs `SEP_GLOBAL_BASE_ADDR` / `SEP_REGION_SIZE`, exposed at both the `sep.sv` and `smu.sv` boundaries as `sep_global_base_addr_o` / `sep_region_size_o` (both 56-bit). `SEP_REGION_SIZE` is a 32-bit CSR field, zero-extended to 56-bit at the SEP system CSR; the xbar input takes only the lower 32 bits. `SEP_REGION_SIZE` defines the SEP global/chiplet aperture for SMU routing and SEP inbound global-to-local remap. It is distinct from the SEP CPU's architectural local-alias span. `SEP_REGION_SIZE` currently resets to `0x0100_0000` (16 MiB). Tied to '0' at the `smu.sv` boundary when the `SEP` parameter is 0.
- `smc_global_base_addr_i` / `smc_region_size_i` come from the SMC CPU control CSRs `GLOBAL_BASE` / `REGION_SIZE`, exposed at both the `smc.sv` and `smu.sv` boundaries as `smc_global_base_o` (56-bit) / `smc_region_size_o` (32-bit). `REGION_SIZE` resets to `0x0100_0000` (16 MiB).

SEP, SMC, and `smu_axi_xbar` all run on `clk_smu_i`, so these quasi-static CSR fields need no clock-domain crossing. The crossbar's address arithmetic is done on 57-bit widened sums. `ext_out` is a catch-all port that carries any outbound SEP/SMC manager request not matched by the programmed apertures; it has no fixed lower-bound address. Unmatched inbound (`ext_in`) requests return a decode error and do not reach `ext_out`.

### Firmware programming order

Firmware must program `SEP_GLOBAL_BASE_ADDR` / `SEP_REGION_SIZE` and SMC `GLOBAL_BASE` / `REGION_SIZE` during early boot, **before any cross-subsystem AXI traffic is issued**, and must ensure the two apertures do not overlap and both fall within the chiplet-visible SMU envelope assigned by the integrator. The SMU crossbar does not itself constrain envelope size.

See the *OCAH Memory Map* section of the Technical Reference Manual for that responsibility split.

## 1.6.4. SMU Default Parameters

The SMU exposes a set of configurable parameters that control DTP JTAG features, identification, cross trigger, and pipeline depths. The table below lists each parameter with its default value.

| Parameter                | Value    | Description                                                                                                                                                   |
|--------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JTAG_BSR_ENABLE          | 1        | Boundary scan enabled                                                                                                                                         |
| JTAG_EXTEST_TRAIN_ENABLE | 1        | EXTEST_TRAIN instruction enabled                                                                                                                              |
| JTAG_EXTEST_PULSE_ENABLE | 1        | EXTEST_PULSE instruction enabled                                                                                                                              |
| JTAG_INTEST_ENABLE       | 1        | INTEST instruction enabled                                                                                                                                    |
| JTAG_CLAMP_ENABLE        | 1        | CLAMP instruction enabled                                                                                                                                     |
| JTAG_HIGHZ_ENABLE        | 1        | HIGHZ instruction enabled                                                                                                                                     |
| JTAG_RUNBIST_ENABLE      | 1        | RUNBIST instruction enabled                                                                                                                                   |
| JTAG_TMP_ENABLE          | 1        | TMP controller enabled                                                                                                                                        |
| JTAG_IC_RESET_ENABLE     | 1        | Enables the external IC_RESET TDR slice (SMU drives this into the DTP's JTAG_IC_RESET_EXT_ENABLE). SMC and SEP slices are enabled unconditionally by the SMU. |
| JTAG_SMC_DBG_ENABLE      | 1        | SMC debug interface enabled                                                                                                                                   |
| JTAG_STAP_IO_ENABLE      | 1        | I/O STAP enabled                                                                                                                                              |
| JTAG_NUM_EXTRA_STAPS     | 1        | Number of additional STAPs                                                                                                                                    |
| JTAG_IDCODE_MFR_ID       | 110h000  | Manufacturer ID                                                                                                                                               |
| JTAG_IDCODE_PART_NUM     | 160h0000 | Part number                                                                                                                                                   |
| JTAG_IDCODE_SI_REV       | 40h0     | Silicon revision                                                                                                                                              |
| JTAG_OCH_VER             | 80h00    | OCAH IP version                                                                                                                                               |
| XTRIG_INT_CT_MODE        | 80h00    | Internal CT protocol mode bits (0=pulse sync, 1=req/ack)                                                                                                      |
| SMC_OTP_RD_PL_DEPTH      | 20h3     | SMC OTP read pipeline depth                                                                                                                                   |
| SMC_OTP_WR_PL_DEPTH      | 20h3     | SMC OTP write pipeline depth                                                                                                                                  |
| SMC_RD_PL_DEPTH          | 20h3     | SMC fabric read pipeline depth                                                                                                                                |
| SMC_WR_PL_DEPTH          | 20h3     | SMC fabric write pipeline depth                                                                                                                               |
| SEP_ABR_SRAM_LATENCY     | 1        | ABR 1R1W registered-read latency, in clocks                                                                                                                   |
| SEP_ABR_MASKING_EN       | 1        | Adams Bridge 2-share DOM masking                                                                                                                              |

SEP-related DTP configuration is not exposed as an SMU config-struct field. The SMU derives SEP debug/STAP enablement from the module parameter `SEP` and wires the corresponding DTP interfaces internally to the SEP block when `SEP=1`.

## 1.6.5. IC\_RESET TDR Structure

The DTP primary TAP implements the IEEE 1149.1-2013 §17 IC\_RESET TDR as three independently enabled slices plus a single `reset_hold` bit nearest TDO. At the SMU the scan order from TDI to TDO is:

TDI → SMC slice → SEP slice → external slice → reset\_hold → TDO

The SMU enables the SMC slice unconditionally, the SEP slice when `SEP=1` (otherwise that slice has width 0 and is omitted), and the external slice when `JTAG_IC_RESET_ENABLE` is set. Each slice is a packed struct with `.ovrd` and `.val` halves

of equal width; `jtag_ptap` sizes the slice from  $\$bits(type)/2$  and interleaves the halves into the `{reset_enable, reset_control}` pair IEEE §17 requires per port. Declaration order within each half is the TDI-to-TDO scan order: the first field is nearest TDI.

| Slice                   | Ports                                 | Defined by                                           | Consumed by                                     |
|-------------------------|---------------------------------------|------------------------------------------------------|-------------------------------------------------|
| SMC                     | 68                                    | OCAH ( <code>smc_pkg::jtag_smc_reset_ctrl_t</code> ) | SMC reset unit; SMU-internal, no SMU port       |
| SEP                     | 7 when <code>SEP=1</code> ,<br>else 0 | OCAH ( <code>sep_pkg::jtag_sep_reset_ctrl_t</code> ) | SEP reset controller; SMU-internal, no SMU port |
| External                | $\$bits(ic\_reset\_ext\_t)/2$         | Adopter ( <code>ic_reset_ext_t</code> )              | SMU output <code>jtag_ic_reset_ext_o</code>     |
| <code>reset_hold</code> | 1 bit                                 | IEEE 1149.1 §17                                      | Nearest TDO; not a reset port                   |

The TDR length is  $2 * (SMC\ ports + SEP\ ports + external\ ports) + 1$ . Programming order and `reset_enable` polarity are in the DTP JTAG chapter of the Technical Reference Manual.

## SMC slice (TDI to TDO)

The SMC slice covers the SMC reset domains and the 32 managed-subsystem (SS) cold and warm resets. Vector index `[31]` of each SS field is nearer TDI than `[0]`.

| Port                               | Reset domain                       |
|------------------------------------|------------------------------------|
| <code>ss_warm_reset_n[31:0]</code> | Per-subsystem warm reset           |
| <code>ss_cold_reset_n[31:0]</code> | Per-subsystem cold reset           |
| <code>cold_reset_n</code>          | SMC cold reset                     |
| <code>cool_reset_n</code>          | SMC cool reset                     |
| <code>warm_reset_n</code>          | SMC warm reset                     |
| <code>fuse_reset_n</code>          | Fuse reset (nearest the SEP slice) |

## SEP slice (TDI to TDO)

When `SEP=1` the SEP slice covers the SEP main reset and the crypto, Key Manager, and TRNG resets consumed by `sep_reset_ctrl`. `trng_jtag_rst_n` is nearest TDI.

| Port                         | Reset domain                             |
|------------------------------|------------------------------------------|
| <code>trng_jtag_rst_n</code> | TRNG                                     |
| <code>sep_reset_n</code>     | SEP main reset                           |
| <code>kmac_jtag_rst_n</code> | KMAC                                     |
| <code>hmac_jtag_rst_n</code> | HMAC                                     |
| <code>aes_jtag_rst_n</code>  | AES                                      |
| <code>otbn_jtag_rst_n</code> | OTBN                                     |
| <code>km_jtag_rst_n</code>   | Key Manager (nearest the external slice) |

## External slice (adopter-defined)

The external slice is reserved for reset overrides on logic instantiated outside the SMU. OCAH does not name its ports. Set the SMU type parameter `ic_reset_ext_t` to a packed struct with `.ovrd` and `.val` halves of equal width, and route `jtag_ic_reset_ext_o` to that logic. The parameter default (`jtag_tap_pkg::jtag_ic_reset_default_t`) is a single-override stub. Clear `JTAG_IC_RESET_ENABLE` when no external overrides are required.

Declare a new external override at each half's MSB so already-assigned TDR positions do not shift. Extra subsystem reset controls beyond the SMC slice use this typed port.

## 1.6.6. DTP Parameter Pass-Through

The SMU passes through most DTP parameters to allow integrator customization. Type parameters are set internally based on SMC package types.

### JTAG Configuration Parameters

| Parameter                 | SMU Default | Pass-through | Notes                                                                                                                                                    |
|---------------------------|-------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| JTAG_BSR_ENABLE           | 1           | Yes          | Boundary scan enabled                                                                                                                                    |
| JTAG_EXTTEST_TRAIN_ENABLE | 1           | Yes          |                                                                                                                                                          |
| JTAG_EXTTEST_PULSE_ENABLE | 1           | Yes          |                                                                                                                                                          |
| JTAG_INTTEST_ENABLE       | 1           | Yes          |                                                                                                                                                          |
| JTAG_CLAMP_ENABLE         | 1           | Yes          |                                                                                                                                                          |
| JTAG_HIGHZ_ENABLE         | 1           | Yes          |                                                                                                                                                          |
| JTAG_RUNBIST_ENABLE       | 1           | Yes          |                                                                                                                                                          |
| JTAG_TMP_ENABLE           | 1           | Yes          |                                                                                                                                                          |
| JTAG_IC_RESET_ENABLE      | 1           | No           | SMU routes this to the DTP's JTAG_IC_RESET_EXT_ENABLE; SMC and SEP slices are enabled inside the SMU instantiation (SMC unconditionally, SEP when SEP=1) |
| JTAG_SMC_DBG_ENABLE       | 1           | Yes          |                                                                                                                                                          |
| JTAG_SEP_DBG_ENABLE       | 1 (SEP=1)   | No           | Derived internally from the SMU module parameter SEP; not exposed in smu_cfg_t                                                                           |
| JTAG_STAP_IO_ENABLE       | 1           | Yes          |                                                                                                                                                          |
| JTAG_NUM_EXTRA_STAPS      | 1           | Yes          | Port array size equals parameter value; when set to 0, array size is 1 (tie-off required)                                                                |

### JTAG ID Parameters

| Parameter            | SMU Default | Pass-through | Notes            |
|----------------------|-------------|--------------|------------------|
| JTAG_IDCODE_MFR_ID   | 11h000      | Yes          | Manufacturer ID  |
| JTAG_IDCODE_PART_NUM | 16h0000     | Yes          | Part number      |
| JTAG_IDCODE_SI_REV   | 4h0         | Yes          | Silicon revision |
| JTAG_OCH_VER         | 8h00        | Yes          | OCAH IP version  |

### Cross Trigger Parameters

| Parameter         | SMU Default | DTP Internal | Pass-through | Notes                                  |
|-------------------|-------------|--------------|--------------|----------------------------------------|
| XTRIG_INT_CT_MODE | [7:0]       | [9:0]        | Yes          | 8-bit on SMU; DTP bits [1:0]=0 for SMC |

The SMU parameter width is two less than DTP's for cross trigger (8 vs 10) because internal cross trigger ports [1:0] and clock stop request #0 are connected to the SMC internally. The SMU remaps DTP ports [9:2] to SMU external ports [7:0].

## Pipeline Depth Parameters

| Parameter           | SMU Default | Pass-through | Notes                                                     |
|---------------------|-------------|--------------|-----------------------------------------------------------|
| SMC_OTP_RD_PL_DEPTH | 20h3        | Yes          | SMC OTP read pipeline depth                               |
| SMC_OTP_WR_PL_DEPTH | 20h3        | Yes          | SMC OTP write pipeline depth                              |
| SEP_OTP_RD_PL_DEPTH | 20h3        | No           | Fixed to 20h3 in SMU; used when SEP=1, ignored when SEP=0 |
| SEP_OTP_WR_PL_DEPTH | 20h3        | No           | Fixed to 20h3 in SMU; used when SEP=1, ignored when SEP=0 |
| SMC_RD_PL_DEPTH     | 20h3        | Yes          | SMC fabric read pipeline depth                            |
| SMC_WR_PL_DEPTH     | 20h3        | Yes          | SMC fabric write pipeline depth                           |

## Type Parameters

The following type parameters are set internally by the SMU and are not exposed for integrator configuration.

| Parameter           | Set To                                  | Notes                                              |
|---------------------|-----------------------------------------|----------------------------------------------------|
| jtag_tap_ctrl_t     | prim_jtag_pkg::jtag_tap_ctrl_t          | JTAG TAP control struct                            |
| jtag_scan_ctrl_t    | prim_jtag_pkg::jtag_scan_ctrl_t         | JTAG scan control struct                           |
| smc_jtag_axi_req_t  | smc_pkg::smc_jtag_56_64_2_12_axi_req_t  | SMC debug AXI request                              |
| smc_jtag_axi_resp_t | smc_pkg::smc_jtag_56_64_2_12_axi_resp_t | SMC debug AXI response                             |
| smc_otp_axil_req_t  | smc_efuse_pkg::efuse_axil_req_t         | SMC OTP AXI-Lite request                           |
| smc_otp_axil_resp_t | smc_efuse_pkg::efuse_axil_resp_t        | SMC OTP AXI-Lite response                          |
| sep_otp_axil_req_t  | smc_efuse_pkg::efuse_axil_req_t         | SEP OTP AXI-Lite request (error slave when SEP=0)  |
| sep_otp_axil_resp_t | smc_efuse_pkg::efuse_axil_resp_t        | SEP OTP AXI-Lite response (error slave when SEP=0) |
| xtrig_axil_req_t    | smc_pkg::smc_axil_32_32_req_t           | Cross trigger CSR request                          |
| xtrig_axil_resp_t   | smc_pkg::smc_axil_32_32_resp_t          | Cross trigger CSR response                         |

## 1.6.7. DTP Port Mapping

The following tables define the complete DTP port list as connected within the SMU. Ports marked "SMU external port" are exposed on the SMU boundary for integrator connection. Ports marked with a subsystem name (e.g. SMC, SEP) are wired internally within the SMU.

### Clock and Reset

| Port          | Direction | Width | SMU Connection                                  |
|---------------|-----------|-------|-------------------------------------------------|
| clk_i         | input     | 1     | SMU clk_smu_i (shared with SMC clk_smc_i)       |
| rst_n_i       | input     | 1     | SMC rst_primary_smc_clk_no                      |
| pwr_on_rst_ni | input     | 1     | SMU rst_cold_ni (power-on reset for JTAG logic) |

### Primary JTAG TAP Interface

| Port                        | Direction | Width           | SMU Connection    |
|-----------------------------|-----------|-----------------|-------------------|
| jtag_ptap_client_tap_ctrl_i | input     | jtag_tap_ctrl_t | SMU external port |
| jtag_ptap_client_tdi_i      | input     | 1               | SMU external port |
| jtag_ptap_client_tdo_o      | output    | 1               | SMU external port |

| Port                       | Direction | Width | SMU Connection    |
|----------------------------|-----------|-------|-------------------|
| jtag_ptap_client_tdo_oen_o | output    | 1     | SMU external port |

## Boundary Scan Interface

| Port                      | Direction | Width            | SMU Connection    |
|---------------------------|-----------|------------------|-------------------|
| jtag_bsr_host_scan_ctrl_o | output    | jtag_scan_ctrl_t | SMU external port |
| jtag_bsr_host_scan_in_i   | input     | 1                | SMU external port |
| jtag_bsr_host_scan_out_o  | output    | 1                | SMU external port |

## I/O STAP Interface

| Port                         | Direction | Width           | SMU Connection    |
|------------------------------|-----------|-----------------|-------------------|
| jtag_stap_io_host_tap_ctrl_o | output    | jtag_tap_ctrl_t | SMU external port |
| jtag_stap_io_host_tdi_i      | input     | 1               | SMU external port |
| jtag_stap_io_host_tdo_o      | output    | 1               | SMU external port |
| jtag_stap_io_host_tdo_oen_o  | output    | 1               | SMU external port |

## SMC Debug STAP Interface

| Port                          | Direction | Width           | SMU Connection                                                                                                         |
|-------------------------------|-----------|-----------------|------------------------------------------------------------------------------------------------------------------------|
| jtag_stap_smc_host_tap_ctrl_o | output    | jtag_tap_ctrl_t | SMC smc_cpu_jtag_{TCK,TMS}_i; trst_n inverted drives SMC smc_cpu_jtag_reset_i when JTAG_SMC_DBG_ENABLE=1, else tie-off |
| jtag_stap_smc_host_tdi_i      | input     | 1               | SMC smc_cpu_jtag_TDO_data_o when JTAG_SMC_DBG_ENABLE=1, else 1'b0                                                      |
| jtag_stap_smc_host_tdo_o      | output    | 1               | SMC smc_cpu_jtag_TDI_i when JTAG_SMC_DBG_ENABLE=1, else unused                                                         |
| jtag_stap_smc_host_tdo_oen_o  | output    | 1               | Internal only; not consumed by SMC                                                                                     |

## SEP Debug STAP Interface

| Port                          | Direction | Width           | SMU Connection                                    |
|-------------------------------|-----------|-----------------|---------------------------------------------------|
| jtag_stap_sep_host_tap_ctrl_o | output    | jtag_tap_ctrl_t | SEP jtag_{tck,tms,trst_n} when SEP=1, else unused |
| jtag_stap_sep_host_tdi_i      | input     | 1               | SEP jtag_tdo when SEP=1, else 1'b0                |
| jtag_stap_sep_host_tdo_o      | output    | 1               | SEP jtag_tdi when SEP=1, else unused              |
| jtag_stap_sep_host_tdo_oen_o  | output    | 1               | Internal only; not consumed by SEP at SMU level   |

## Extra STAP Interfaces

| Port                            | Direction | Width                                          | SMU Connection    |
|---------------------------------|-----------|------------------------------------------------|-------------------|
| jtag_stap_extra_host_tap_ctrl_o | output    | jtag_tap_ctrl_t[JTAG_NUM_EXTRA_STAP_PORTS-1:0] | SMU external port |
| jtag_stap_extra_host_tdi_i      | input     | [JTAG_NUM_EXTRA_STAP_PORTS-1:0]                | SMU external port |
| jtag_stap_extra_host_tdo_o      | output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0]                | SMU external port |
| jtag_stap_extra_host_tdo_oen_o  | output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0]                | SMU external port |

## Extended STAP Scan Interface

| Port                       | Direction | Width            | SMU Connection    |
|----------------------------|-----------|------------------|-------------------|
| jtag_stap_host_scan_ctrl_o | output    | jtag_scan_ctrl_t | SMU external port |
| jtag_stap_host_scan_in_i   | input     | 1                | SMU external port |
| jtag_stap_host_scan_out_o  | output    | 1                | SMU external port |

## DFD iJTAG Interface

| Port                      | Direction | Width            | SMU Connection    |
|---------------------------|-----------|------------------|-------------------|
| jtag_dfd_host_scan_ctrl_o | output    | jtag_scan_ctrl_t | SMU external port |
| jtag_dfd_host_scan_in_i   | input     | 1                | SMU external port |
| jtag_dfd_host_scan_out_o  | output    | 1                | SMU external port |

## DFT iJTAG Interface

| Port                      | Direction | Width            | SMU Connection    |
|---------------------------|-----------|------------------|-------------------|
| jtag_dft_host_scan_ctrl_o | output    | jtag_scan_ctrl_t | SMU external port |
| jtag_dft_host_scan_in_i   | input     | 1                | SMU external port |
| jtag_dft_host_scan_out_o  | output    | 1                | SMU external port |

## SMC Fabric Debug AXI Interface

| Port               | Direction | Width               | SMU Connection         |
|--------------------|-----------|---------------------|------------------------|
| axi_smc_dbg_req_o  | output    | smc_jtag_axi_req_t  | SMC jtag_axi_in_req_i  |
| axi_smc_dbg_resp_i | input     | smc_jtag_axi_resp_t | SMC jtag_axi_in_resp_o |

## SMC OTP Debug AXI-Lite Interface

| Port                     | Direction | Width               | SMU Connection               |
|--------------------------|-----------|---------------------|------------------------------|
| axil_smc_otp_jtag_req_o  | output    | smc_otp_axil_req_t  | SMC axil_smc_otp_jtag_req_i  |
| axil_smc_otp_jtag_resp_i | input     | smc_otp_axil_resp_t | SMC axil_smc_otp_jtag_resp_o |

## SEP OTP Debug AXI-Lite Interface

When `SEP=1`, this interface connects to the SEP `axil_sep_otp_jtag_*` ports. When `SEP=0`, the SMU instantiates `u_sep_otp_axil_err_slv`, which responds with DECERR to all transactions.

| Port                     | Direction | Width               | SMU Connection                                                                                                                        |
|--------------------------|-----------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| axil_sep_otp_jtag_req_o  | output    | sep_otp_axil_req_t  | SEP <code>axil_sep_otp_jtag_req_i</code> when <code>SEP=1</code> , else AXI-Lite error slave ( <code>u_sep_otp_axil_err_slv</code> )  |
| axil_sep_otp_jtag_resp_i | input     | sep_otp_axil_resp_t | SEP <code>axil_sep_otp_jtag_resp_o</code> when <code>SEP=1</code> , else AXI-Lite error slave ( <code>u_sep_otp_axil_err_slv</code> ) |

## Clock Control

| Port        | Direction | Width | SMU Connection                                          |
|-------------|-----------|-------|---------------------------------------------------------|
| stop_clks_o | output    | 1     | SMU external port: dtp_stop_clks_o (to PLL clock gates) |

## Boot Stall Control

| Port                   | Direction | Width | SMU Connection             |
|------------------------|-----------|-------|----------------------------|
| jtag_boot_stall_ovrd_o | output    | 1     | SMC boot_stall_jtag_ovrd_i |
| jtag_boot_stall_o      | output    | 1     | SMC boot_stall_jtag_val_i  |

## JTAG Reset Control

| Port                | Direction | Width                          | SMU Connection                                                                                                                       |
|---------------------|-----------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| jtag_ic_reset_smc_o | output    | smc_pkg::jtag_smc_reset_ctrl_t | SMC jtag_smc_reset_ctrl_i (packed struct with .ovrd override enables and .val active-low override values)                            |
| jtag_ic_reset_sep_o | output    | sep_pkg::jtag_sep_reset_ctrl_t | SEP jtag_sep_reset_ctrl_i (packed struct with .ovrd override enables and .val active-low override values); SMU-internal, no SMU port |
| jtag_ic_reset_ext_o | output    | ic_reset_ext_t                 | SMU jtag_ic_reset_ext_o external port (integrator-defined packed struct; carries reset overrides for logic outside the SMU)          |

## JTAG State Outputs

| Port                     | Direction | Width                      | SMU Connection    |
|--------------------------|-----------|----------------------------|-------------------|
| jtag_ptap_state_o        | output    | tap_state_e                | SMU external port |
| jtag_ptap_inst_decoded_o | output    | jtag_instruction_decoded_e | SMU external port |

## Cross Trigger CSR AXI-Lite Interface

| Port              | Direction | Width             | SMU Connection          |
|-------------------|-----------|-------------------|-------------------------|
| axil_xtrig_req_i  | input     | xtrig_axil_req_t  | SMC axil_dtp_csr_req_o  |
| axil_xtrig_resp_o | output    | xtrig_axil_resp_t | SMC axil_dtp_csr_resp_i |

## Cross Trigger Matrix Interface

| Port                | Direction | Width                  | SMU Connection                                                               |
|---------------------|-----------|------------------------|------------------------------------------------------------------------------|
| xtrig_ctm_src_req_o | output    | [XTRIG_NUM_INT_CT-1:0] | [1:0]: SMU internal (SMC), [9:2]: SMU external xtrig_ctm_src_req_o[7:0]      |
| xtrig_ctm_src_ack_i | input     | [XTRIG_NUM_INT_CT-1:0] | [1:0]: Tie-off (SMC), [9:2]: SMU external xtrig_ctm_src_ack_i[7:0]           |
| xtrig_ctm_dst_req_i | input     | [XTRIG_NUM_INT_CT-1:0] | [1:0]: SMC xtrigger_ss_o[1:0], [9:2]: SMU external xtrig_ctm_dst_req_i[7:0]  |
| xtrig_ctm_dst_ack_o | output    | [XTRIG_NUM_INT_CT-1:0] | [1:0]: Unused (SMC connection), [9:2]: SMU external xtrig_ctm_dst_ack_o[7:0] |

Internal cross trigger ports [1:0] are reserved for SMC (both bits of `xtrigger_ss_o`). The SMU exposes ports [9:2] as external ports [7:0], so the SMU `XTRIG_NUM_INT_CT` parameter is two less than DTP's (8 vs 10). The SMC uses pulse-synchronization mode, so ack signals for ports [1:0] are unused: `xtrig_ctm_src_ack_i[1:0]` should be tied to `2'b00` and `xtrig_ctm_dst_ack_o[1:0]` are left unconnected. The usage of ack signals depends on the `XTRIG_INT_CT_MODE` parameter: when a mode bit is 0 (pulse synchronization), the corresponding ack is unused; when 1 (req/ack handshaking), the ack must be connected to the cross trigger endpoint.

## Clock Stop Request Interface

| Port                              | Direction | Width                        | SMU Connection                                                                                                                 |
|-----------------------------------|-----------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <code>xtrig_clk_stop_req_i</code> | input     | [XTRIG_NUM_CLK_STOP_REQ-1:0] | [0]: SMC<br><code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code> , [8:1]:<br>SMU external <code>xtrig_clk_stop_req_i</code> [7:0] |

Clock stop request port #0 is reserved for SMC. The SMU exposes ports [8:1] as external ports [7:0], so the SMU `XTRIG_NUM_CLK_STOP_REQ` parameter should be one less than DTP's (8 vs 9).

## Cross Trigger Port GPIO Interface

The 16 CTP signals from DTP are ported through the SMU and are direct `smu_wrapper` external ports (connected through the `.*` SMU instantiation). The adopter must connect them to GPIO pad logic; no tie-off or dangling is present in the wrapper itself.

| Port                                     | Direction | Width  | SMU Connection                               |
|------------------------------------------|-----------|--------|----------------------------------------------|
| <code>xtrig_ctp_req_out_dout_o</code>    | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_req_out_dout_en_o</code> | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_req_out_din_i</code>     | input     | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_req_out_din_en_o</code>  | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_req_in_dout_o</code>     | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_req_in_dout_en_o</code>  | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_req_in_din_i</code>      | input     | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_req_in_din_en_o</code>   | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_ack_in_dout_o</code>     | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_ack_in_dout_en_o</code>  | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_ack_in_din_i</code>      | input     | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_ack_in_din_en_o</code>   | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_ack_out_dout_o</code>    | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_ack_out_dout_en_o</code> | output    | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_ack_out_din_i</code>     | input     | [15:0] | SMU external port; connect to GPIO pad logic |
| <code>xtrig_ctp_ack_out_din_en_o</code>  | output    | [15:0] | SMU external port; connect to GPIO pad logic |

## 1.6.8. SMC Port Mapping

The following tables define the complete SMC port list as connected within the SMU. Ports marked "SMU external port" are exposed on the SMU boundary. Ports marked with "DTP" or "SEP" are wired internally.

### Clocks

| Port                      | Direction | Width | SMU Connection                                                   |
|---------------------------|-----------|-------|------------------------------------------------------------------|
| <code>clk_smc_i</code>    | input     | 1     | SMU <code>clk_smu_i</code> (shared with DTP <code>clk_i</code> ) |
| <code>clk_periph_i</code> | input     | 1     | SMU external port                                                |
| <code>clk_ref_i</code>    | input     | 1     | SMU external port                                                |

## Resets

| Port                                | Direction | Width | SMU Connection                               |
|-------------------------------------|-----------|-------|----------------------------------------------|
| <code>rst_cold_ni</code>            | input     | 1     | SMU external port                            |
| <code>rst_primary_ref_clk_no</code> | output    | 1     | SMU external port                            |
| <code>rst_primary_smc_clk_no</code> | output    | 1     | SMU external port + DTP <code>rst_n_i</code> |

## System AXI Input

| Port                           | Direction | Width                                         | SMU Connection    |
|--------------------------------|-----------|-----------------------------------------------|-------------------|
| <code>sys_axi_in_req_i</code>  | input     | <code>smc_sys_in_56_64_6_12_axi_req_t</code>  | SMU external port |
| <code>sys_axi_in_resp_o</code> | output    | <code>smc_sys_in_56_64_6_12_axi_resp_t</code> | SMU external port |

## JTAG AXI Input

| Port                            | Direction | Width                                       | SMU Connection                      |
|---------------------------------|-----------|---------------------------------------------|-------------------------------------|
| <code>jtag_axi_in_req_i</code>  | input     | <code>smc_jtag_56_64_2_12_axi_req_t</code>  | DTP <code>axi_smc_dbg_req_o</code>  |
| <code>jtag_axi_in_resp_o</code> | output    | <code>smc_jtag_56_64_2_12_axi_resp_t</code> | DTP <code>axi_smc_dbg_resp_i</code> |

## SMC OTP JTAG AXI-Lite

| Port                                  | Direction | Width                          | SMU Connection                            |
|---------------------------------------|-----------|--------------------------------|-------------------------------------------|
| <code>axil_smc_otp_jtag_req_i</code>  | input     | <code>efuse_axil_req_t</code>  | DTP <code>axil_smc_otp_jtag_req_o</code>  |
| <code>axil_smc_otp_jtag_resp_o</code> | output    | <code>efuse_axil_resp_t</code> | DTP <code>axil_smc_otp_jtag_resp_i</code> |

## SEP AXI Input

When `SEP=0`, the SEP AXI subordinate input is tied off to '0' and the response output is left unconnected. When `SEP=1`, the SMU drives this path from the SEP-to-SMC dedicated AXI connection.

| Port                           | Direction | Width                                         | SMU Connection                              |
|--------------------------------|-----------|-----------------------------------------------|---------------------------------------------|
| <code>sep_axi_in_req_i</code>  | input     | <code>smc_sep_in_56_64_6_12_axi_req_t</code>  | Tied off to '0' (no incoming transactions)  |
| <code>sep_axi_in_resp_o</code> | output    | <code>smc_sep_in_56_64_6_12_axi_resp_t</code> | Unconnected (no transactions to respond to) |

## AXI Output

| Port                           | Direction | Width                                          | SMU Connection    |
|--------------------------------|-----------|------------------------------------------------|-------------------|
| <code>output_axi_req_o</code>  | output    | <code>smc_sys_out_56_64_8_12_axi_req_t</code>  | SMU external port |
| <code>output_axi_resp_i</code> | input     | <code>smc_sys_out_56_64_8_12_axi_resp_t</code> | SMU external port |

## DTP CSR Interface

| Port                             | Direction | Width                              | SMU Connection                     |
|----------------------------------|-----------|------------------------------------|------------------------------------|
| <code>axil_dtp_csr_req_o</code>  | output    | <code>smc_axil_32_32_req_t</code>  | DTP <code>axil_xtrig_req_i</code>  |
| <code>axil_dtp_csr_resp_i</code> | input     | <code>smc_axil_32_32_resp_t</code> | DTP <code>axil_xtrig_resp_o</code> |

## Adopter Peripheral Window AXI-Lite

| Port                             | Direction | Width                              | SMU Connection                               |
|----------------------------------|-----------|------------------------------------|----------------------------------------------|
| <code>smc_external_req_o</code>  | output    | <code>smc_axil_32_32_req_t</code>  | SMU external port (to adopter peripherals)   |
| <code>smc_external_resp_i</code> | input     | <code>smc_axil_32_32_resp_t</code> | SMU external port (from adopter peripherals) |

## eFuse Interface

| Port                                   | Direction | Width                             | SMU Connection    |
|----------------------------------------|-----------|-----------------------------------|-------------------|
| <code>efuse_bank_ctrl_req_o</code>     | output    | <code>efuse_axil_req_t</code>     | SMU external port |
| <code>efuse_bank_ctrl_resp_i</code>    | input     | <code>efuse_axil_resp_t</code>    | SMU external port |
| <code>efuse_shim_command_req_o</code>  | output    | <code>efuse_command_req_t</code>  | SMU external port |
| <code>efuse_shim_command_resp_i</code> | input     | <code>efuse_command_resp_t</code> | SMU external port |
| <code>shadow_regs_o</code>             | output    | <code>efuse_map_t</code>          | SMU external port |

## GPIO Data Signals

| Port                                 | Direction | Width                | SMU Connection    |
|--------------------------------------|-----------|----------------------|-------------------|
| <code>lsio_interface_select_o</code> | output    | [NUM_GPIO_WRAPS-1:0] | SMU external port |
| <code>pad2core_i</code>              | input     | [NUM_GPIO_WRAPS-1:0] | SMU external port |
| <code>core2pad_o</code>              | output    | [NUM_GPIO_WRAPS-1:0] | SMU external port |
| <code>pad2core_en_o</code>           | output    | [NUM_GPIO_WRAPS-1:0] | SMU external port |
| <code>core2pad_en_o</code>           | output    | [NUM_GPIO_WRAPS-1:0] | SMU external port |

## GPIO External Pins

| Port                               | Direction | Width | SMU Connection    |
|------------------------------------|-----------|-------|-------------------|
| <code>rst_cool_n_from_pin_i</code> | input     | 1     | SMU external port |

## ATB Telemetry

| Port                             | Direction | Width                                                      | SMU Connection    |
|----------------------------------|-----------|------------------------------------------------------------|-------------------|
| <code>clk_telemetry_i</code>     | input     | 1                                                          | SMU external port |
| <code>rst_telemetry_ni</code>    | input     | 1                                                          | SMU external port |
| <code>telemetry_atdata_i</code>  | input     | <code>telemetry_data_t[NUM_TELEMETRY_RECEIVERS-1:0]</code> | SMU external port |
| <code>telemetry_atid_i</code>    | input     | <code>atb_id_t[NUM_TELEMETRY_RECEIVERS-1:0]</code>         | SMU external port |
| <code>telemetry_atready_o</code> | output    | [NUM_TELEMETRY_RECEIVERS-1:0]                              | SMU external port |
| <code>telemetry_atvalid_i</code> | input     | [NUM_TELEMETRY_RECEIVERS-1:0]                              | SMU external port |
| <code>telemetry_afvalid_o</code> | output    | [NUM_TELEMETRY_RECEIVERS-1:0]                              | SMU external port |
| <code>telemetry_afeady_i</code>  | input     | [NUM_TELEMETRY_RECEIVERS-1:0]                              | SMU external port |

## DED/WDT

| Port                     | Direction | Width | SMU Connection    |
|--------------------------|-----------|-------|-------------------|
| smc_cluster_ded_o        | output    | 1     | SMU external port |
| smc_wdt_first_timeout_o  | output    | 1     | SMU external port |
| smc_wdt_second_timeout_o | output    | 1     | SMU external port |

## Region Size

| Port              | Direction | Width  | SMU Connection    |
|-------------------|-----------|--------|-------------------|
| smc_region_size_o | output    | [31:0] | SMU external port |

## External Interrupts

| Port                     | Direction | Width                    | SMU Connection                                                         |
|--------------------------|-----------|--------------------------|------------------------------------------------------------------------|
| smc_ext_interrupts_i     | input     | [NUM_EXT_INTERRUPTS-1:0] | SMU external port                                                      |
| sep_mailbox_interrupts_i | input     | [7:0]                    | Internal: from SEP smc_mailbox_interrupt_o when SEP=1, else tied to '0 |
| sep_wdt_reset_n_i        | input     | 1                        | Internal: ~sep_wdt_timer_rst_req when SEP=1, else tied to 10b1         |

## Fuse Signals

| Port                       | Direction | Width | SMU Connection    |
|----------------------------|-----------|-------|-------------------|
| smc_fuse_sense_done_o      | output    | 1     | SMU external port |
| smc_fuse_reset_n_delayed_o | output    | 1     | SMU external port |

## DFT Signals

| Port                | Direction | Width | SMU Connection    |
|---------------------|-----------|-------|-------------------|
| skip_mem_repair_o   | output    | 1     | SMU external port |
| ext_boot_seq_done_i | input     | 1     | SMU external port |

## Lifecycle State

| Port            | Direction | Width                  | SMU Connection                                                         |
|-----------------|-----------|------------------------|------------------------------------------------------------------------|
| lc_state_o      | output    | [2*LC_STATE_WIDTH-1:0] | SEP lifecycle state when SEP=1, else 80hf0; ported through SMU         |
| lc_sigint_err_o | output    | 1                      | SEP lifecycle integrity error when SEP=1, else SMC efuse_lc_sigint_err |

## Mailbox Interrupts

| Port                         | Direction | Width               | SMU Connection    |
|------------------------------|-----------|---------------------|-------------------|
| smc_ext_mailbox_interrupts_o | output    | [NUM_MAILBOXES-1:0] | SMU external port |

## Reset Unit Signals

| Port                             | Direction | Width                           | SMU Connection    |
|----------------------------------|-----------|---------------------------------|-------------------|
| <code>cfg_flr_pf_active_i</code> | input     | 1                               | SMU external port |
| <code>isolate_req_o</code>       | output    | [31:0]                          | SMU external port |
| <code>ss_reset_complete_i</code> | input     | [31:0]                          | SMU external port |
| <code>ss_config_o</code>         | output    | [31:0]                          | SMU external port |
| <code>ss_reset_ctrl_o</code>     | output    | <code>reset_ctrl_t[31:0]</code> | SMU external port |
| <code>sync_irq_o</code>          | output    | 1                               | SMU external port |

## CPU Memory Signals

| Port                                       | Direction | Width                                  | SMU Connection    |
|--------------------------------------------|-----------|----------------------------------------|-------------------|
| <code>smc_rom_intf_req_o</code>            | output    | <code>rom_req_t</code>                 | SMU external port |
| <code>smc_rom_intf_rsp_i</code>            | input     | <code>rom_rsp_t</code>                 | SMU external port |
| <code>smc_scratch_ram_intf_req_o</code>    | output    | <code>scratch_ram_req_t[31:0]</code>   | SMU external port |
| <code>smc_scratch_ram_intf_rsp_i</code>    | input     | <code>scratch_ram_rsp_t[31:0]</code>   | SMU external port |
| <code>smc_l1_icache_tag_intf_req_o</code>  | output    | <code>l1_icache_tag_req_t[3:0]</code>  | SMU external port |
| <code>smc_l1_icache_tag_intf_rsp_i</code>  | input     | <code>l1_icache_tag_rsp_t[3:0]</code>  | SMU external port |
| <code>smc_l1_icache_data_intf_req_o</code> | output    | <code>l1_icache_data_req_t[7:0]</code> | SMU external port |
| <code>smc_l1_icache_data_intf_rsp_i</code> | input     | <code>l1_icache_data_rsp_t[7:0]</code> | SMU external port |
| <code>smc_l1_dcache_tag_intf_req_o</code>  | output    | <code>l1_dcache_tag_req_t[3:0]</code>  | SMU external port |
| <code>smc_l1_dcache_tag_intf_rsp_i</code>  | input     | <code>l1_dcache_tag_rsp_t[3:0]</code>  | SMU external port |
| <code>smc_l1_dcache_data_intf_req_o</code> | output    | <code>l1_dcache_data_req_t[3:0]</code> | SMU external port |
| <code>smc_l1_dcache_data_intf_rsp_i</code> | input     | <code>l1_dcache_data_rsp_t[3:0]</code> | SMU external port |

## Memory Init

| Port                                      | Direction | Width | SMU Connection    |
|-------------------------------------------|-----------|-------|-------------------|
| <code>smc_disable_sram_auto_init_i</code> | input     | 1     | SMU external port |
| <code>smc_init_mem_done_o</code>          | output    | 1     | SMU external port |

## System Timer OCTS Interface

| Port                       | Direction | Width  | SMU Connection    |
|----------------------------|-----------|--------|-------------------|
| <code>timer_count_o</code> | output    | [63:0] | SMU external port |

## Reset Overrides (from DTP)

| Port                           | Direction | Width                                       | SMU Connection                                                                                                                                 |
|--------------------------------|-----------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>jtag_reset_ctrl_i</code> | input     | <code>smc_pkg::jtag_smc_reset_ctrl_t</code> | DTP <code>jtag_ic_reset_smc_o</code> (packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values) |

## DFD Signals

| Port                           | Direction | Width | SMU Connection    |
|--------------------------------|-----------|-------|-------------------|
| <code>cla_gpio_enable_o</code> | output    | 1     | SMU external port |

| Port                                              | Direction | Width | SMU Connection                                                                  |
|---------------------------------------------------|-----------|-------|---------------------------------------------------------------------------------|
| <code>tdr_dbg_ctrl_clock_stop_en_i</code>         | input     | 1     | DTP <code>cla_clock_stop_en_o</code>                                            |
| <code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code> | output    | 1     | DTP <code>xtrig_clk_stop_req_i[0]</code> (SMU connects SMC output to DTP input) |

## Cross Trigger Interface

| Port                       | Direction | Width                            | SMU Connection                                                                            |
|----------------------------|-----------|----------------------------------|-------------------------------------------------------------------------------------------|
| <code>xtrigger_ss_i</code> | input     | <code>smc_pkg::xtrigger_t</code> | SMU internal: DTP <code>xtrig_ctm_src_req_o[1:0]</code> → SMC. Not an SMU external port.  |
| <code>xtrigger_ss_o</code> | output    | <code>smc_pkg::xtrigger_t</code> | SMU internal: SMC → DTP <code>xtrig_ctm_dst_req_i[1:0]</code> . Not an SMU external port. |

## Trace Memory

| Port                          | Direction | Width                                               | SMU Connection    |
|-------------------------------|-----------|-----------------------------------------------------|-------------------|
| <code>trace_mem_req_o</code>  | output    | <code>SinkMemPktIn_s[TRC_RAM_INSTANCES-1:0]</code>  | SMU external port |
| <code>trace_mem_resp_i</code> | input     | <code>SinkMemPktOut_s[TRC_RAM_INSTANCES-1:0]</code> | SMU external port |

## External Debug Bus

| Port                         | Direction | Width   | SMU Connection    |
|------------------------------|-----------|---------|-------------------|
| <code>ext_debug_bus_i</code> | input     | [127:0] | SMU external port |

## Test Mode

| Port                    | Direction | Width | SMU Connection    |
|-------------------------|-----------|-------|-------------------|
| <code>test_en_i</code>  | input     | 1     | SMU external port |
| <code>scan_rst_i</code> | input     | 1     | SMU external port |

## 1.6.9. SMC-DTP Internal Interconnection

The following table summarizes the key internal connections between the DTP and SMC within the SMU. These connections are not visible on the SMU boundary.

| Connection               | DTP Port                              | SMC Port                              | Direction | Notes                                                                                                                   |
|--------------------------|---------------------------------------|---------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------|
| AXI Debug Bus            | <code>axi_smc_dbg_req_o</code>        | <code>jtag_axi_in_req_i</code>        | DTP → SMC |                                                                                                                         |
|                          | <code>axi_smc_dbg_resp_i</code>       | <code>jtag_axi_in_resp_o</code>       | SMC → DTP |                                                                                                                         |
| AXI OTP Bus              | <code>axil_smc_otp_jtag_req_o</code>  | <code>axil_smc_otp_jtag_req_i</code>  | DTP → SMC |                                                                                                                         |
|                          | <code>axil_smc_otp_jtag_resp_i</code> | <code>axil_smc_otp_jtag_resp_o</code> | SMC → DTP |                                                                                                                         |
| DTP CSR Bus              | <code>axil_xtrig_req_i</code>         | <code>axil_dtp_csr_req_o</code>       | SMC → DTP |                                                                                                                         |
|                          | <code>axil_xtrig_resp_o</code>        | <code>axil_dtp_csr_resp_i</code>      | DTP → SMC |                                                                                                                         |
| JTAG SMC Reset Overrides | <code>jtag_ic_reset_smc_o</code>      | <code>jtag_reset_ctrl_i</code>        | DTP → SMC | Packed <code>smc_pkg::jtag_smc_reset_ctrl_t</code> struct carrying all SMC reset override enables and active-low values |

| Connection          | DTP Port                              | SMC Port                                          | Direction | Notes     |
|---------------------|---------------------------------------|---------------------------------------------------|-----------|-----------|
| DTP Reset           | <code>rst_n_i</code>                  | <code>rst_primary_smc_clk_no</code>               | SMC → DTP |           |
| Cross Trigger [1:0] | <code>xtrig_ctm_dst_req_i[1:0]</code> | <code>xtrigger_ss_o[1:0]</code>                   | SMC → DTP | Both bits |
| Clock Stop #0       | <code>xtrig_clk_stop_req_i[0]</code>  | <code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code> | SMC → DTP |           |
| Boot Stall          | <code>jtag_boot_stall_ovrd_o</code>   | <code>boot_stall_jtag_ovrd_i</code>               | DTP → SMC |           |
|                     | <code>jtag_boot_stall_o</code>        | <code>boot_stall_jtag_val_i</code>                | DTP → SMC |           |

## 1.6.10. SEP-Conditional Wiring

SEP-related interfaces are handled internally within the SMU. Their behavior depends on the module parameter `SEP`.

| Module/Signal       | Implementation                                                                                                        | Description                                                              |
|---------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| SMC SEP AXI Input   | <code>sep_axi_in_req_i</code> = SEP traffic when <code>SEP=1</code> , else '0                                         | Dedicated SEP-to-SMC subordinate path                                    |
| DTP SEP OTP JTAG    | <code>SEP_axil_sep_otp_jtag_*</code> when <code>SEP=1</code> , else <code>u_sep_otp_axil_err_slv</code>               | SEP OTP debug AXI-Lite path                                              |
| DTP SEP STAP return | <code>SEP_jtag_tdo</code> when <code>SEP=1</code> , else <code>sep_stap_tdo_to_dtp = 10b0</code>                      | SEP JTAG STAP return path                                                |
| SMU                 | <code>sep_mailbox_interrupts</code>                                                                                   | Internal SEP-to-SMC mailbox interrupts when <code>SEP=1</code> , else '0 |
| SMC                 | <code>sep_wdt_reset_n_i</code> = <code>~sep_wdt_timer_rst_req</code> when <code>SEP=1</code> , else <code>10b1</code> | SEP watchdog reset input to SMC                                          |

## Other Internal Tie-offs

| Module | Port                                  | Tie-off Value      | Reason                   |
|--------|---------------------------------------|--------------------|--------------------------|
| DTP    | <code>xtrig_ctm_src_ack_i[1:0]</code> | <code>20b00</code> | SMC uses pulse-sync mode |

`xtrig_ctm_src_ack_i[9:2]` (exposed as SMU `xtrig_ctm_src_ack_i[7:0]`) is ported out through the SMU. The integrator should tie off to '0 when using pulse-sync mode (`XTRIG_INT_CT_MODE` bit = 0), or connect to actual ack signals when using req/ack mode (bit = 1).

## 1.6.11. SMU Port Interface

The SMU exposes a single consolidated boundary that an adopter connects to at the chip or chiplet level. The following table defines all SMU top-level ports including clock, reset, JTAG, cross trigger, SMC-originated interfaces (AXI, GPIO, eFuse, PLL, PVT, memory), and SEP passthrough interfaces (memory macros, eFuse, crypto SRAM, SPI, AXI extension). Ports prefixed with `sep_` are only active when the module parameter `SEP=1`.

Table 11. SMU Port Declaration

| Interface Name           | Type  | Direction | Width | Description                                                                                                  | Connection          | Default value (if applicable) |
|--------------------------|-------|-----------|-------|--------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------|
| <code>clk_smu_i</code>   | logic | Input     | 1     | Primary SMU clock domain. Shared by SMC and DTP system logic.                                                | PLL output          | N/A                           |
| <code>clk_ref_i</code>   | logic | Input     | 1     | Reference clock domain for timing-sensitive operations.                                                      | PLL reference clock | N/A                           |
| <code>rst_cold_ni</code> | logic | Input     | 1     | Cold reset (active low). Asynchronous assertion, synchronous deassertion. Also serves as DTP power-on reset. | System cold reset   | N/A                           |

| Interface Name                   | Type                            | Direction | Width                           | Description                                                            | Connection              | Default value (if applicable) |
|----------------------------------|---------------------------------|-----------|---------------------------------|------------------------------------------------------------------------|-------------------------|-------------------------------|
| rst_cold_stable_ref_clk_no       | logic                           | Output    | 1                               | Stable cold reset synchronized to reference clock domain (active low). | Downstream subsystems   | N/A                           |
| powergood_i                      | logic                           | Input     | 1                               | Power-good indication from supply supervisor.                          | Power supply supervisor | N/A                           |
| jtag_ptap_client_tap_ctrl_i      | prim_jtag_pkg::jtag_tap_ctrl_t  | Input     | -                               | Primary JTAG TAP control (TCK, TMS, TRST_N).                           | External JTAG pads      | N/A                           |
| jtag_ptap_client_tdi_i           | logic                           | Input     | 1                               | Primary JTAG TDI.                                                      | External JTAG pads      | N/A                           |
| jtag_ptap_client_tdo_o           | logic                           | Output    | 1                               | Primary JTAG TDO.                                                      | External JTAG pads      | N/A                           |
| jtag_ptap_client_tdo_oen_o       | logic                           | Output    | 1                               | Primary JTAG TDO output enable.                                        | External JTAG pads      | N/A                           |
| jtag_bsr_host_scan_ctrl_o        | prim_jtag_pkg::jtag_scan_ctrl_t | Output    | -                               | Boundary scan control.                                                 | Boundary scan chain     | N/A                           |
| jtag_bsr_host_scan_in_i          | logic                           | Input     | 1                               | Boundary scan data in.                                                 | Boundary scan chain     | N/A                           |
| jtag_bsr_host_scan_out_o         | logic                           | Output    | 1                               | Boundary scan data out.                                                | Boundary scan chain     | N/A                           |
| jtag_stap_io_host_tap_ctrl_o     | prim_jtag_pkg::jtag_tap_ctrl_t  | Output    | -                               | I/O STAP TAP control.                                                  | I/O STAP host           | N/A                           |
| jtag_stap_io_host_tdi_i          | logic                           | Input     | 1                               | I/O STAP TDI.                                                          | I/O STAP host           | Tie to '0 if unused           |
| jtag_stap_io_host_tdo_o          | logic                           | Output    | 1                               | I/O STAP TDO.                                                          | I/O STAP host           | N/A                           |
| jtag_stap_io_host_tdo_oen_o      | logic                           | Output    | 1                               | I/O STAP TDO output enable.                                            | I/O STAP host           | N/A                           |
| jtag_stap_extra_host_tap_ctrl_o  | prim_jtag_pkg::jtag_tap_ctrl_t  | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TAP control array.                                          | Extra STAP hosts        | Tie to '0 if unused           |
| jtag_stap_extra_host_tdi_i       | logic                           | Input     | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDI array.                                                  | Extra STAP hosts        | Tie to '0 if unused           |
| jtag_stap_extra_host_tdo_o       | logic                           | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDO array.                                                  | Extra STAP hosts        | N/A                           |
| jtag_stap_extra_host_tdo_oen_o   | logic                           | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDO output enable array.                                    | Extra STAP hosts        | N/A                           |
| jtag_stap_host_scan_ctrl_o       | prim_jtag_pkg::jtag_scan_ctrl_t | Output    | -                               | Extended STAP scan control.                                            | STAP scan chain         | N/A                           |
| jtag_stap_host_scan_in_i         | logic                           | Input     | 1                               | Extended STAP scan data in.                                            | STAP scan chain         | N/A                           |
| jtag_stap_host_scan_out_o        | logic                           | Output    | 1                               | Extended STAP scan data out.                                           | STAP scan chain         | N/A                           |
| jtag_dfd_host_scan_ctrl_o        | prim_jtag_pkg::jtag_scan_ctrl_t | Output    | -                               | DFD iJTAG scan control.                                                | DFD iJTAG chain         | N/A                           |
| jtag_dfd_host_scan_in_i          | logic                           | Input     | 1                               | DFD iJTAG scan data in.                                                | DFD iJTAG chain         | N/A                           |
| jtag_dfd_host_scan_out_o         | logic                           | Output    | 1                               | DFD iJTAG scan data out.                                               | DFD iJTAG chain         | N/A                           |
| jtag_dft_secure_host_scan_ctrl_o | prim_jtag_pkg::jtag_scan_ctrl_t | Output    | -                               | Secure DFT iJTAG scan control.                                         | Secure DFT iJTAG chain  | N/A                           |

| Interface Name                  | Type                                          | Direction | Width                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Connection                         | Default value (if applicable) |
|---------------------------------|-----------------------------------------------|-----------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------|
| jtag_dft_secure_host_scan_in_i  | logic                                         | Input     | 1                                  | Secure DFT iJTAG scan data in.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Secure DFT iJTAG chain             | N/A                           |
| jtag_dft_secure_host_scan_out_o | logic                                         | Output    | 1                                  | Secure DFT iJTAG scan data out.                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Secure DFT iJTAG chain             | N/A                           |
| jtag_dft_host_scan_ctrl_o       | prim_jtag_pkg::jtag_scan_ctrl_t               | Output    | -                                  | Non-secure DFT iJTAG scan control.                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Non-secure DFT iJTAG chain         | N/A                           |
| jtag_dft_host_scan_in_i         | logic                                         | Input     | 1                                  | Non-secure DFT iJTAG scan data in.                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Non-secure DFT iJTAG chain         | N/A                           |
| jtag_dft_host_scan_out_o        | logic                                         | Output    | 1                                  | Non-secure DFT iJTAG scan data out.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Non-secure DFT iJTAG chain         | N/A                           |
| dtp_stop_clks_o                 | logic                                         | Output    | 1                                  | Clock stop output from DTP (JTAG DEBUG_CONTROL or CLA clock-stop).                                                                                                                                                                                                                                                                                                                                                                                                                             | PLL clock gates                    | N/A                           |
| jtag_ptap_state_o               | jtag_tap_pkg::tap_state_e                     | Output    | -                                  | Current PTAP TAP state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Debug infrastructure               | N/A                           |
| jtag_ptap_inst_decoded_o        | jtag_inst_reg_pkg::jtag_instruction_decoded_e | Output    | -                                  | Current decoded PTAP instruction.                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Debug infrastructure               | N/A                           |
| jtag_ic_reset_ext_o             | ic_reset_ext_t                                | Output    | $\$bits(ic\_reset\_ext\_t)$        | External JTAG IC_RESET override slice. Integrator-defined packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values. Driven by the DTP's <code>jtag_ic_reset_ext_o</code> when <code>Cfg.JTAG_IC_RESET_ENABLE=1</code> , otherwise tied to '0. An integrator needing JTAG reset overrides for logic outside the SMU sets <code>ic_reset_ext_t</code> to their own override struct and routes this port to it; the SMU does not define that type. | Integrator-defined reset overrides | N/A                           |
| xtrig_ctm_src_req_o             | logic                                         | Output    | $[XTRIG\_NUM\_INT\_CT-1:0]$        | Cross trigger source requests (SMU external ports [7:0]; DTP [1:0] reserved for SMC).                                                                                                                                                                                                                                                                                                                                                                                                          | Cross trigger network              | N/A                           |
| xtrig_ctm_src_ack_i             | logic                                         | Input     | $[XTRIG\_NUM\_INT\_CT-1:0]$        | Cross trigger source acknowledges. Unused in pulse-sync mode (mode bit = 0).                                                                                                                                                                                                                                                                                                                                                                                                                   | Cross trigger network              | Tie to '0 in pulse-sync mode  |
| xtrig_ctm_dst_req_i             | logic                                         | Input     | $[XTRIG\_NUM\_INT\_CT-1:0]$        | Cross trigger destination requests.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cross trigger network              | Tie to '0 if unused           |
| xtrig_ctm_dst_ack_o             | logic                                         | Output    | $[XTRIG\_NUM\_INT\_CT-1:0]$        | Cross trigger destination acknowledges. Unused in pulse-sync mode.                                                                                                                                                                                                                                                                                                                                                                                                                             | Cross trigger network              | N/A                           |
| xtrig_clk_stop_req_i            | logic                                         | Input     | $[XTRIG\_NUM\_CLK\_STOP\_REQ-1:0]$ | Clock stop requests (SMU external ports [7:0]; port [0] reserved for SMC internally).                                                                                                                                                                                                                                                                                                                                                                                                          | Clock stop requesters              | Tie to '0 if unused           |
| xtrig_ctp_req_out_dout_o        | logic                                         | Output    | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-out data output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | GPIO pad ring                      | N/A                           |
| xtrig_ctp_req_out_dout_en_o     | logic                                         | Output    | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-out data output enable.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | GPIO pad ring                      | N/A                           |
| xtrig_ctp_req_out_din_i         | logic                                         | Input     | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-out data input.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GPIO pad ring                      | Tie to '0 if unused           |
| xtrig_ctp_req_out_din_en_o      | logic                                         | Output    | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-out data input enable.                                                                                                                                                                                                                                                                                                                                                                                                                                                             | GPIO pad ring                      | N/A                           |
| xtrig_ctp_req_in_dout_o         | logic                                         | Output    | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-in data output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GPIO pad ring                      | N/A                           |
| xtrig_ctp_req_in_dout_en_o      | logic                                         | Output    | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-in data output enable.                                                                                                                                                                                                                                                                                                                                                                                                                                                             | GPIO pad ring                      | N/A                           |
| xtrig_ctp_req_in_din_i          | logic                                         | Input     | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-in data input.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GPIO pad ring                      | Tie to '0 if unused           |

| Interface Name                | Type                               | Direction | Width               | Description                                                          | Connection            | Default value (if applicable) |
|-------------------------------|------------------------------------|-----------|---------------------|----------------------------------------------------------------------|-----------------------|-------------------------------|
| xtrig_ctp_req_in_din_en_o     | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP request-in data input enable.                                    | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_in_dout_o       | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-in data output.                                              | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_in_dout_en_o    | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-in data output enable.                                       | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_in_din_i        | logic                              | Input     | [XTRIG_NUM_CTP-1:0] | CTP ack-in data input.                                               | GPIO pad ring         | Tie to '0 if unused           |
| xtrig_ctp_ack_in_din_en_o     | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-in data input enable.                                        | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_out_dout_o      | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data output.                                             | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_out_dout_en_o   | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data output enable.                                      | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_out_din_i       | logic                              | Input     | [XTRIG_NUM_CTP-1:0] | CTP ack-out data input.                                              | GPIO pad ring         | Tie to '0 if unused           |
| xtrig_ctp_ack_out_din_en_o    | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data input enable.                                       | GPIO pad ring         | N/A                           |
| rst_primary_ref_clk_no        | logic                              | Output    | 1                   | Primary reset synchronized to reference clock domain (active low).   | Downstream subsystems | N/A                           |
| rst_primary_smc_clk_no        | logic                              | Output    | 1                   | Primary reset synchronized to SMC clock domain (active low).         | Downstream subsystems | N/A                           |
| smu_axi_in_req_i              | smu_axi_xbar_pkg::axi_56_64_req_t  | Input     | -                   | SMU AXI crossbar inbound request (from SMN).                         | SMN fabric            | Tie to '0 if unused           |
| smu_axi_in_resp_o             | smu_axi_xbar_pkg::axi_56_64_resp_t | Output    | -                   | SMU AXI crossbar inbound response.                                   | SMN fabric            | N/A                           |
| smu_axi_out_req_o             | smu_axi_xbar_pkg::axi_out_req_t    | Output    | -                   | SMU AXI crossbar outbound request (to SMN).                          | SMN fabric            | N/A                           |
| smu_axi_out_resp_i            | smu_axi_xbar_pkg::axi_out_resp_t   | Input     | -                   | SMU AXI crossbar outbound response.                                  | SMN fabric            | Tie to '0 if unused           |
| smc_external_req_o            | smc_pkg::smc_axil_32_req_t         | Output    | -                   | SMC AXI-Lite external port request for adopter peripheral expansion. | Adopter peripherals   | N/A                           |
| smc_external_resp_i           | smc_pkg::smc_axil_32_resp_t        | Input     | -                   | SMC AXI-Lite external port response.                                 | Adopter peripherals   | Tie to DECERR if unused       |
| smc_efuse_bank_ctrl_req_o     | smc_pkg::smc_axil_32_req_t         | Output    | -                   | SMC eFuse bank control AXI-Lite request.                             | eFuse shim            | N/A                           |
| smc_efuse_bank_ctrl_resp_i    | smc_pkg::smc_axil_32_resp_t        | Input     | -                   | SMC eFuse bank control AXI-Lite response.                            | eFuse shim            | N/A                           |
| smc_efuse_shim_command_req_o  | smc_efuse_pkg::fuse_command_req_t  | Output    | -                   | SMC eFuse command interface request.                                 | eFuse shim            | N/A                           |
| smc_efuse_shim_command_resp_i | smc_efuse_pkg::fuse_command_resp_t | Input     | -                   | SMC eFuse command interface response.                                | eFuse shim            | N/A                           |
| smc_shadow_regs_o             | smc_efuse_pkg::efuse_map_t         | Output    | -                   | SMC eFuse shadow register output.                                    | External systems      | N/A                           |
| pll_clk_obs_i                 | logic                              | Input     | 1                   | PLL clock observation input.                                         | PLL shim              | N/A                           |
| pll_clk_obs_en_i              | logic                              | Input     | 1                   | PLL clock observation enable.                                        | Control logic         | N/A                           |
| pvt_process_clk_obs_i         | logic                              | Input     | 1                   | PVT process monitor clock observation input.                         | PVT shim              | N/A                           |

| Interface Name             | Type                                     | Direction | Width                         | Description                                                                                          | Connection                 | Default value (if applicable) |
|----------------------------|------------------------------------------|-----------|-------------------------------|------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------|
| pvt_process_clk_obs_en_i   | logic                                    | Input     | 1                             | PVT process monitor clock observation enable.                                                        | Control logic              | N/A                           |
| cat_therm_i                | logic                                    | Input     | 1                             | CAT thermal sensor input.                                                                            | CAT thermal sensor         | Tie to '0 if unused           |
| lsio_interface_select_o    | logic                                    | Output    | [NUM_GPIO_WRAP S-1:0]         | LSIO interface selection.                                                                            | GPIO padring               | N/A                           |
| pad2core_i                 | logic                                    | Input     | [NUM_GPIO_WRAP S-1:0]         | GPIO input data from pad ring to core.                                                               | GPIO padring → SMU         | N/A                           |
| core2pad_o                 | logic                                    | Output    | [NUM_GPIO_WRAP S-1:0]         | GPIO output data from core to pad ring.                                                              | SMU → GPIO padring         | N/A                           |
| pad2core_en_o              | logic                                    | Output    | [NUM_GPIO_WRAP S-1:0]         | GPIO pad-to-core path enable.                                                                        | SMU → GPIO padring         | N/A                           |
| core2pad_en_o              | logic                                    | Output    | [NUM_GPIO_WRAP S-1:0]         | GPIO core-to-pad path enable.                                                                        | SMU → GPIO padring         | N/A                           |
| rst_cool_n_from_pin_i      | logic                                    | Input     | 1                             | Cool reset input from GPIO pin.                                                                      | GPIO pin                   | N/A                           |
| clk_telemetry_i            | logic                                    | Input     | 1                             | Telemetry clock domain for ATB data collection.                                                      | Telemetry clock source     | N/A                           |
| rst_telemetry_n_i          | logic                                    | Input     | 1                             | Telemetry reset (active low).                                                                        | Telemetry reset source     | N/A                           |
| telemetry_atdata_i         | telemetry_receiver_pkg::telemetry_data_t | Input     | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB telemetry data input array.                                                                      | ATB sources                | N/A                           |
| telemetry_atid_i           | telemetry_receiver_pkg::atb_id_t         | Input     | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB telemetry ID input array.                                                                        | ATB sources                | N/A                           |
| telemetry_atready_o        | logic                                    | Output    | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB telemetry ready output.                                                                          | ATB sources                | N/A                           |
| telemetry_atvalid_i        | logic                                    | Input     | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB telemetry valid input.                                                                           | ATB sources                | N/A                           |
| telemetry_atflushvalid_o   | logic                                    | Output    | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB flush valid output.                                                                              | ATB sources                | N/A                           |
| telemetry_atflushready_i   | logic                                    | Input     | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB flush ready input.                                                                               | ATB sources                | N/A                           |
| smc_cluster_double_error_o | logic                                    | Output    | 1                             | Cluster double error detection output.                                                               | External systems           | N/A                           |
| smc_wdt_first_timeout_o    | logic                                    | Output    | 1                             | First watchdog timeout signal.                                                                       | Reset unit                 | N/A                           |
| smc_wdt_second_timeout_o   | logic                                    | Output    | 1                             | Second watchdog timeout signal.                                                                      | External systems           | N/A                           |
| smc_region_size_o          | logic                                    | Output    | [31:0]                        | SMC address region size output.                                                                      | External systems           | N/A                           |
| smc_ext_interrupts_i       | logic                                    | Input     | [Cfg.NUM_INT_T O_SMC-1:0]     | External interrupt inputs to SMC. Active-high level or a pulse long enough to be sampled on clk_smc. | External interrupt sources | Tie to '0 if unused           |
| smc_fuse_sense_done_o      | logic                                    | Output    | 1                             | SMC fuse sense completion output.                                                                    | Memory repair logic        | N/A                           |
| smc_fuse_reset_n_delayed_o | logic                                    | Output    | 1                             | Delayed fuse reset output.                                                                           | External systems           | N/A                           |

| Interface Name                | Type                             | Direction | Width                   | Description                                                                                                                                                                            | Connection                      | Default value (if applicable) |
|-------------------------------|----------------------------------|-----------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------|
| skip_mem_repair_o             | logic                            | Output    | 1                       | Skip memory repair output.                                                                                                                                                             | Memory repair logic             | N/A                           |
| ext_boot_seq_done_i           | logic                            | Input     | 1                       | External boot sequence completion input (memory repair and shadow register override done); gates reset release.                                                                        | External boot/repair controller | N/A                           |
| lc_state_o                    | logic                            | Output    | [2*LC_STATE_WIDTH-1:0]  | Lifecycle state output (driven by SEP when SEP=1, else 8'hf0).                                                                                                                         | External systems                | N/A                           |
| lc_sigint_err_o               | logic                            | Output    | 1                       | Lifecycle signal integrity error.                                                                                                                                                      | External systems                | N/A                           |
| smc_ndmreset_request_i        | logic                            | Input     | [CPU_CLUSTER_COUNT-1:0] | Non-debug-module reset requests.                                                                                                                                                       | Debug infrastructure            | Tie to '0 if unused           |
| smc_ndmreset_process_o        | logic                            | Output    | [CPU_CLUSTER_COUNT-1:0] | Non-debug-module reset process indicators.                                                                                                                                             | Debug infrastructure            | N/A                           |
| smc_ext_mailbox_interrupts_o  | logic                            | Output    | [NUM_MAILBOXES-1:0]     | External mailbox interrupt outputs.                                                                                                                                                    | External targets                | N/A                           |
| cfg_flr_pf_active_i           | logic                            | Input     | 1                       | Function Level Reset (FLR) PF active signal.                                                                                                                                           | PCIe / system                   | Tie to '0 if unused           |
| isolate_req_o                 | logic                            | Output    | [31:0]                  | Isolation request output for 32 subsystems.                                                                                                                                            | Subsystems                      | N/A                           |
| ss_reset_complete_i           | logic                            | Input     | [31:0]                  | Subsystem reset completion status.                                                                                                                                                     | Subsystems                      | 32'hFFFFFFFF                  |
| ss_config_o                   | logic                            | Output    | [31:0]                  | Subsystem configuration output.                                                                                                                                                        | Subsystems                      | N/A                           |
| ss_reset_ctrl_o               | smc_reset_unit_pkg::reset_ctrl_t | Output    | [31:0]                  | Subsystem reset control output array.                                                                                                                                                  | Subsystems                      | N/A                           |
| sync_irq_o                    | logic                            | Output    | 1                       | Software-controlled global synchronisation bit (SYNC_REG.sync in the SMC reset unit). Not an interrupt aggregate output.                                                               | External systems                | N/A                           |
| gpio_interrupt_o              | logic                            | Output    | [NUM_GPIO_WRAPS-1:0]    | Raw GPIO interrupt vector from the padding, one bit per GPIO wrap (bonded + unbonded). Passed through from SMC. Software reads GPIO status registers to identify the interrupt source. | SMC → external consumers        | N/A, leave floating if unused |
| uart_interrupt_o              | logic                            | Output    | [NUM_UART-1:0]          | Raw UART interrupt outputs, one bit per UART instance, in the peripheral clock domain. Passed through from SMC.                                                                        | SMC → external consumers        | N/A, leave floating if unused |
| smc_rom_intf_req_o            | rom_req_t                        | Output    | -                       | SMC ROM memory interface request.                                                                                                                                                      | Boot ROM                        | N/A                           |
| smc_rom_intf_rsp_i            | rom_rsp_t                        | Input     | -                       | SMC ROM memory interface response.                                                                                                                                                     | Boot ROM                        | N/A                           |
| smc_scratch_ram_intf_req_o    | scratch_ram_req_t                | Output    | [31:0]                  | SMC scratch RAM interface requests.                                                                                                                                                    | Scratch RAM banks               | N/A                           |
| smc_scratch_ram_intf_rsp_i    | scratch_ram_rsp_t                | Input     | [31:0]                  | SMC scratch RAM interface responses.                                                                                                                                                   | Scratch RAM banks               | N/A                           |
| smc_l1_icache_tag_intf_req_o  | l1_icache_tag_req_t              | Output    | [3:0]                   | SMC L1 I-cache tag interface requests.                                                                                                                                                 | I-cache tag banks               | N/A                           |
| smc_l1_icache_tag_intf_rsp_i  | l1_icache_tag_rsp_t              | Input     | [3:0]                   | SMC L1 I-cache tag interface responses.                                                                                                                                                | I-cache tag banks               | N/A                           |
| smc_l1_icache_data_intf_req_o | l1_icache_data_req_t             | Output    | [7:0]                   | SMC L1 I-cache data interface requests.                                                                                                                                                | I-cache data banks              | N/A                           |
| smc_l1_icache_data_intf_rsp_i | l1_icache_data_rsp_t             | Input     | [7:0]                   | SMC L1 I-cache data interface responses.                                                                                                                                               | I-cache data banks              | N/A                           |

| Interface Name                | Type                               | Direction | Width                   | Description                                                                   | Connection          | Default value (if applicable) |
|-------------------------------|------------------------------------|-----------|-------------------------|-------------------------------------------------------------------------------|---------------------|-------------------------------|
| smc_l1_dcache_tag_intf_req_o  | l1_dcache_tag_req_t                | Output    | [3:0]                   | SMC L1 D-cache tag interface requests.                                        | D-cache tag banks   | N/A                           |
| smc_l1_dcache_tag_intf_rsp_i  | l1_dcache_tag_rsp_t                | Input     | [3:0]                   | SMC L1 D-cache tag interface responses.                                       | D-cache tag banks   | N/A                           |
| smc_l1_dcache_data_intf_req_o | l1_dcache_data_req_t               | Output    | [3:0]                   | SMC L1 D-cache data interface requests.                                       | D-cache data banks  | N/A                           |
| smc_l1_dcache_data_intf_rsp_i | l1_dcache_data_rsp_t               | Input     | [3:0]                   | SMC L1 D-cache data interface responses.                                      | D-cache data banks  | N/A                           |
| smc_disable_sram_auto_init_i  | logic                              | Input     | 1                       | Disable SRAM automatic initialization.                                        | Control logic       | 1'b0                          |
| smc_init_mem_done_o           | logic                              | Output    | 1                       | Memory initialization completion signal.                                      | Control logic       | N/A                           |
| chiplet_is_primary_i          | logic                              | Input     | 1                       | Chiplet is primary (for System Timer OCTS).                                   | GPIO padring straps | N/A                           |
| timer_count_o                 | logic                              | Output    | [63:0]                  | 64-bit system timer count output (OCTS).                                      | External systems    | N/A                           |
| trace_mem_req_o               | dfd_trace_mem_pkg::SinkMemPktIn_s  | Output    | [TRC_RAM_INSTANCES-1:0] | Trace memory interface requests.                                              | Trace memory        | N/A                           |
| trace_mem_resp_i              | dfd_trace_mem_pkg::SinkMemPktOut_s | Input     | [TRC_RAM_INSTANCES-1:0] | Trace memory interface responses.                                             | Trace memory        | N/A                           |
| test_en_i                     | logic                              | Input     | 1                       | DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs. | Test controller     | 1'b0                          |
| scan_rst_ni                   | logic                              | Input     | 1                       | Scan reset (active-low). Bypasses reset synchronizers during DFT.             | Test controller     | 1'b1                          |
| mem_repair_done_i             | logic                              | Input     | 1                       | Memory repair done indicator.                                                 | DFT controller      | 1'b0                          |
| mem_repair_success_i          | logic                              | Input     | 1                       | Memory repair success indicator.                                              | DFT controller      | 1'b0                          |
| mem_repair_abort_i            | logic                              | Input     | 1                       | Memory repair abort indicator.                                                | DFT controller      | 1'b0                          |
| mbist_done_i                  | logic                              | Input     | 1                       | MBIST done indicator.                                                         | DFT controller      | 1'b0                          |
| mbist_pass_i                  | logic                              | Input     | 1                       | MBIST pass indicator.                                                         | DFT controller      | 1'b0                          |
| mbist_abort_i                 | logic                              | Input     | 1                       | MBIST abort indicator.                                                        | DFT controller      | 1'b0                          |
| sep_sram_req_o                | sep_pkg::sep_sram_req_t            | Output    | -                       | SEP scratch SRAM request passthrough.                                         | SEP SRAM macros     | N/A                           |
| sep_sram_rsp_i                | sep_pkg::sep_sram_rsp_t            | Input     | -                       | SEP scratch SRAM response passthrough.                                        | SEP SRAM macros     | N/A                           |
| sep_boot_rom_req_o            | sep_pkg::sep_sram_req_t            | Output    | -                       | SEP boot ROM request passthrough.                                             | SEP ROM macro       | N/A                           |
| sep_boot_rom_rsp_i            | sep_pkg::sep_sram_rsp_t            | Input     | -                       | SEP boot ROM response passthrough.                                            | SEP ROM macro       | N/A                           |
| sep_cpu_tcm_req_o             | sep_pkg::sep_cpu_tcm_req_t         | Output    | -                       | SEP CPU TCM request passthrough (ICCM + DCCM).                                | SEP SRAM macros     | N/A                           |
| sep_cpu_tcm_rsp_i             | sep_pkg::sep_cpu_tcm_rsp_t         | Input     | -                       | SEP CPU TCM response passthrough.                                             | SEP SRAM macros     | N/A                           |
| sep_efuse_bank_ctrl_req_o     | sep_efuse_pkg::efuse_axil_req_t    | Output    | -                       | SEP eFuse bank control AXI-Lite request passthrough.                          | SEP eFuse shim      | N/A                           |
| sep_efuse_bank_ctrl_rsp_i     | sep_efuse_pkg::efuse_axil_rsp_t    | Input     | -                       | SEP eFuse bank control AXI-Lite response passthrough.                         | SEP eFuse shim      | N/A                           |

| Interface Name                 | Type                                           | Direction | Width | Description                                                                                                                       | Connection                 | Default value (if applicable) |
|--------------------------------|------------------------------------------------|-----------|-------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------|
| sep_efuse_shim_command_req_o   | sep_efuse_pkg::fuse_command_req_t              | Output    | -     | SEP eFuse command interface request passthrough.                                                                                  | SEP eFuse shim             | N/A                           |
| sep_efuse_shim_command_resp_i  | sep_efuse_pkg::fuse_command_resp_t             | Input     | -     | SEP eFuse command interface response passthrough.                                                                                 | SEP eFuse shim             | N/A                           |
| sep_crypto_pka_imem_sram_req_o | sep_crypto_pkg::sep_crypto_pka_imem_sram_req_t | Output    | -     | SEP OTBN PKA instruction memory SRAM request passthrough.                                                                         | SRAM macro                 | N/A                           |
| sep_crypto_pka_imem_sram_rsp_i | sep_crypto_pkg::sep_crypto_pka_imem_sram_rsp_t | Input     | -     | SEP OTBN PKA instruction memory SRAM response passthrough.                                                                        | SRAM macro                 | N/A                           |
| sep_crypto_pka_dmem_sram_req_o | sep_crypto_pkg::sep_crypto_pka_dmem_sram_req_t | Output    | -     | SEP OTBN PKA data memory SRAM request passthrough.                                                                                | SRAM macro                 | N/A                           |
| sep_crypto_pka_dmem_sram_rsp_i | sep_crypto_pkg::sep_crypto_pka_dmem_sram_rsp_t | Input     | -     | SEP OTBN PKA data memory SRAM response passthrough.                                                                               | SRAM macro                 | N/A                           |
| sep_km_rom_mem_req_o           | km_intf_pkg::km_rom_mem_req_t                  | Output    | -     | SEP Key Manager ROM memory request passthrough.                                                                                   | ROM macro                  | N/A                           |
| sep_km_rom_mem_rsp_i           | km_intf_pkg::km_rom_mem_rsp_t                  | Input     | -     | SEP Key Manager ROM memory response passthrough.                                                                                  | ROM macro                  | N/A                           |
| sep_km_sram_mem_req_o          | km_intf_pkg::km_sram_mem_req_t                 | Output    | -     | SEP Key Manager SRAM memory request passthrough.                                                                                  | SRAM macro                 | N/A                           |
| sep_km_sram_mem_rsp_i          | km_intf_pkg::km_sram_mem_rsp_t                 | Input     | -     | SEP Key Manager SRAM memory response passthrough.                                                                                 | SRAM macro                 | N/A                           |
| sep_external_req_o             | sep_pkg::sep_32_64_12_axi_req_t                | Output    | -     | SEP AXI extension port request passthrough.                                                                                       | Adopter peripherals        | N/A                           |
| sep_external_resp_i            | sep_pkg::sep_32_64_12_axi_resp_t               | Input     | -     | SEP AXI extension port response passthrough.                                                                                      | Adopter peripherals        | Tie to DECERR if unused       |
| sep_cpu_trace_o                | sep_pkg::sep_cpu_trace_t                       | Output    | -     | SEP CPU trace output.                                                                                                             | Trace infrastructure       | N/A                           |
| sep_lockstep_ctrl_i            | sep_pkg::sep_lockstep_ctrl_t                   | Input     | -     | SEP CPU lockstep controls, passed through to the SEP. Inert unless the SEP core is built with <b>RV_LOCKSTEP_ENABLE</b> .         | Chip-level RAS control     | Tie to '0 if unused           |
| sep_lockstep_status_o          | sep_pkg::sep_lockstep_status_t                 | Output    | -     | SEP CPU lockstep status. Reads 0 when the SEP is absent ( <b>SEP=0</b> ) or the core is built without <b>RV_LOCKSTEP_ENABLE</b> . | Chip-level RAS aggregation | N/A                           |
| lcc_demote_state_1_o           | logic                                          | Output    | [1:0] | SEP lifecycle demote state 1.                                                                                                     | External lifecycle logic   | N/A                           |
| lcc_demote_state_2_o           | logic                                          | Output    | [1:0] | SEP lifecycle demote state 2.                                                                                                     | External lifecycle logic   | N/A                           |
| sep_fuse_sense_done_o          | logic                                          | Output    | 1     | SEP fuse sense done output.                                                                                                       | System integration         | N/A                           |
| clk_sep_wdt_i                  | logic                                          | Input     | 1     | SEP watchdog timer clock passthrough.                                                                                             | Low-frequency clock source | N/A                           |
| secure_tm_req_i                | logic                                          | Input     | -     | Secure test mode request, passthrough to SEP.                                                                                     | GPIO padding / tie-off     | N/A                           |

# Chapter 2. System Management Network (SMN) Integration

## WARNING

This section is currently under development. Content will be added to describe SMN integration procedures, network topology, and management interfaces.

# Chapter 3. Project Defines

Compile-time defines that apply across OCAH rather than to a single block.

## 3.1. SIMULATION

**SIMULATION** selects the verification view of OCAH RTL. Set it on every simulation compile and on no other kind of build:

```
+define+SIMULATION    # VCS, Xcelium
-DSIMULATION          # Verilator, Icarus
```

OCAH's own DV flows set it; a compile driven by an adopter's testbench does not.

### WARNING

Do not let **SIMULATION** reach synthesis, formal, or a gate-level flow. It enables **initial** blocks, **\$readmemh**, **\$random** and DPI exports; a tool that skips those rather than rejecting them yields a netlist that does not correspond to the RTL you verified.

Without the define, these testbench hooks are not compiled in and the plusargs are silently ignored:

| Plusarg                                                           | Effect                                                                                                                                         |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| +skip_fuse_sense                                                  | <b>efuse_shadow_regs</b> reports fuse sense complete immediately rather than sensing the bank through the adopter's SHIM.                      |
| +smc_shadow_reg_preload=<file>,<br>+sep_shadow_reg_preload=<file> | Loads the shadow registers from a hex image. Takes effect only alongside +skip_fuse_sense.                                                     |
| +rom_bin64=<file>, +rom_hex=<file>                                | Preloads <b>OCAH4CORECluster_rom_ext</b> , the SMC CPU ROM simulation model. Applies only to integrations that use the reference memory swaps. |

The define also zeroes the RMA and security-disable token and digest bits in **efuse\_token\_processing** at time 0, which have no reset and otherwise power up X, and routes OpenTitan assertion failures in the SEP **prim\_assert** shim to **\$error** on non-UVM runs.

Two model families it enables make simulation deliberately non-deterministic: **prim\_flop\_2sync** and its Caliptra and Adams Bridge equivalents insert **prim\_cdc\_rand\_delay** ahead of the synchronizer, and **prim\_lfsr** and **prim\_double\_lfsr** randomize their default seed on simulators other than Verilator (+**prim\_lfsr\_use\_default\_seed**=1 pins it). Two runs of one test are therefore not identical unless the simulator seed is fixed; record the seed with any failure you intend to reproduce.

# Chapter 4. Prim Cell Replacement and Don't-Touch Constraints

OCAH primitives (`prim_stdbuf`, `prim_and2`, `prim_clkgater`, the `prim_flop_*sync*` family, and so on) ship as behavioral models with a reference technology mapping. Adopters replace them with cells from their own library.

Two rules govern those replacements. Both are the adopter's responsibility — OCAH provides the instance names, but the names have no effect until the adopter's synthesis and place-and-route flows constrain them.

## 4.1. Naming Requirement

**Every library cell instantiated inside a replacement primitive must have `d0nt_` in its instance name.**

The marker is deliberately misspelled with a zero. `dont_touch` is a reserved keyword in several tools and a common substring in ordinary comments and signal names; `d0nt_` occurs only where the protection is intended, so a name-based match cannot pick up unrelated cells.

Clock-path cells take the extended prefix `d0nt_clk`, which selects the stricter rule below. The `d0nt_*` naming convention appears in the eFuse IP (`efuse_token_digest_sha256`, `efuse_triple_redundant_comparator`) as the primary example. The adopter is expected to carry the same convention into any technology-mapped primitive wrapper they deliver. The intended name assignments are:

| Instance name               | Rule        | Primitive                                         |
|-----------------------------|-------------|---------------------------------------------------|
| <code>d0nt_clkgate</code>   | Don't touch | <code>prim_clkgater</code>                        |
| <code>d0nt_clkbuf</code>    | Don't touch | <code>prim_clock_buf</code>                       |
| <code>d0nt_clkinv</code>    | Don't touch | <code>prim_clock_inv</code>                       |
| <code>d0nt_clkmux</code>    | Don't touch | <code>prim_clock_mux2</code>                      |
| <code>d0nt_clknand</code>   | Don't touch | <code>prim_clock_nand2</code>                     |
| <code>d0nt_clkxor</code>    | Don't touch | <code>prim_xor2</code>                            |
| <code>d0nt_stdbuf</code>    | Size only   | <code>prim_stdbuf</code>                          |
| <code>d0nt_stdinv</code>    | Size only   | <code>prim_flop_*sync_*</code> reset inverter     |
| <code>d0nt_an</code>        | Size only   | <code>prim_and2</code>                            |
| <code>d0nt_ao222</code>     | Size only   | <code>prim_ao222</code>                           |
| <code>d0nt_wrap_sync</code> | Size only   | <code>prim_flop_*sync_*</code> synchronizer flops |

Keep the same names when substituting your own cells. Adopters who rename must apply the equivalent constraints to whatever names they use.

## 4.2. Constraint Requirement

In synthesis, apply `set_size_only` to every cell matching `d0nt_`, then `set_dont_touch` to every cell matching `d0nt_clk`. Every step after synthesis needs `size_only` alone — on all `d0nt_` cells, clock cells included.

```
# Order matters: d0nt_clk* also matches d0nt_*, so the clock rule is applied
# second and overrides size-only for those cells.
```

```
set_size_only [get_cells -hierarchical -filter "full_name =~ *d0nt_*"]
set_dont_touch [get_cells -hierarchical -filter "full_name =~ *d0nt_clk*"]
```

**size\_only** keeps the instance in the netlist—the tool may not delete, merge, absorb, restructure, or buffer through it—while still allowing it to be swapped for another library cell of the same logic function, typically to change drive strength or threshold-voltage flavor for timing, area, and EM closure. That is the intent for every replacement cell: these primitives exist because the structure matters, not because the logic function could not be inferred. Synthesis will otherwise remove them—an inverter pair that cancels, redundant AND gates that always agree, a buffer driving a single load are all dead weight to an optimizer.

Clock cells need the stronger **dont\_touch** because that swap is exactly the problem. Under **size\_only** the tool treats the logic function as the contract and every other property of the cell as free to change. On the clock path the function is the least of the reasons the cell was chosen. Clock buffers, inverters, muxes, and gates are picked for electrical characteristics the function does not capture—matched rise and fall times above all, and with them duty-cycle preservation, balanced input-to-output delay, and low delay variation across PVT. A functionally identical cell pulled from the general-purpose library guarantees none of that. The netlist still says "inverter", functional simulation still passes, and the duty-cycle distortion shows up only in silicon or in a timing run late enough to be expensive.

**dont\_touch** removes the tool's discretion: the cell the designer selected is the cell that reaches the netlist. Sizing and insertion-delay decisions on the clock path belong to clock tree synthesis, which owns them explicitly, rather than to a logic-optimization pass making a substitution it believes is free.

### 4.2.1. After Synthesis

The **dont\_touch** on **d0nt\_clk\*** is a synthesis-only constraint. Once synthesis has handed off the netlist, release the clock cells to **size\_only** and carry that—and nothing stronger—through placement, clock tree synthesis, routing, and optimization:

```
# Post-synthesis: uniform size_only, clock cells included.
set_dont_touch [get_cells -hierarchical -filter "full_name =~ *d0nt_clk*"] false
set_size_only [get_cells -hierarchical -filter "full_name =~ *d0nt_*"]
```

The stricter rule exists to stop logic optimization from choosing the cell. Downstream, clock cell sizing is CTS's job: it cannot balance skew or close transition violations with the cells it needs to resize frozen, and freezing them there buys nothing. Keeping **size\_only** in place still prevents deletion, merging, and restructuring for the rest of the flow.

Restrict the physical flow to the library's clock-cell list so a resize stays inside the same clock cell family. Sizing a **CLKBUF\_D4** up to a **CLKBUF\_D8** keeps the matched rise and fall behavior the cell was chosen for; substituting a general-purpose buffer of equivalent drive does not.

Additional requirements:

- **Carry size\_only through PnR.** The **dont\_touch** is synthesis-only, but **size\_only** is not. A cell protected in synthesis and left entirely unprotected downstream is not protected.
- **Verify the match is non-zero.** A typo in the filter, or a flow that renames instances before the constraint is read, silently protects nothing. Check the collection size and error out if it is empty.
- **Preserve the names.** Uniquification, flattening, and name mapping must not strip the **d0nt\_** substring. If your flow renames instances, derive the collections before flattening and carry them forward.

- **Restore protection after ECOs.** Flows that relax preservation constraints for clock-tree or filler ECOs — clearing `eco_honor_dont_touch`, for example — must restore `size_only` before any subsequent optimization pass.
- **Check the final netlist**, not just the post-synthesis one.

Flows other than Design Compiler and Fusion Compiler must apply the equivalent `size_only` and `dont_touch` attributes to the same collections.

## 4.3. Simulation

These constraints have no functional effect and no simulation impact. RTL simulation compiles the technology-independent primitive models (`not(synth)`). Library cells are not on that filelist, so a simulator never elaborates them.

Confirming that the protected cells survived is therefore a gate-level netlist check. RTL simulation cannot detect their loss.

# Chapter 5. CDC Timing Constraints

OCAH ships synthesis constraints that bound the enumerated synchronization structures in the SMC, SEP, DTP and SMU. Independent CDC sign-off is still required for the assembled SoC. This section explains what those constraints do, how they are produced, and what an adopter has to change when integrating OCAH into their own flow and cell library.

The collateral ships with the harness. `integration/constraint/<block>/synth/` groups it by system while preserving the original filenames and relative path depth:

- `constraints.sdc` — the block’s full timing intent, including the clock definitions and the asynchronous clock groups. It sources the two files below.
- `<block>_cdc_max_delay.tcl` — generated-clock periods and any hand-written overrides.
- `<block>_cdc_max_delay_generated.tcl` — the per-instance calls.

The shared `async_clock_groups.tcl` and `cdc_max_delay_procs.tcl` helpers are under `integration/constraint/shared/`. These are symlink indexes: the canonical per-system files remain under `hw/sys/<block>/synth/`, and the canonical shared helpers remain under `flows/synth/constraints/`. Regenerate the index with `python3 scripts/collect_integration.py` after changing the collateral inventory.

## 5.1. Why crossings are bounded rather than timed

A clock-domain crossing has no fixed phase relationship, so it cannot be timed against a capture edge. Its correctness still depends on delay, though: every CDC element assumes the data reaches the receiver within a bounded time.

OCAH therefore keeps the crossings visible to the tool and gives each one an explicit upper bound, sized from the clock that captures it. This asserts no phase relationship — it states only how much delay the crossing tolerates while its CDC element’s guarantee still holds.

## 5.2. What is constrained

Each enumerated CDC element in each block is given its own `set_max_delay`, sized from the clock that actually captures it as read from the netlist. Muxed domains resolve to several clocks; the tightest period wins.

The elements covered are `prim_sync2/3/4` and their reset and set variants, the pulp `sync` shim, `prim_sync_reset`, `prim_sync3_pulse`, `prim_fifo_async`, `avsbu_async_fifo`, `cdc_fifo_gray`, `axi_cdc`, `prim_reg_cdc`, `prim_sync_reqack_data`, `prim_sync_data_autohs`, `cdc_2phase`, `dmi_wrapper` and Rocket `AsyncQueue`.

The bound depends on what protects the crossing:

| Kind of path             | Bound                    | Reason                                                                                                                                     |
|--------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Synchronizer data input  | $0.5 * T_{dst}$          | Leaves half a destination period of settling margin before the capture edge the synchronizer relies on.                                    |
| Protocol-protected data  | $1.0 * T_{dst}$          | Conservative. The requirement is set by the pointer synchronizer’s depth — see below.                                                      |
| Two-phase handshake nets | $\min(T_{src}, T_{dst})$ | <code>cdc_2phase</code> runs the handshake in both directions, so the tighter domain governs. This is the module’s own stated requirement. |

$T_{dst}$  rather than  $\min(T_{src}, T_{dst})$  in the general case because the gating synchronizer is clocked by the destination, so its settling window is  $S * T_{dst}$  and  $T_{src}$  does not enter the guarantee. This makes slow-to-fast and fast-to-slow equally safe.

### 5.2.1. Why protocol-protected data still needs a bound

The payload of a gray-pointer FIFO or a req/ack handshake never passes through a synchronizer — only the pointer or request does. So this is not a metastability constraint, and it is reasonable to ask why such a path needs a bound at all, or why a single blanket bound covering every crossing in the design would not be enough.

Because the protocol guarantees **ordering**, not **arrival**. The read side will not look at an entry until the write pointer has crossed, which takes  $SYNC\_STAGES$  destination cycles, where  $SYNC\_STAGES$  is the depth of the pointer synchronizer. Nothing in the protocol ensures the data physically arrived within that window; that is a timing property, and only a  $max\_delay$  states it. Miss it and the consumer samples the entry’s previous contents. There is no violation report and no metastability event, just wrong data.

So the requirement is  $data\ path < SYNC\_STAGES * T_{dst}$ . It is not a fixed number of nanoseconds: it scales with the clock that captures the crossing, and the destination periods within one design span an order of magnitude.

| Destination period | Guarantee at 3 stages |
|--------------------|-----------------------|
| 1 ns               | 3 ns                  |
| 5 ns               | 15 ns                 |
| 10 ns              | 30 ns                 |

That is why the bound is applied per instance rather than as one global value. A single blanket  $max\_delay$  has to be chosen once for the whole design: loose enough to be feasible on the slowest domains, it permits several times more delay than a fast crossing’s pointer scheme tolerates; tight enough for the fastest domain, it over-constrains every slow crossing and burns effort closing paths that were never at risk. Sizing each crossing from its own capturing clock is the only bound that tracks the requirement.

The blanket default that the clock-group layer applies (see [The two layers](#)) is deliberately on the loose side, and for slow destination domains it is already sufficient — there the explicit per-instance constraint is redundant. It is the fast crossings that need it.

$1.0 * T_{dst}$  is therefore conservative by a factor of two to three. It is kept simple deliberately, but if wide payloads such as AXI data will not close, raising  $cdc\_data\_delay$  to  $2.0 * T_{dst}$  stays inside a three-stage guarantee with a cycle to spare. An adopter whose synchronizer cells use a different depth should recompute from their own  $SYNC\_STAGES$  rather than inheriting this factor.

Enumeration and clock resolution run once, offline, against an elaborated design, and their output is an explicit per-block file of constraint calls that is reviewed and checked in. The flow itself does no discovery. This matters to an integrator because those files are written against the OCAH hierarchy and OCAH’s clock names. If you instantiate a block as delivered, you do not need to regenerate or hand-edit them: set `::cdc_hier_prefix` to where the block sits in your hierarchy, and optionally `::cdc_clock_alias` to map OCAH clock names onto your own. Regeneration is needed only when the block is reconfigured in a way that changes which CDC elements exist, since no prefix can conjure constraints for elements the file never listed (see [Integration checklist](#) item 2).

## 5.3. The two layers

The per-instance bounds take effect only on top of a correctly configured clock-group layer. That layer decides whether the tool considers inter-domain paths at all, so it must be established first; the per-instance exceptions then refine it.

**Layer 1** is `set_async_clock_groups`, defined in `flows/synth/constraints/async_clock_groups.tcl` and called from each block's `constraints.sdc`. It declares the asynchronous groups **with** `-allow_paths` and applies a loose 20 ns / -20 ns default to every inter-group clock pair.

#### IMPORTANT

`-allow_paths` is a prerequisite, not a refinement. In DC and PrimeTime exception priority, `set_false_path` outranks `set_max_delay` regardless of which is more specific, so a bare `set_clock_groups -asynchronous` silently masks **every** per-instance constraint. There is no warning; the exceptions simply have no effect.

The 20 ns default is what makes `-allow_paths` safe to enable. Once paths are allowed, every crossing without a specific constraint would otherwise be timed against a meaningless launch-capture relationship and fail. The default bounds them loosely so that only deliberately tightened paths appear as real requirements. It is emitted per ordered clock pair, never group-wide, because a 20 ns bound inside a group would mask genuine setup failures on synchronous paths.

**Layer 2** is the per-instance `set_max_delay` output of the three steps above, with two deliberate carve-outs:

- **Paths from primary inputs revert to the default.** A synchronizer input reachable from a top-level port carries that port's `set_input_delay`, which is a budget figure rather than real logic and can alone exceed the tightened bound. Only the parent block sees both ends, so there is nothing to tighten at block level.
- **Same-clock instances are skipped.** A two-clock element wired to one clock is not a crossing, and applying `set_max_delay -ignore_clock_latency` there would replace a proper setup check with a flat budget that discards skew and uncertainty. These are emitted commented out. OCAH has five, where OpenTitan AES, KMAC and OTBN instantiate `prim_sync_reqack_data` on EDN and OTP interfaces that are separate domains upstream but share a clock in SEP.

## 5.4. What is covered

| Block | CDC elements constrained | Exceptions emitted |
|-------|--------------------------|--------------------|
| SMC   | 365                      | 412                |
| SEP   | 18                       | 21                 |
| DTP   | 250                      | 270                |
| SMU   | 758                      | 838                |

SMU is a flat build containing the other three, so its instance paths carry `u_smc/`, `gen_sep.u_sep/` and `u_dtp/` prefixes. It is not the sum of the parts: DTP contributes 375 elements inside SMU against 250 standalone, because it instantiates three JTAG2AXI bridges in the SMU configuration and two on its own.

## 5.5. Integration checklist

Five things must change, three should be checked. None of them requires editing per-instance data by hand; every change is either a one-line policy edit or a regeneration.

### 5.5.1. 1. Repoint the constraints at the pin of the cell inside the prim wrapper

This is the most important change.

As shipped, the procs target the prim wrapper's own port, `prim_sync3/i_d`. That port exists only as long as the wrapper survives as a hierarchical boundary, and a prim cell is ordinary RTL: nothing stops synthesis from ungrouping it and

dissolving its ports.

#### WARNING

Do not rely on the wrapper port surviving. Anchor the constraint to the pin of the cell instantiated **inside** the wrapper. When you swap in your own synchronizer, that cell is a real library cell or macro—a leaf that cannot be flattened—so its pins persist through optimisation regardless of what happens to the RTL hierarchy around it.

```
proc set_cdc_max_delay_prim_sync3 { inst dst_clk {delay {}} } {  
    cdc_emit "prim_sync3 $inst" [cdc_sync_delay $dst_clk $delay] to \  
        {} [get_pins "$inst/u_my_sync_cell/D" -quiet]  
}
```

`$inst` stays the wrapper instance path the generator recorded; you extend it down to the leaf. If your cell is instantiated in a generate loop or per bit, the path needs a glob. Note that generate scopes are separated by `..`, not `/`:

```
[get_pins "$inst/g_bits*.u_my_sync_cell/D" -quiet]
```

Anchoring this way makes hierarchy preservation a safety margin rather than a load-bearing dependency.

If you keep the prim wrappers unchanged, the shipped port names work as-is, but your flow must then preserve those wrapper boundaries. A `destination empty` warning appearing after `compile` where there was none at constraint-load time means a boundary was dissolved.

If you go further and replace a wrapper with a module of your own name, the enumeration that produced `<block>_cdc_max_delay_generated.tcl` knows nothing about it and those crossings end up with no per-instance bound at all. Nothing warns—an element that was never listed cannot report itself missing. Add the calls for it to `<block>_cdc_max_delay.tcl` by hand; the procs take an instance path and the capturing clock regardless of the module they were named after.

### 5.5.2. 2. Point the files at your hierarchy, or regenerate

The checked-in `<block>_cdc_max_delay_generated.tcl` files are written against the OCAH hierarchy, with paths relative to the block's own top and clocks named as OCAH names them. Two ways to make them apply to your design.

**If you instantiate the block as delivered**, reuse the files and set two variables. Both default to no-ops, so leaving them alone reproduces the shipped behaviour exactly.

```
# where the block sits in your hierarchy, with a trailing slash  
set ::cdc_hier_prefix "u_chiplet/u_mgmt/u_smc/"  
  
# optional: map an OCAH clock name onto the clock you actually drive it with  
set ::cdc_clock_alias(SMCLK) MY_SYS_CLK
```

The prefix is prepended to every instance path. The alias is applied wherever a clock name reaches the tool, so OCAH's names become labels in your design. You can also skip the alias and supply the period directly with `set ::cdc_clock_period(SMCLK) <ps>`, which is enough for the synchronizer constraints; the alias additionally matters for

the data-path constraints, which name the destination clock as a timing endpoint and so need it to exist.

**If you reconfigure the block**, regenerate instead. Changing parameters that add or remove CDC elements makes the shipped file stale in the silent direction—the constraints that should exist for new elements simply will not, and nothing warns. The prefix does not help with that.

A wrong prefix fails loudly: every path resolves to nothing, so you get a warning per constraint against a zero exception count, and **SUMMARY 0 exceptions** is itself reported as a setup failure.

Regeneration is also the right answer whenever a parent build contains the block. Instance counts there can differ from the standalone build, so a parent’s file is not the child’s file with a prefix.

### 5.5.3. 3. Set the default bound

The 20 ns / -20 ns figures are a starting point. Both are **info exists**-guarded in `flows/synth/constraints/async_clock_groups.tcl`, so setting them before that file is sourced overrides without editing it. `set_async_clock_groups` also accepts per-call overrides, which is the cleaner way to give one block a different default:

```
set_async_clock_groups { {CLK_A} {CLK_B} } -max_delay 8000 -min_delay -8000
```

Two constraints on the choice. The default must exceed every per-instance bound you tighten, because it also governs paths from primary inputs into synchronizers; if it is tighter than  $0.5 * T_{dst}$  it becomes the binding constraint there. And tighter is not automatically better: this is a net for crossings you have not enumerated, so set it loose enough that those do not fail and rely on layer 2 for the paths that matter.

The minimum is negative deliberately. With `-allow_paths` a crossing has no meaningful hold relationship, and a zero floor would report hold violations where hold is not a real requirement.

### 5.5.4. 4. Declare your own clock groups

`set_async_clock_groups` is called from each block’s `constraints.sdc` with that block’s group list. Each inner list is one asynchronous group and globs are allowed. Globs resolve when the call runs, so it has to come after every `create_clock` and `create_generated_clock` it is meant to cover.

```
set_async_clock_groups {  
    {CLK_A CLK_A_DERIVED_*}  
    {CLK_B}  
    {JTAG_TCK}  
} -exclude {{PATTERN_X* PATTERN_Y*}}
```

`-exclude` takes pairs of patterns and suppresses that part of the cross product, for clock pairs that already have a `-logically_exclusive` or `-physically_exclusive` relationship that `-allow_paths` would contradict. Omit it unless you hit a **TIM-184** conflict.

### 5.5.5. 5. Declare periods for generated clocks

A generated clock exposes no readable `period` attribute in Design Compiler, so any generated clock that clocks a CDC element needs an explicit entry in `<block>_cdc_max_delay.tcl`:

```
set ::cdc_clock_period(MY_DIV_CLK) [expr {4 * $clock_periods(MY_MASTER_PERIOD)}]
```

Omitting one is loud — a **no period for clock** warning per affected element — so a clean run means you have them all. Keep them in step with your **create\_generated\_clock** divide factors; nothing cross-checks them.

### 5.5.6. Also check

**Hierarchy preservation.** The reference flow applies **set\_ungroup false** and **set\_boundary\_optimization false** to the CDC designs before **compile**. How much this matters depends on item 1: with leaf cell pins it is a safety margin, with wrapper ports it is load-bearing. Either way, your flow needs the equivalent, and it needs to name your cells rather than OCAH's.

**Delay factors.**  $0.5 * T_{dst}$  and  $1.0 * T_{dst}$  are policy, not physics. If your methodology wants more settling margin, change **cdc\_sync\_delay** and **cdc\_data\_delay** in **flows/synth/constraints/cdc\_max\_delay\_procs.tcl**. Per-instance overrides are also available without editing procs: every proc takes an optional trailing delay in picoseconds, and re-stating a call in **<block>\_cdc\_max\_delay.tcl** after the **source** wins.

**Tool differences.** Two Design Compiler specific choices to revisit under PrimeTime. **get\_attribute <pin> clocks** is used because **get\_clocks -of\_objects** is PrimeTime-only, and generated clocks may expose a readable **period** in PrimeTime, making item 5 unnecessary.

## 5.6. Confirming it worked

Load the constraints and check the summary:

```
INFO: cdc_max_delay: SUMMARY <n> exceptions, 0 warnings
```

Zero warnings is necessary but not sufficient. A pin pattern that matches nothing warns, but one that matches the **wrong** thing does not. Two stronger checks:

- **Predict the count.** Multiply each proc's number of emitted exceptions by its number of calls and compare against the summary. A mismatch localises the failure, because the size of the shortfall identifies which proc did not fire.
- **Confirm the precedence.** Layer 2 is on pins and layer 1 on clocks, and the design assumes the more specific specification wins. Verify once in your tool:

```
report_timing -to [get_pins <a synchronizer>/i_d] -nworst 1
```

The **max\_delay** line must show your  $0.5 * T_{dst}$  value rather than the default. If it shows the default, layer 2 is inert and nothing else in this section applies.

**SUMMARY 0 exceptions** is treated as a setup failure and warns as such. It means the constraints did not load, not that the design has no crossings.

## 5.7. Scope and limitations

The 20 ns default is loose by design. Against a 1 ns clock it permits twenty clock periods of delay into a synchronizer

input. It is a safety net that keeps `-allow_paths` usable, not a real constraint; any crossing that matters belongs in layer 2.

Enumeration works from a fixed list of module names, so a hand-written two-flop synchronizer that uses none of them will not be found. Structural detection of crossings is what a static CDC tool does, and this flow does not attempt it. These constraints time crossings already known to exist; they are not a substitute for CDC signoff.

Because the instance list is generated offline, removals are loud but additions are silent. A stale instance path warns, but a synchronizer added without regenerating produces no warning: the constraint never exists, the path falls back to the default, and it passes timing. Regenerate and diff the result against what is checked in whenever the RTL changes.